

DISCRETE MATHEMATICS

Handwritten Notes

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(Hinglish Weekday)

Other Notes Uploaded Below

<https://t.me/pwgatenotes>

Graph Theory

Basics of Graph:

- * Point/joint/node $\rightarrow$ vertex
- * line/arc/branch $\rightarrow$ edge

$\rightarrow$ Graph $G = (V, E)$

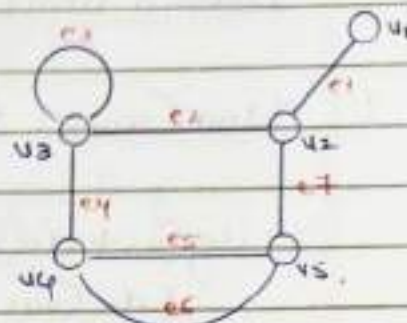
Set of vertices $\rightarrow$ Set of edges

$$V = \{u_1, u_2, u_3, u_4, u_5\}$$

$$E = \{e_1, e_2, \dots, e_7\}$$

$$e_1 \rightarrow (u_1, u_2) \mid (u_2, u_1)$$

* Each edge must be associated with unordered pair of vertex.



$$G = (V, E, \psi)$$

$$V = \{ \dots \}$$

$$E = \{ \dots \}$$

* Each edge must be associated with unordered pair of vertices.

$$\psi : E \rightarrow V \times V$$

* Incident point: Meeting point of vertex and edge

* End vertex: each edge is associated with unordered pair of vertex called as end vertex.

* Loop/self-loop: if end vertices are same, edge is called loop.

* Parallel edges: two or more edges associated with same end vertex.

$$\text{eg: } e_5 \rightarrow (u_4, u_5) \\ e_6 \rightarrow (u_4, u_5)$$

* Degree/valency ($d(u_i)$): no. of edges incident with vertex.

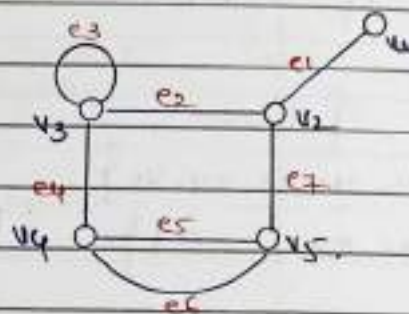
- Pendant vertex: degree one vertex is called pendant vertex.
- Isolated vertex: degree '0' vertex is called isolated vertex.
- New Graph: Set of isolated vertex.

$$\rightarrow \deg(v_1) + \deg(v_2) + \deg(v_3) + \deg(v_4) + \deg(v_5)$$

$$= 1 + 3 + 4 + 3 + 3$$

$$= 14 = 2 \times 7$$

no. of edges



[Theorem 1]

Sum of degree of all vertices is equal to twice the no. of edges.

[Theorem 2]

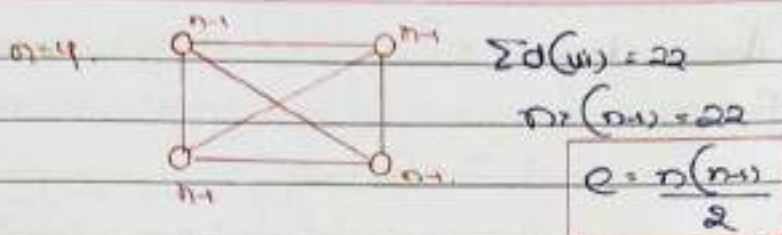
Sum of degree of all vertices will be even.

[Theorem 3]: No. of odd degree vertices in a graph can always be even.



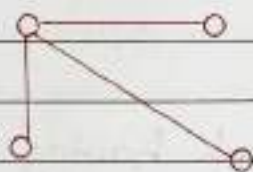
	loop	'p' edge	
Simple Graph	x	x	
Multi Graph	x	✓	
Pseudo Graph	✓	✓	

[Theorem 3]: Maximum no. of edges in Simple Graph $\leq \frac{n(n-1)}{2}$



[Theorem 4]: Maximum degree in Simple graph $\leq n-1$.

$n=4$



Note: if degrees of all vertices are $n-1$, then it will have exactly $\frac{n(n-1)}{2}$ edges.

eg: $n=4$

Maximum edges $= \frac{4 \cdot 3}{2} = 6$

n = total vertices

$\rightarrow$ Total no. of Graphs $= 2^{\frac{n(n-1)}{2}}$

Total no. of graphs $= 2^{\frac{4 \cdot 3}{2}} = 2^6$

Q1. How many graphs are possible with 4 vertices and 1 edge?

Q2. How many graphs possible with n vertices and e edges.

$\rightarrow \frac{n(n-1)}{2} = e$
 C_e

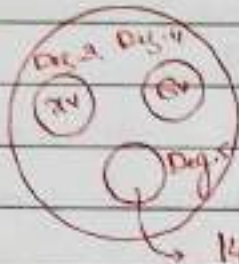
Q3. How many graphs are possible with 4 vertices and at least 2 edges

Ans. $m_1 \rightarrow 6C_2 + 6C_3 + 6C_4 + 6C_5 + 6C_6$
 $m_2 \rightarrow 2^6 - 6C_0 - 6C_1$

Q4. A certain graph as 'G' has order 14 and size 27. The degree of each vertex of G is 3, 4 or 5. There are 6 vertices of degree 4. How many vertices of G have degree 3 and how

how many have degree 5?

-Ans. $n=14$, $E=27$.



$$14 - 6 - x$$

$$(8-x)v$$

$$1 \quad \sum d(v_i) = 2E$$

$$6 \times 4 + x \times 3 + (8-x) \times 5 = 2 \times 27$$

$$\underline{x=5}$$

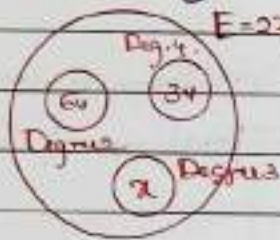
eg. Consider a graph having 27 edges

$6v \rightarrow \text{Degree} \rightarrow 2$

$3v \rightarrow \text{Degree} \rightarrow 4$

Remaining vertices $\rightarrow \text{Degree} \rightarrow 3$

Solⁿ:



$$E=27$$

$$\sum d(v_i) = 2E$$

$$6 \times 2 + 3 \times 4 + x \times 3 = 2 \times 27$$

$$12 + 12 + 3x = 54$$

$$3x = 54 - 24 = 30$$

$$3x = 30$$

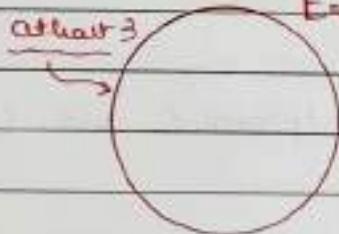
$$\underline{x=10}$$

Total vertices

$$= 6 + 3 + 10$$

$$= \underline{19}$$

eg: Consider a graph having 15 edges, degree of each vertex is atleast 3. What can be the number of vertices?



$$E=15$$

$$\delta(G) = 3 \quad n = ?$$

$$\delta(G) \leq \frac{2E}{n} \leq \Delta(G) \leq n-1$$

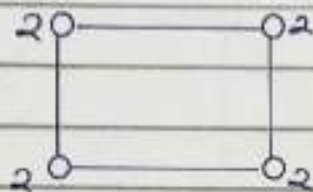
$$\delta(G) \leq \frac{2E}{n} \Rightarrow 3 \leq \frac{2 \times 15}{n} \Rightarrow n \leq 10$$

Maximum no. of vertices = 10.

Minimum degree ($\delta(G)$)
Maximum degree ($\Delta(G)$)

Case I:

$$\delta(G) = \frac{2e}{n} = \Delta(G)$$



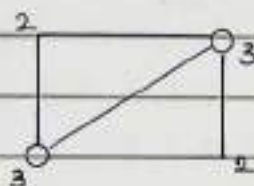
$$\delta(G) = 2$$

$$\Delta(G) = 2$$

$$\frac{2e}{n} = 2$$

$$\text{avg. degree} = \frac{2+2+2+2}{\text{Total vertices}} = \frac{\sum d(u_i)}{n} = 2$$

eg:



$$\delta(G) = 2$$

$$\Delta(G) = 3$$

Case II:

$$\delta(G) < \frac{2e}{n} < \Delta(G)$$

Theorem 5:

$$\delta(G) \leq \frac{2e}{n} \leq \Delta(G) \leq n-1$$

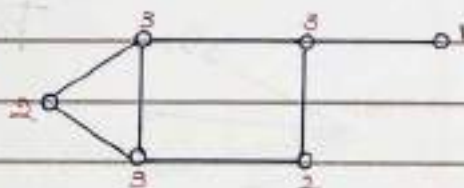
eg: G is Graph $\delta(G) \geq 3$, each vertex is having degree at least 3
Maximum value of n is 16.

$$E = 25 \quad \delta(G) = 3$$

$$\delta(G) \leq \frac{2e}{n} = 3 \leq \frac{2 \cdot 25}{n} \Rightarrow n \leq \frac{50}{3} \therefore n = 16$$

Degree Sequence:

Writing degree of all vertices either in increasing or decreasing order is called "Degree Sequence".



$$\rightarrow 3, 3, 3, 2, 2, 1 \quad \text{or} \quad 1, 2, 2, 3, 3, 3$$

eg: How many will be the no. of edges in $\delta(G) = 5, 2, 2, 2, 1$?

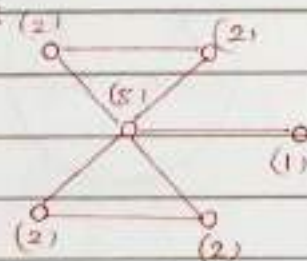
$$\text{m1: } \sum d(u_i) = 2e$$

$$5+2+2+2+1$$

$$14 = 2e$$

$$e = 7$$

m2:



$$e = 7$$

Q2. What will be edge in 2's of 3, 3, 3, 1?

m1:

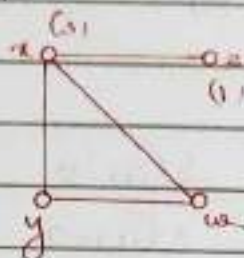
$$\sum d(u_i) = 2e$$

$$3 + 3 + 3 + 1 = 2e$$

$$10 = 2e$$

$$e = 5$$

m2:



Degree Sequence $\rightarrow$ Simple Graph

$\rightarrow$ Graphical Sequence

5, 2, 2, 2, 2, 1 $\rightarrow$ Graphical Sequence

3, 3, 3, 1 $\rightarrow$ No Simple Graph

Graphical?

(a)

5, 4, 3, 2, 1

Not Graphical

Reason 1: Total vertices = 5, $n=5$

$$\Delta(G) \leq n-1$$

$$\Delta(G) \leq 4$$

Reason 2: Theorem 2

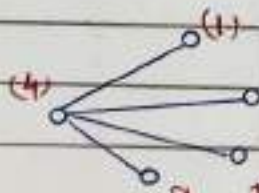
(5) 4, (3) 2, (1)

(b)

4, 4, 3, 2, 1

Not Graphical

Total vertices = 5



2 Demand: 3

Total vertices = n

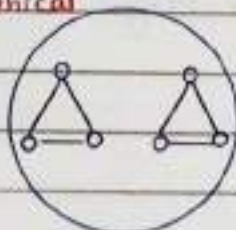
$$\rightarrow n-1, n-1, \dots, 1$$

$\rightarrow$ not graphical (2, (n-1) and a vertex cannot be graphical)

(c)

2, 2, 2, 2, 2, 2

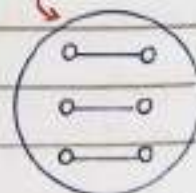
Graphical



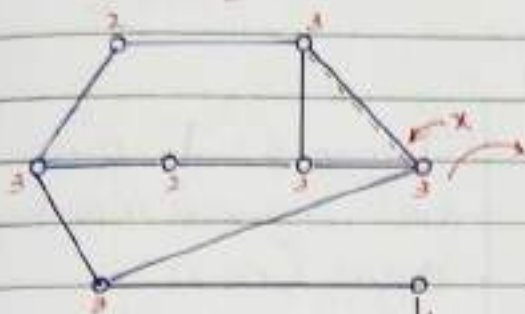
(d)

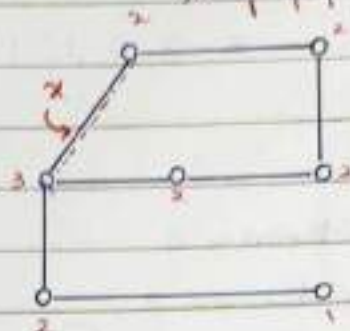
1, 1, 1, 1, 1, 1

Graphical



Havel-Hakimi

$$\begin{array}{ccccccc} 3 & 3 & 3 & 3 & 3 & 2 & 2 & 1 \\ & 2 & 2 & 2 & & & & \end{array}$$


$$\begin{array}{ccccccc} 2 & 2 & 2 & 2 & 2 & 0 & 0 & 1 \\ & 1 & 1 & 1 & 1 & & & \end{array}$$


removing
that
vertex

$$\begin{array}{ccccccc} 2 & 2 & 1 & 1 & 1 & 1 \\ & 1 & 1 & 1 & 1 & 0 \end{array}$$

$$\begin{array}{ccccccc} 1 & 1 & 1 & 1 & 0 \end{array}$$

$$\begin{array}{ccccccc} 1 & 1 & 0 & 0 \end{array}$$

$$\begin{array}{cc} 0 & 0 \end{array}$$

$$\begin{array}{cc} 0 & 0 \end{array}$$

$$\begin{array}{cc} 0 & 0 \end{array}$$

$$\begin{array}{cc} 0 & 0 \end{array}$$

$$\begin{array}{cc} 0 & 0 \end{array}$$

$$\begin{array}{cc} 0 & 0 \end{array}$$

$$\begin{array}{ccccccc} 3 & 3 & 3 & 3 & 3 & 2 & 2 & 1 \\ & 2 & 2 & 2 & 2 & & & \end{array}$$

$$\begin{array}{ccccccc} 3 & 2 & 2 & 2 & 2 & 2 & 1 \\ & 1 & 1 & 1 & 1 & 1 & \end{array} \quad (\text{ordering})$$

$$\begin{array}{ccccccc} 2 & 2 & 1 & 1 & 1 & 1 \\ & 1 & 0 & & & \end{array} \quad (\text{ordering})$$

$$\begin{array}{ccccccc} 1 & 1 & 1 & 1 & 0 \\ & 0 & & & \end{array} \quad (\text{ordering})$$

$$\begin{array}{ccccccc} 1 & 1 & 0 & 0 \end{array}$$

if this can be a simple graph, its previous ones also can.

eg Graph Pra:

a. 7 6 5 (4 4) 3 2 1 ✓

b. 6 6 6 6 (3 3) 2 2 ✗

c. 7 (6 6) (4 4) 3 (2 2) ✓

d. 8 (7 7) 6 4 2 (1 1) ✗

a.
$$\begin{array}{ccccccc} 7 & 6 & 5 & 4 & 4 & 3 & 2 & 1 \\ & 5 & 4 & 3 & 3 & 2 & 1 & 0 \\ & 3 & 2 & 2 & 1 & 0 & 0 & 0 \\ & 1 & 1 & 0 & 0 & 0 & 0 & 0 \end{array}$$

b.
$$\begin{array}{ccccccc} 6 & 6 & 6 & 6 & 3 & 3 & 2 & 2 \\ & 5 & 5 & 2 & 2 & 2 & 1 & \end{array}$$

$$\begin{array}{ccccccc} 4 & 4 & 1 & 1 & 1 & 1 \\ & 3 & 0 & 0 & 0 & 1 \end{array}$$
 } no
Simple
graph

$$\begin{array}{ccccccc}
 (c) & 7 & 6 & 6 & 4 & 4 & 3 & 2 & 2 \\
 & & 5 & 5 & 3 & 3 & 2 & 1 & 1 \\
 & & & 5 & 2 & 2 & 1 & 1 & 0 \times
 \end{array}$$

$$\begin{array}{ccccccc}
 (d) & 8 & 7 & 7 & 6 & 4 & 2 & 1 & 1 \times \\
 & & & & n=8 \\
 & & & & \Delta(4) \leq 7.
 \end{array}$$

Steps:

1. Check theorem 2
2. Check theorem 3
3. $0-1, 0-1, \dots, 1$ (not possible)
4. All degrees are distinct (graph not possible)
5. Hamey Harimi.

Theorem 6: In Simple graph at least 2 vertex will have same degree ($n \geq 2$).

Q1. $0 \leq x \leq 5$ for which value of x it is graphical?

$$x: 1, 2, 3, 5, 5.$$

$$\begin{array}{ccccccc}
 5, & 5, & 3, & 2, & 1, & x. \\
 n-1, & n-1, & \dots, & 1.
 \end{array}$$

$\therefore$ for no values it will be graphical.

Total vertices = 6. } not graphical

Q2. $0 \leq x \leq 7$ for which value of x it is graphical?

$$7 \ 6 \ 5 \ 4 \ 3 \ 2 \ 1 \ x$$

x cannot be odd

$$x=0 \quad 7 \ 6 \ 5 \ 4 \ 3 \ 2 \ 1 \ 0 \times \text{ (all distinct)}$$

$$\begin{array}{ccccccc}
 x=2 & 7 & 6 & 5 & 4 & 3 & 2 & 2 & 1 \\
 & 5 & 4 & 3 & 2 & 1 & 1 & 0
 \end{array}$$

$$\begin{array}{ccccccc}
 x=4 & 7 & 6 & 5 & 4 & 4 & 3 & 2 & 1 \\
 & 5 & 4 & 3 & 3 & 2 & 1 & 0
 \end{array}$$

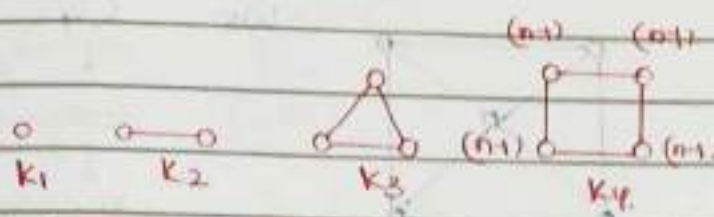
$$\begin{array}{ccccccc}
 & 3 & 2 & 2 & 1 & 0 & 0 \\
 & & 1 & 1 & 0 & 0 & 0.
 \end{array}$$

$$3 \ 2 \ 1 \ 0 \ 0 \ 0. \times \text{ (not possible)}$$

} Graphical $\checkmark$

$\therefore$ for $x=4$ it is graphical

H.W $\left\{ \begin{array}{l} x, 7, 7, 5, 5, 4, 3, 2 \\ 0 \leq x \leq 7 \\ \text{value for } x \text{ for which it is graphical} \end{array} \right.$

Complete Graph: (K_n) ($n \geq 1$)

* Degree of all vertices are $n-1$.

$$\sum d(v) = 2e$$

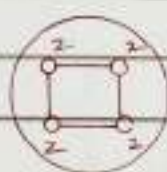
$$n(n-1) = 2e$$

$$\boxed{\delta(G) = \frac{2e}{n} = \Delta(G) = n-1}$$

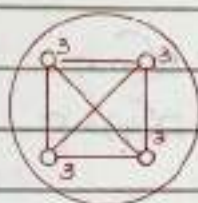
$$\boxed{e = \frac{n(n-1)}{2}}$$

Regular Graph ($\delta(G) = \frac{2e}{n} = \Delta(G)$)

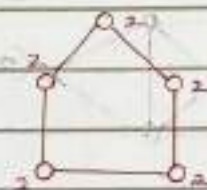
→ If degrees of all vertices are same then it is called Regular Graph.



2-Regular



3-Regular



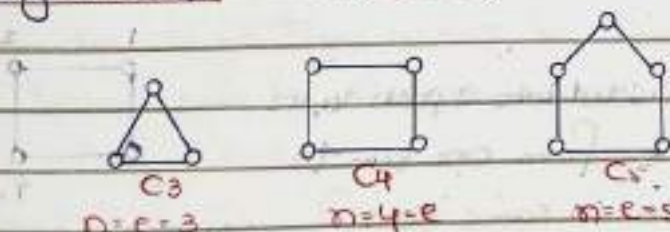
2-Regular

$$\delta(G) = \frac{2e}{n} = \Delta(G) = k$$

k-Regular Graph

Point about Complete Graph:

- * $n-1$ Regular Graph is a Complete Graph
- * all K_n are Regular Graph. (True)
- * all Regular Graph are K_n (False)

Cycle Graph: (C_n) ($n \geq 3$)

* Degree of all vertices are 2

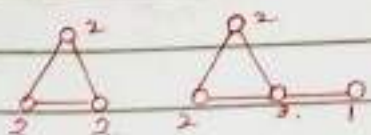
$$\sum d(v) = 2e$$

$$n \cdot 2 = 2e$$

$$* \underline{n=e}$$

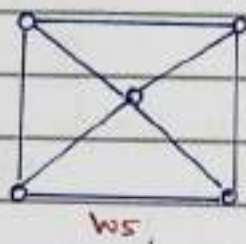
* All Cycle Graph are Regular graph

* All regular graph are Cycle graph.

Graph Contrary cycle

* if Graph is $C_n \rightarrow n=e$ (True)

* if $n=e \rightarrow$ then it is cycle graph (False)

Wheel Graph (W_n) ($n \geq 4$)

W_5
↓
 W_n

C_4
↓
4 edges + 4 edges
↓
 $n-1$ edges + $n-1$ edges

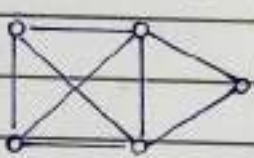
→ if G is $W_n \rightarrow e = 2(n-1)$ (True)

$$e(W_n) = n-1 + n-1$$

$$e(W_n) = 2(n-1)$$

→ if G is having $e = 2(n-1) \rightarrow G$ is wheel graph. (False)

eg.

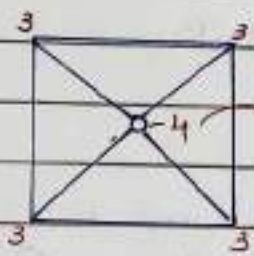


$\rightarrow n=5$

$$e = 2(n-1)$$

$$= 2(4) = 8$$

eg.:



$(n-1)$
 W_5

W_5

4, 3, 3, 3, 3

W_6

5, 3, 3, 3, 3, 3

5 vertices

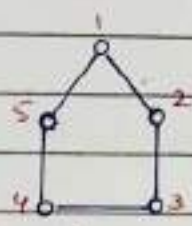
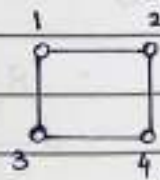
Properties of wheel graph
- all degrees are 3 (corner)
- middle vertex has degree $(n-1)$.

Bipartite Graph

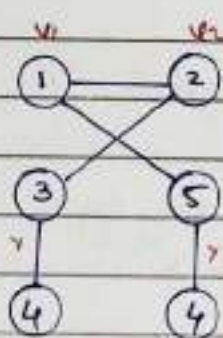
$$G = (V, E)$$

* Vertices can be divided into 2 parts V_1, V_2

* Each edge must be from one set to another set but not in same set.

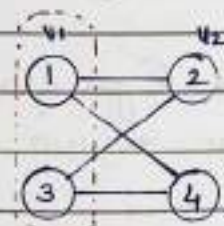


Not Bipartite Graph.

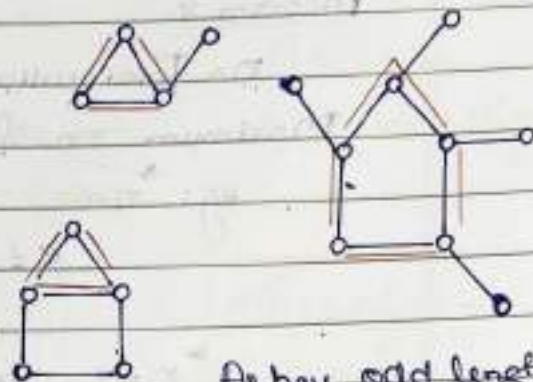
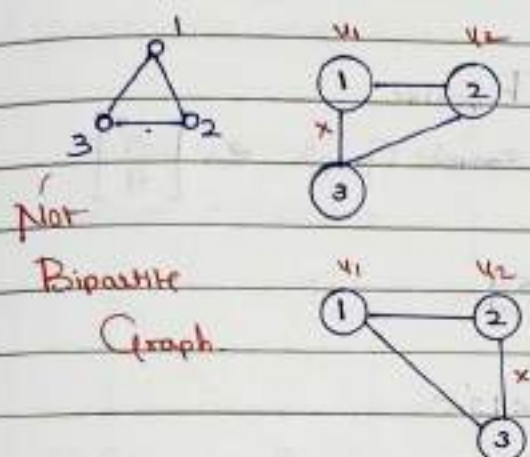


Because 4 is connected

to both 3 & 5 so cannot do partition.



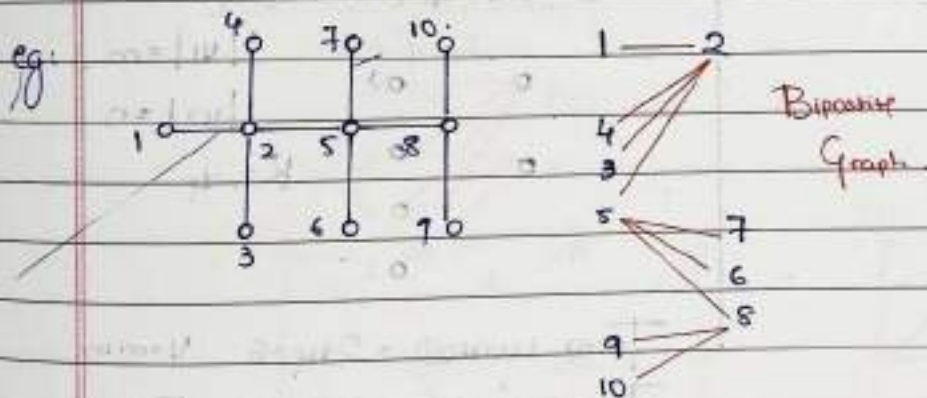
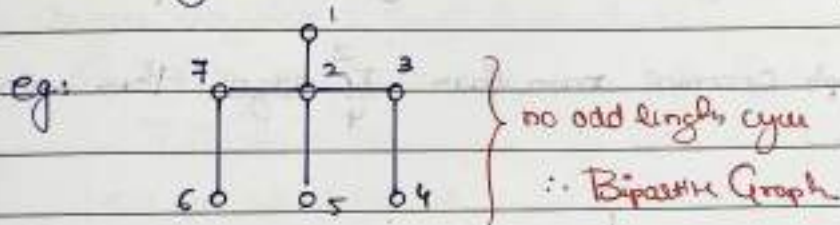
Bipartite Graph.



As have odd length cycles. ~~So not~~
Bipartite Graph.

Theorem 7.

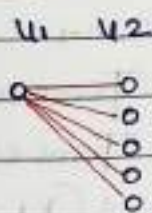
Bipartite graph does not contain odd length cycle.



Consider $n \rightarrow$ Total vertex in Bipartite Graph
What will be the maximum no. of edges?

$$n=6$$

Case 1:



Case 2:



Case 3:



Theorem 8 n = Total vertices

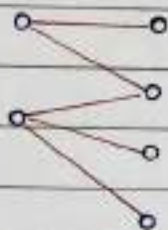
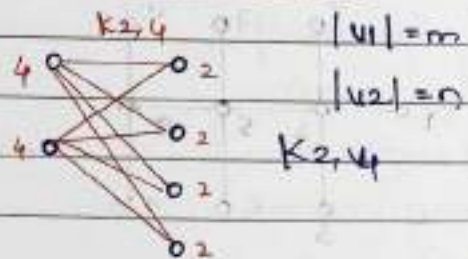
Bipartite

Maximum no. of edges in Simple Graph $e \leq \lfloor \frac{n^2}{4} \rfloor$ eg: $n=6$

$$e \leq \frac{6^2}{4} = e \leq 9$$

$$n=7 \quad e \leq \frac{7^2}{4} = e \leq 12.5$$

$$e \leq 12$$

 $\rightarrow$ Maximum no. of edges in bipartite graph $e \leq \frac{n^2}{4}$ $\rightarrow$ If bipartite graph contains more than $\frac{n^2}{4}$ edges then it contains odd length cycle.Bipartite GraphComplete Bipartite Graph ($K_{m,n}$)Total vertices = $2+4=6$ $V=m+n$ Total edges = $2 \times 4 = 8$ $E=m \cdot n$

$\Delta(K_{2,4}) = 4$

$\Delta(K_{m,n}) = \max(m, n)$

$\delta(K_{2,4}) = 2$

$\delta(K_{m,n}) = \min(m, n)$

Complete Bipartite GraphStar Graph ($K_{1,n-1}$)

Draw Star Graph of 6

vertices

$n=6$

$k_{1,5}$



$k_{1,n-1}$

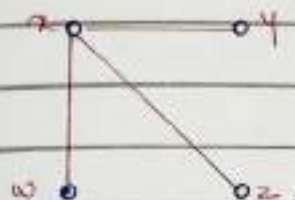
Total vertices = n Total edges = $n-1$

$\Delta(K_{1,n-1}) = n-1$

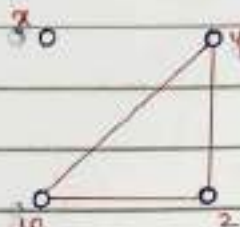
$\delta(K_{1,n-1}) = 1$

Compliment Graph ($\bar{G}$)

Edge $\rightarrow$ Present
edge $\rightarrow$ absent



edge $\rightarrow$ absent
edge $\rightarrow$ Present



$$G + \bar{G} = K_n$$

$$e(G) + e(\bar{G}) = \frac{n(n-1)}{2}$$

$$e(\bar{G}) = \frac{n(n-1)}{2} - e(G)$$

Q. What will be edge in the Compliment of the graph having degree sequence: 5, 2, 2, 2, 2, 1?

Ans. $G \rightarrow 5, 2, 2, 2, 2, 1$

$$\sum d(u_i) = 2e$$

$$5 + 2 + 2 + 2 + 2 + 1 = 2e$$

$$14 = 2e$$

$$e = 7$$

$$e(\bar{G}) = ?$$

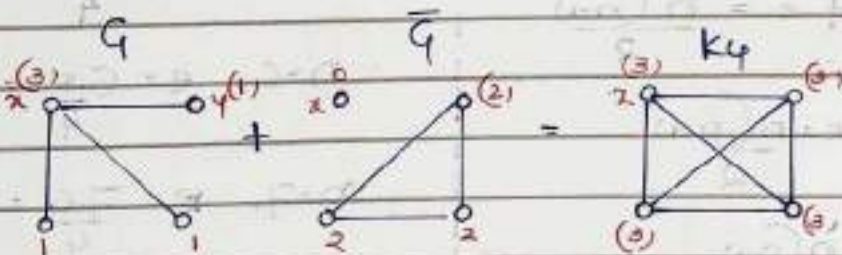
$$e(G) = 7$$

$$\Rightarrow e(\bar{G}) + e(G) = \frac{n(n-1)}{2}$$

$$\text{Total vertices} = 6$$

$$\Rightarrow 7 + e(\bar{G}) = \frac{6 \cdot 5}{2}$$

$$\Rightarrow e(\bar{G}) = 15 - 7 = 8$$



$$K_4 \quad 3 \quad 3 \quad 3 \quad 3$$

$$G \quad 3 \quad 1 \quad 1 \quad 1 \quad n=4$$

$$\bar{G} \quad 0 \quad 2 \quad 2 \quad 2$$

General: Note:

$$\rightarrow K_n \quad n-1 \quad n-1 \quad n-1 \quad \dots \quad n-1$$

$$G \quad d_1 \quad d_2 \quad d_3 \quad \dots \quad d_n$$

$$\bar{G} \quad n-1-d_1 \quad n-1-d_2 \quad \dots \quad n-1-d_n$$

eg: What will be no. of edges in the Compliment of the graph having DE 5, 2, 2, 2, 2, 1.

$\rightarrow$ Total vertices = 6.

$$K_6 \quad 5 \quad 5 \quad 5 \quad 5 \quad 5 \quad 5$$

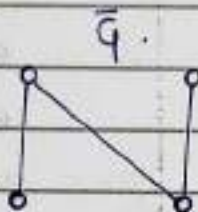
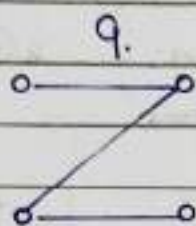
$$G \quad 5 \quad 2 \quad 2 \quad 2 \quad 2 \quad 1$$

$$\bar{G} \quad 0 \quad 3 \quad 3 \quad 3 \quad 3 \quad 4$$

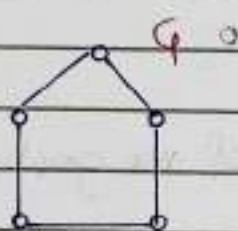
$$\sum d(u_i) = 2e$$

$$0 + 3 + 3 + 3 + 3 + 4 = 2e$$

$$e = 8$$

Self-Complement Graph, $G \cong \bar{G}$ 

Graph is same as its own complement



$\left\{ \begin{array}{l} n=5 \\ e=4 \end{array} \right.$

$\left\{ \begin{array}{l} n=5 \\ e=6 \end{array} \right.$

→ G is same as $\bar{G}$

$$G + \bar{G} = K_n$$

$$e(G) + e(\bar{G}) = \frac{n(n-1)}{2}$$

$$e + e = \frac{n(n-1)}{2}$$

$$2e = \frac{n(n-1)}{2}$$

$$e = \frac{n(n-1)}{4}$$

$$n=4 \quad e = \frac{4(4-1)}{4} = \frac{4 \cdot 3}{4} = 3 \checkmark$$

$$n=5 \quad e = \frac{5(5-1)}{4} = \frac{5 \cdot 4}{4} = 5 \checkmark$$

$$n=6 \quad e = \frac{6(6-1)}{4} = \frac{6 \cdot 5}{4} = \frac{15}{2} = 7.5 \times$$

$$n=7 \quad e = \frac{7(7-1)}{4} = \frac{7 \cdot 6}{4} = \frac{42}{4} \times \text{Should be divisible by 4}$$

→ $\frac{n}{4}$ or $\frac{n-1}{4}$ then we get self-complement graph
divisible by 4.

→ $a \equiv b \pmod{n}$

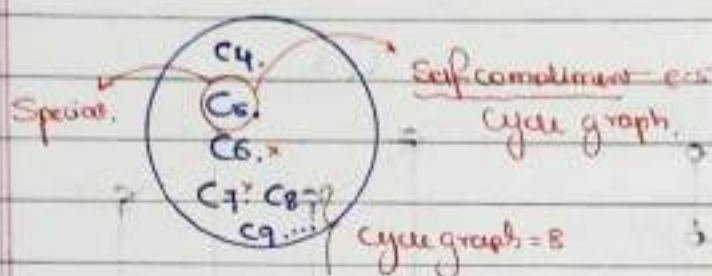
↳ $a \div n = b \div n$

1 and 5 are having same remainder with respect to 4

↳ $1 \equiv 5 \pmod{4}$

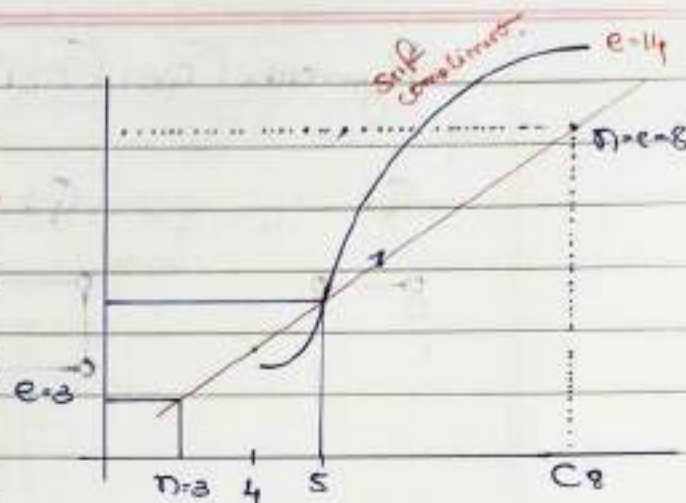
or

$$a \equiv b \pmod{n} = \frac{a-b}{n} \in \mathbb{Z}$$

Cycle graph.

Self complement 4.

* Only for C_5 we get Self Complement graph ($n=5$).

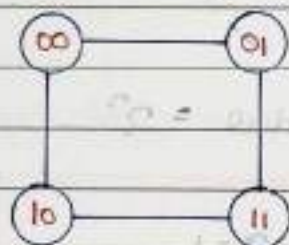


Q1. Graph vertices are represented as a n -bit signal, and 2 vertices are connected, if their bit position changed by 1. bit error will be the total edges?

Solⁿ: $n=2$ bit

Total vertices
= $2^2 = 4$

00
01
10
11



11

01

10

11

00

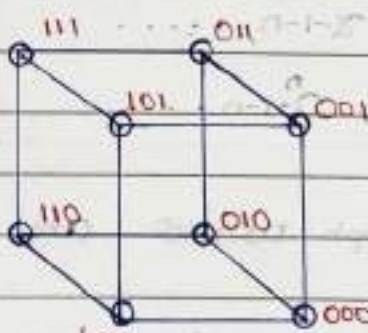


Degree of each vertex will be 2.

 $n=3$ bit

Total vertices
= $2^3 = 8$

000
001
010
011
100
101
110
111



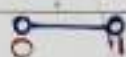
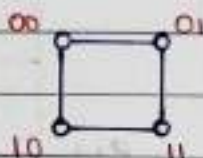
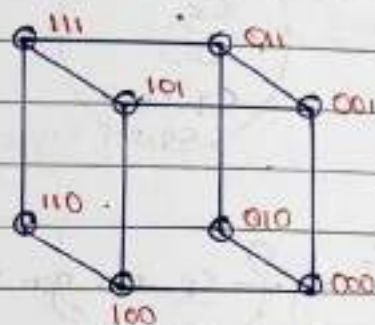
Degree of each vertex will be 3.

 $n \rightarrow$ bit signalTotal vertices (x) = 2^n Degree of each vertex = n

$$\sum d(v_i) = 2e$$

$$2^n \cdot n = 2e$$

$$e = n \cdot 2^{n-1}$$

Hypercube (Q_n) ($n \geq 1$) Q_1  Q_2  Q_3 

$$\rightarrow e(Q) = n \cdot 2^{n-1}$$

$$= \text{Total vertices} = \alpha = 2^n$$

$$= e(Q) + e(\bar{Q}) = kx$$

$$= n \cdot 2^{n-1} + e(\bar{Q}) = \frac{\alpha(\alpha-1)}{2}$$

$$e(\bar{Q}) = \frac{2^n(2^n-1)}{2} - n \cdot 2^{n-1}$$

$$\Rightarrow n \cdot 2^{n-1} + e(\bar{Q}) = \frac{2n(2^n-1)}{2}$$

$$\rightarrow \alpha = \text{Total vertices} = 2^n$$

$$Kx \rightarrow \alpha-1, \alpha-1, \alpha-1$$

$$Q \rightarrow n, n, n, \dots, n$$

$$\bar{Q} \rightarrow \alpha-1-n, \alpha-1-n, \dots$$

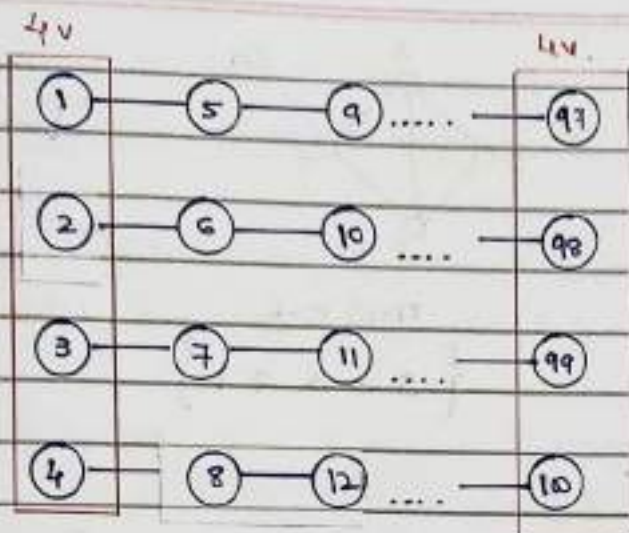
$$2^{n-1}-n, 2^{n-1}-n, \dots$$

Q2. Consider a graph vertices are represented as a no from $\{1, \dots, 100\}$

two vertices are connected $|a-b|=4$ eg $|1-5|=4$ or their difference is 4. like $|1-5|=4$, $|5-9|=4$

What will be the total edge in this?

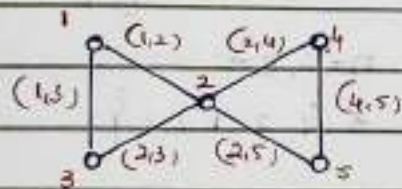
Ans:



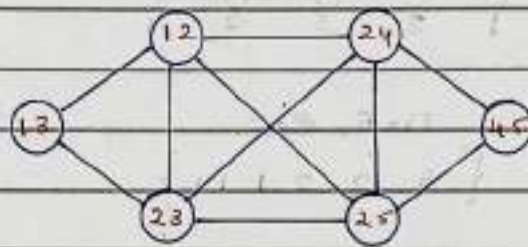
$$4 \times 1 + 4 \times 1 + 9 \times 2 = 2e$$

$$e = 96$$

Line Graph - $L(G)$

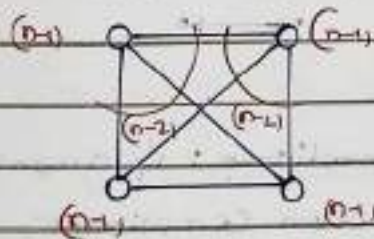


G

 $L(G)$

Line Graph
↓
edge graph.

Q3. What can be the degree sequence of $L(K_n)$?



∴ Degree of each vertex in $L(K_n)$ is $2(n-2)$.

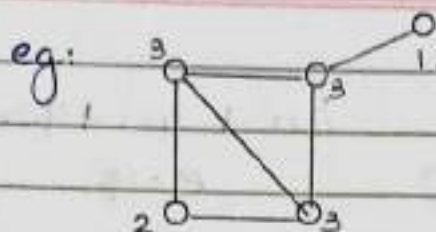
Isomorphic Graph

G_1 and G_2 are isomorphic to each other when they have same incident Property.

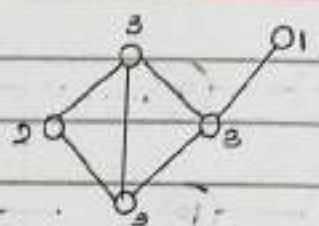
- no. of vertices
- no. of edges
- no. of degree sequence.

These properties are satisfied if two graphs are isomorphic.

If two graphs are isomorphic then they can have same vertices, edges and degree sequence and vice versa need not be true.



$n=5, e=6$
 $\{3, 3, 3, 2, 1\}$

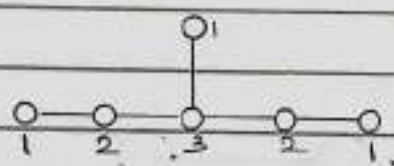


$n=5, e=6$
 $\{3, 3, 3, 2, 1\}$

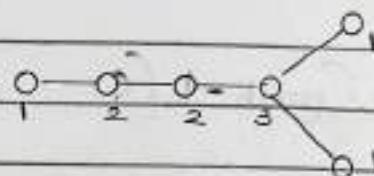
Isomorphic

One Graph but
different representation

eg:



$n=6, e=5$
 $\{3, 2, 2, 1, 1, 1\}$



$n=6, e=5$
 $\{3, 2, 2, 1, 1, 1\}$

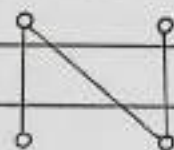
G_1, G_2 are isomorphic to each other.

$f: G_1 \rightarrow G_2$ are 1:1 Correspondence

$f: v_1 \rightarrow v_2$
• •

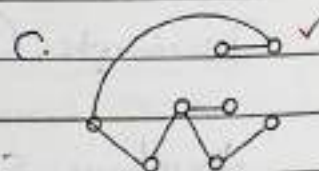
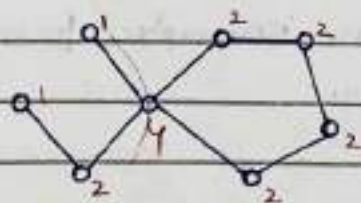
$f: E_1 \rightarrow E_2$
• •

$f: \varphi_1 \rightarrow \varphi_2$
• •

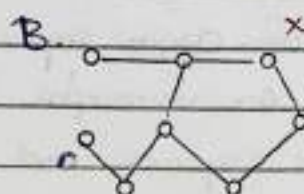


Self Complement: if graph is isomorphic to its own complement.

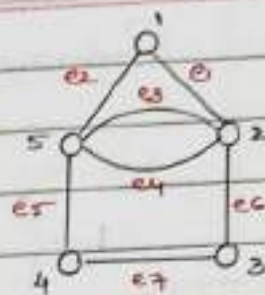
eg:



Cycle length = 5.



Walk: alternating sequence of vertices & edges.
(Repetition V / Repetition E)

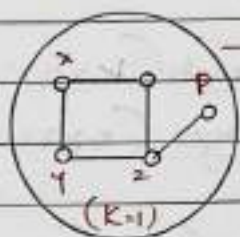


Trail: alternating sequence of vertices & edges
(Repeating V / Repeating E^x)

Path: alternating sequence of vertices & edges
(Repeating V^x / Repeating E^x)

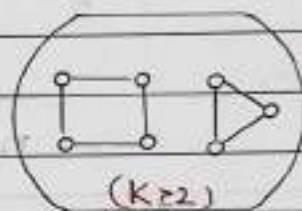
Graph

Connected



Path is available between all pair of vertices

Disconnected:



for atleast two vertices Path is not available

* Disconnected graph contains connected subparts (Component(K)).

Q1. Consider a star graph of 6 vertices, what will be the total edges in the complement of the graph?

Ans. $e(G) + e(\bar{G}) = \frac{n(n-1)}{2}$

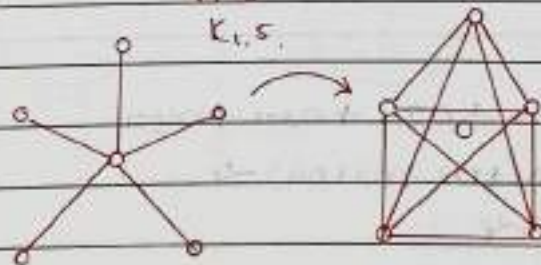
$n=6$

$K_{1,5}$

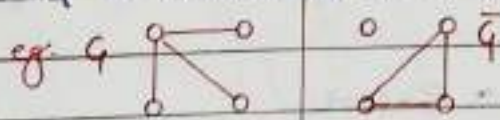
$5+x = \frac{6 \cdot 5}{2}$

$x = 15 - 5$

$x=10$



→ if G is connected then $\bar{G}$ will be connected. (False)



→ if G is disconnected then $\bar{G}$ will be connected. (True)

Connectivity in Graphs

$$1+2+3+4$$

$$1 \text{ or } 2 \text{ or } 3 \text{ or } 4$$

Connected Graph ($k=1$)Range of edges ($k=1$)

- Tree
1. $\begin{cases} n-1 \\ + \\ \text{Connected} \end{cases} \leq e \leq \frac{n(n-1)}{2} \quad (K_n)$
 2. Connected + it does not contain cycle (acyclic graph)
 3. Unique Path is available between all pairs of vertices.
 4. Minimally Connected.

Disconnected Graph ($k \geq 2$)Range of edges ($k \geq 2$)

$$* \quad n-k \leq e \leq \frac{(n-k)(n-k-1)}{2}$$

↓
min no.
of edges

* Forest: Collection of trees



$$\begin{aligned} e &= n_1 - 1 + n_2 - 1 + n_3 - 1 + n_4 - 1 \\ &= (n_1 + n_2 + n_3 + n_4) - 4 \\ &= n - 4 \end{aligned}$$

$$e = n - 4$$

$$e = n - k \quad (\text{min no. of edges})$$

→ Consider undirected graph having 10 vertices and 2 components
 $n=10$, $k=2$
 min no. of edges = $e = n - k$
 $= 10 - 2 = 8$

$$\text{Max no of edges} = e = \frac{(n-k)(n-k+1)}{2} = \frac{(10-2)(10-2+1)}{2} = \underline{\underline{36}}$$

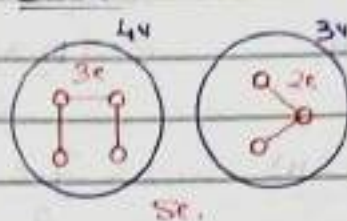
eg: Consider an undirected Graph having 7 vertices and 2 components.
 $n=7, k=2$.

$$e = n - k = 7 - 2 = 5 \text{ (min no. of edges).}$$

Case 1.



Case 2



Case 3



eg: $n=7, k=2$ max no. of edges $e = \frac{(n-k)(n-k+1)}{2} = \frac{5 \times 6}{2} = 15e$

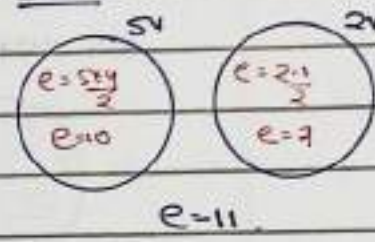
Case 1.



Case 2

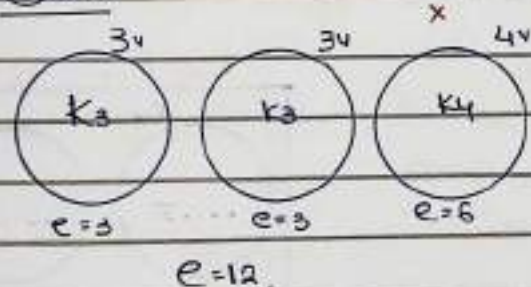


Case 3

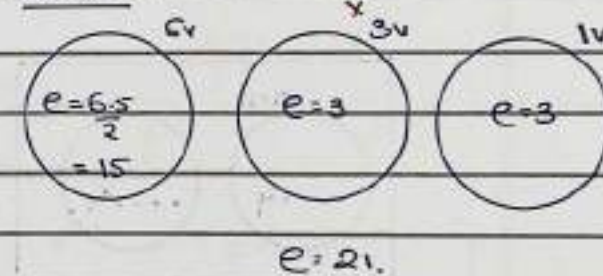


eg: $n=10, k=3$ $e = \frac{(n-k)(n-k+1)}{2} = \frac{(10-3)(10-3+1)}{2} = \frac{7 \times 8}{2} = 28e$

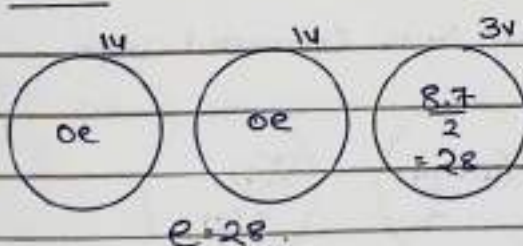
Case 1:



Case 2:



Case 3

 $n=10, k=3$ $e = 28$

2v 2v 6v

3v 3v 4v

...

1v 1v 8v

eg. Consider a disconnected Graph of 10 vertices, what will be the maximum no. of edges?

$$n=10 \quad k \geq 2$$

$$\begin{aligned} k=2 \\ e &= \frac{(n-k)(n-k+1)}{2} \\ &= \frac{(10-2)(10-2+1)}{2} \\ &= \underline{\underline{36e}} \end{aligned}$$

$$\begin{aligned} k=3 \\ e &= \frac{(10-3)(10-3+1)}{2} \\ &= \frac{7 \cdot 8}{2} \\ &= \underline{\underline{28e}} \end{aligned}$$



eg. Consider a disconnected graph of four components, what will be? - max no. of edges?

$$25 \text{ vertices} \rightarrow 1^{\text{st}}$$

$$26v \rightarrow 3$$

$$25 \text{ vertices} \rightarrow 2^{\text{nd}}$$

$$24v \rightarrow 4v$$

$$\begin{aligned} 25v \\ k=25 \\ \frac{25 \cdot 24}{2} \end{aligned}$$

$$\begin{aligned} 25v \\ k=25 \\ \frac{25 \cdot 24}{2} \end{aligned}$$

$$\begin{aligned} 26v \\ k=26 \\ \frac{26 \cdot 25}{2} \end{aligned}$$

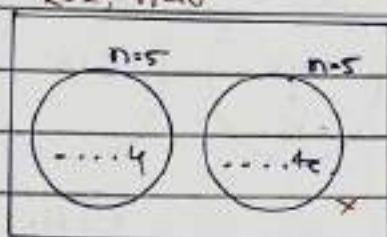
$$\begin{aligned} 24v \\ k=24 \\ \frac{24 \cdot 25}{2} \end{aligned}$$

$$\Rightarrow \underline{\underline{1201}}$$

[True]

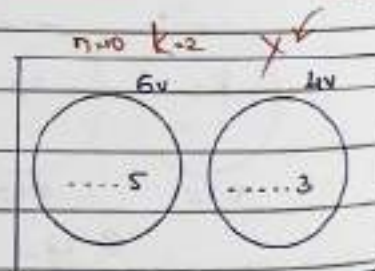
eg. Consider a graph of 10 vertices if $\delta(G) \geq 5$ then it is connected?
 $n=10 \quad \delta(G) \geq 5$ degree of all vertices are atleast 5.
 - Assume: let's take disconnected (wrong assumption)

$$k=2, n=10$$



$\delta(G) \geq 5$
Given

Degree should
be atleast 5.



$\therefore$ Since the assumption of disconnected is wrong, it is connected.

Theorem: If $\delta(G) \geq \frac{n-1}{2}$ then it is connected graph.

- Assume: it is disconnected.

$$\delta(G) = \frac{n-1}{2}$$

$$v_1 = \frac{n-1}{2} + 1$$

$$\delta(G) = \frac{n-1}{2}$$

$$v_2 = \frac{n-1}{2} + 1$$

$$V = v_1 + v_2$$

$$= \frac{n-1}{2} + 1 + \frac{n-1}{2} + 1$$

$$= \frac{2(n-1) + 2}{2} \Rightarrow n-1+1 = n+1$$

Total vertices = n

$\therefore$ Assumption is wrong

It is connected

→ Max. no. of edges

$$= \frac{(n-k)(n-k+1)}{2}$$

$$k=2 \text{ (2 components)}$$

$$= \frac{(n-2)(n-2+1)}{2}$$

$$= e = \frac{(n-2)(n-1)}{2}$$

$$n=5$$

$$e = \frac{(5-1)(5-2)}{2}$$

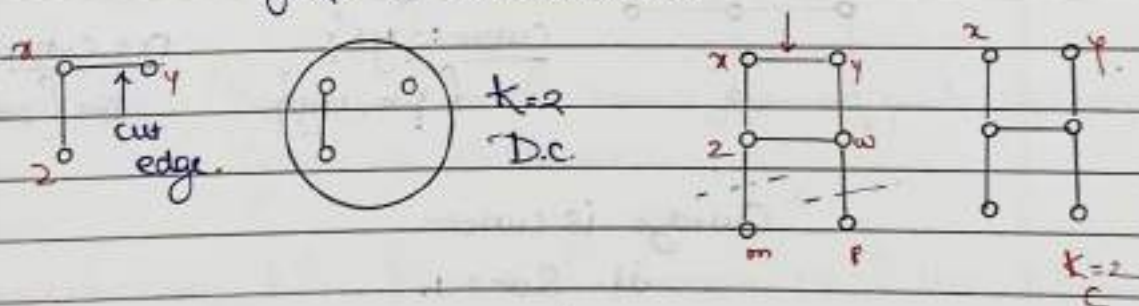
$$e = \frac{(5-1)(5-2)}{2}$$

$$= \frac{4 \cdot 3}{2} \Rightarrow \underline{\underline{6}}$$

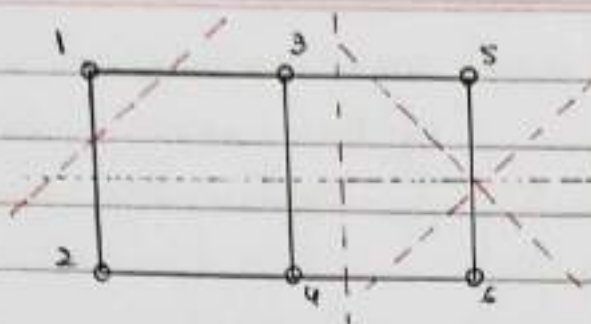
* If a graph contains $\frac{(n-1)(n-2)}{2}$ edges
 $\nearrow$ may be connected
 $\searrow$ may not be connected

* If a graph contains more than $\frac{(n-1)(n-2)}{2}$ no. of edges then it will be connected.

Cut edge/bridge: Removal of single edge from a graph will make graph as disconnected.



* If cut edge exist then it does not belongs to cycle.

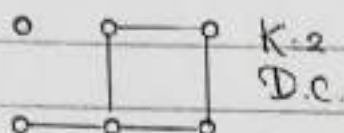


→ no cut edge

1 edge

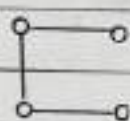
Cut edge set / cut set

Removal of set of edges from a graph will make graph as disconnected



Edge Connectivity / min-cut: $\lambda(G)$

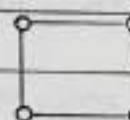
Removal of min. no. of edges from a graph will make graph as disconnected



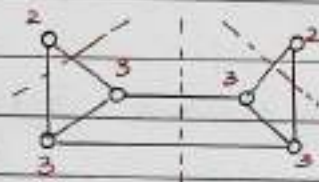
$$\lambda(G) = 1$$



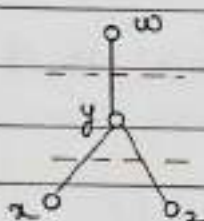
$$\lambda(G) = 2$$



$$\lambda(G) = 2$$

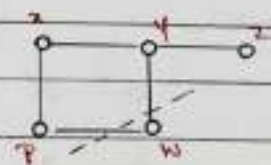


$$\lambda(G) = 2$$



$\{e_1, e_2, e_3\}$

if Subset of any set is cutset then whole set is not cutset.



Cutset: $\{y, z\}$

$\{p, u, w, y\}$

not cutset:

$A = \{p, u, w, y, z\}$

↓
Cutset

* Cut edge is cutset
if $\delta v = 1$.

Subset is cutset
 $A \rightarrow$ not cutset.



$\{2, 3, 3, 4\} \rightarrow$ cutset

$\{2, 3, 3, 4, 4, 7\} \rightarrow$ not cutset

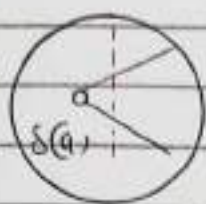
cutset

Size 1

Size 2

cut edge

cutset

Every Graph $\rightarrow S(a)$

edge } min no. of
connecting } edges



$$S(a) = 2 \quad \lambda(a) = 2$$

$$S(a) = \lambda(a) \quad (\text{I})$$



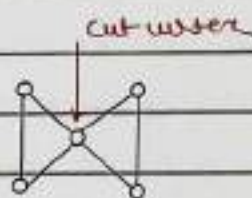
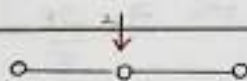
$$S(a) = 2 \quad \lambda(a) = 2$$

$$\lambda(a) < S(a) \quad (\text{II})$$

$$\lambda(a) \leq S(a)$$

Cut vertex / Cut point / Articulation point

Removal of single vertex from a graph will make graph as a disconnected graph.

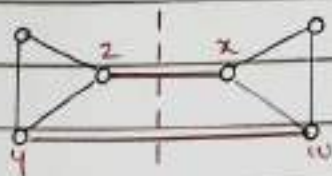


* If cut edge exist then cut vertex also exist for vertex ≥ 2 .

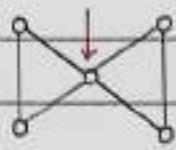
$\rightarrow$ $\circ - \circ$ (above statement is false for this example)

Cut vertex set: Removal of set of vertices from a graph will make graph as disconnected.

Vertex connectivity $k(G)$: Removal of minimum no. of vertices from a graph will make graph as disconnected.

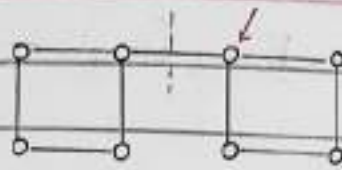


$$k(G) = 2$$



$$\lambda(G) = 2 \quad k(G) = 1$$

$$k(G) < \lambda(G)$$



$$\lambda(G) = 1 \quad k(G) = 1$$

$$k(G) = \lambda(G)$$

$$k(G) \leq \lambda(G)$$

7 edge conn.

$$\lambda(G) \leq \delta(G)$$

$$k(G) \leq \lambda(G)$$

vertex connectivity

$$k(G) \leq \lambda(G) \leq \delta(G)$$

Theorem:

$$\delta(G) \leq 2e/n \leq \Delta(G) \leq n-1$$

$$k(G) \leq \lambda(G) \leq \delta(G) \leq 2e/n \leq \Delta(G) \leq n-1$$

Q1. G is connected graph with 10 vertices and vertex connectivity min no. of edges in G ?

Solⁿ:

$$n=10 \quad k(G)=3$$

$$k(G) \leq \lambda(G) \leq \delta(G) \leq \frac{2e}{n} \leq \Delta(G) \leq n-1$$

$$k(G) \leq \frac{2e}{n} \Rightarrow 3 \leq \frac{2e}{10}$$

$$30 \leq 2e$$

$$15 \leq e$$

$$\Rightarrow 15 \leq 15, 16, 17 \dots$$

Q2. G is bipartite graph of 9 vertices and min no of edges & $k(G)$?

$k=8$

9 vertices

G is bipartite graph + max. no of edges

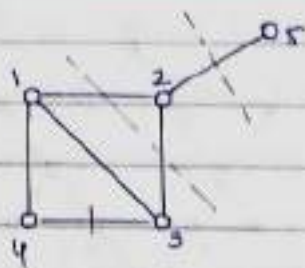
$$K_{4,5} \begin{cases} \lambda(K_{4,5}) = 4 \\ k(K_{4,5}) = 4 \end{cases}$$

$K_{m,n}$

$$k(K_{m,n}) = \min(m,n)$$

$$\lambda(K_{m,n}) = \min(m,n)$$

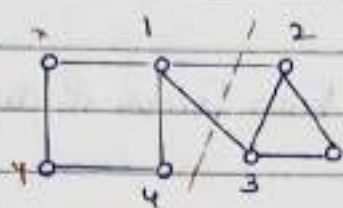
eg: 1

 $\{1, 2, 2, 4\}$ cutset $\{1, 2, 2, 4, 1, 5\}$

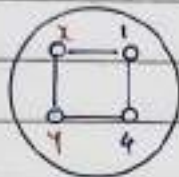
cutset

not cutset.

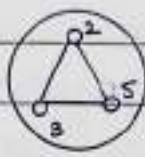
eg: 2

 $\{1, 2, 1, 3\}$ cutset. $\{1, 2, 1, 3, 1, 4\}$ not cutset.

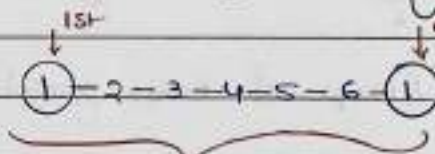
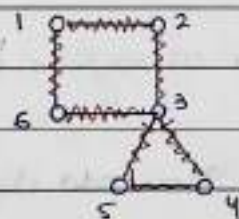
cutset.



V1



V2

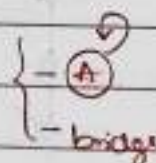
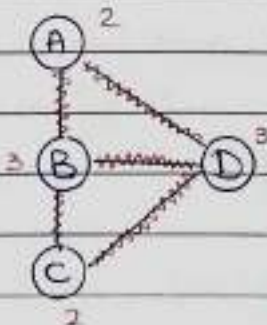
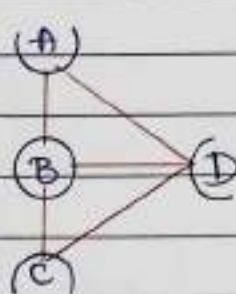
Euler Graph:Trail: It is an alternating sequence of vertices and edges $R_v/R_x E$.Closed Trail: Trail + starting = ending vertex.Euler Circuit: Closed Trail + cover all edges exactly once.

Representation of euler circuit.

 $G = (V, E)$

Euler Graph.

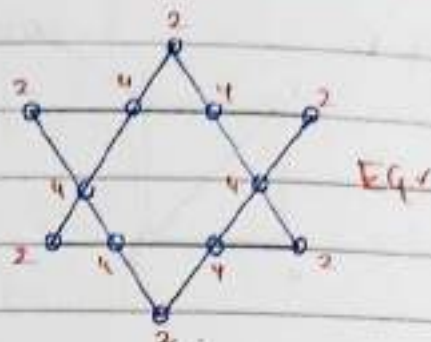
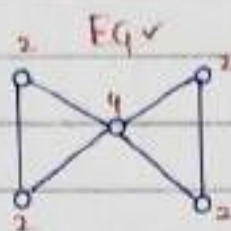
eg:



bridge

Closed ✓

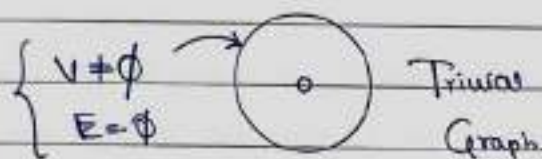
all edges ✓



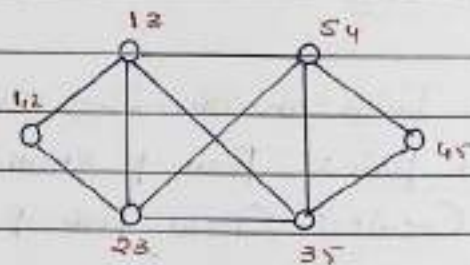
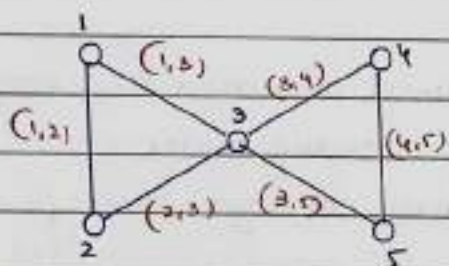
* All vertices degree will be even in Eq.

* Non-trivial connected graph is Euler Graph if degree of all vertices are even.

Trivial $\rightarrow$ Defaut.

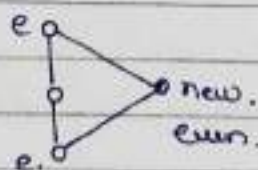


$G(V, E)$
 $\downarrow$ Set $\rightarrow$ Set.



* Line Graph of euler graph will always be Euler Graph

Q1. Consider a connected graph having some odd degree vertices, add new vertex make it connect to all odd degree vertices, this resultant graph would be Euler.

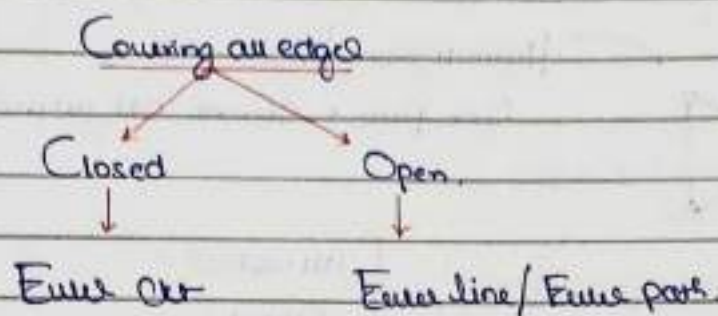


$\therefore$ all degrees are even

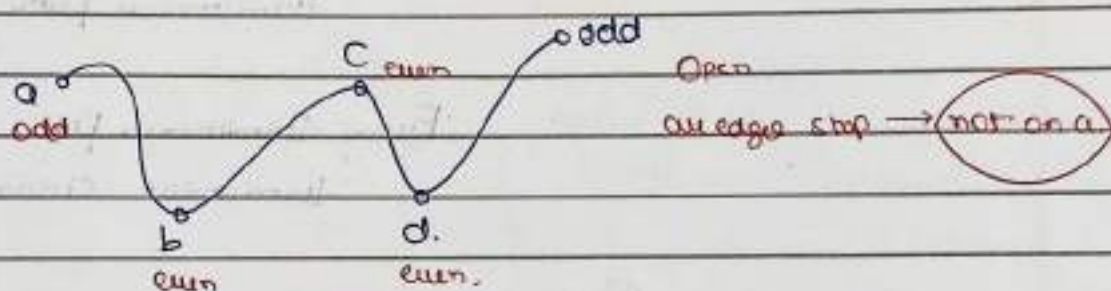
$\therefore$ Euler.

Open Trail: Trail + St + Ending vertex

Euler line: Open Trail + covering all edges exactly once
(Unimodal line / Euler path)

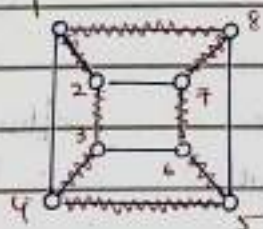
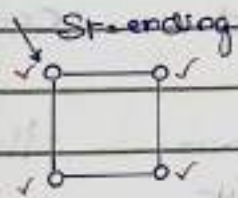


Theorem: Graph contains Euler line if it contains exactly two odd degree vertex.



Closed Path: Path + Starting vertex = ending vertex.

Hamiltonian Circuit: Closed path + cover all vertex exactly once



Graph contains Hamiltonian Cir
then graph is called
Hamiltonian Graph

H.C { Closed
+
all vertex

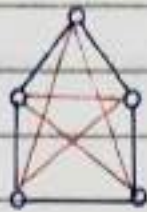
H.C

$1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5 \rightarrow 6 \rightarrow 7 \rightarrow 8 \rightarrow 1$

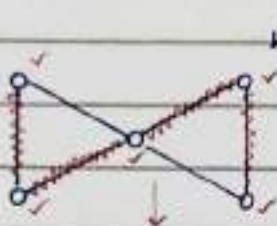
$G = (V, E)$



* All cycle graph are Hamiltonian Graph



All $K_n \rightarrow$ t.t.g. ($n \geq 3$)

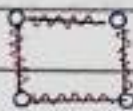


Hamiltonian Path:

One path + cover all vertices exactly once.

Not Hamiltonian Graph

{ all vertices ✓
closed x



all vertices
closed ✓

Every hamiltonian circuit contains
hamiltonian path. ✓

all vertices
open

Every hamiltonian path contains
hamiltonian circuit. (false)

Covering (covering)

All edges

Closed

Open

Euler circuit

Euler Graph.

Euler line

Euler path.

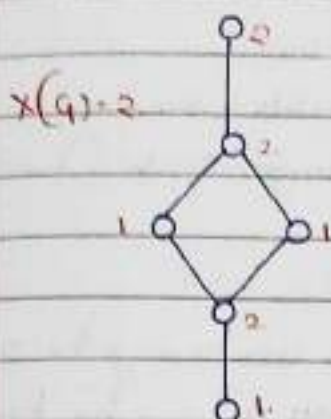
All vertices

Closed

Open

Hamiltonian
Circuit

Hamiltonian
Path



Properly coloring. $\left\{ \begin{array}{l} \text{Paint all vertices with different colors} \\ \text{Such that adjacent should not have same} \\ \text{color.} \end{array} \right.$

$$X(G) = \text{min} + \text{Properly coloring}$$

Chromatic
no.
 $(X(G))$

$\left\{ \begin{array}{l} \text{Paint all the vertices with minimum no. of} \\ \text{colors such that adjacent should not have same} \\ \text{color.} \end{array} \right.$

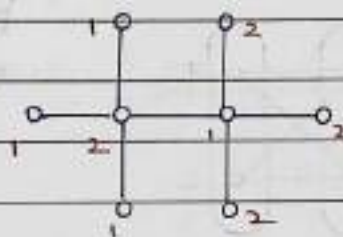
1. $\odot X(G)=1$

$\odot X(G)=1$

2. Tree

$X(\text{Tree})=2$

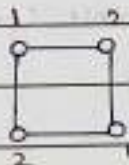
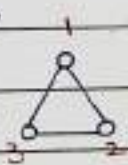
* Every tree will always be 2-colorable. $\checkmark$ (True)



* Every 2-colorable will always be tree. $\times$ (False)

3. Cycle Graph (C_n) ($n \geq 3$)

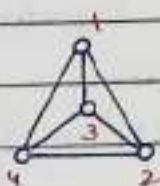
$X(C_3)=3$



$X(C_4)=2$

$$\begin{array}{ll} X(C_n)=2 & n \text{ is even} \\ X(C_n)=3 & n \text{ is odd} \end{array}$$

4. Wheel Graph (W_n) ($n \geq 4$)



$X(W_4)=4$



$X(W_5)=3$

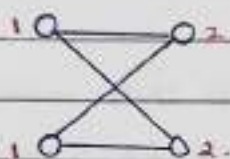
$$\begin{array}{ll} X(W_n)=4 & n \text{ is even} \\ X(W_n)=3 & n \text{ is odd} \end{array}$$

- Every Tree will always be 2-colorable (True)
- Every 2-colorable will always be tree (False)
- Every even length cycle will be 2-colorable (True)
- Every 2-colorable will always be even length cycle (False)

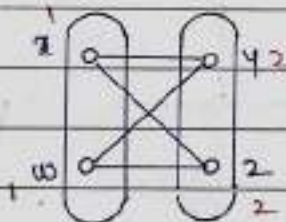
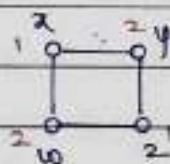
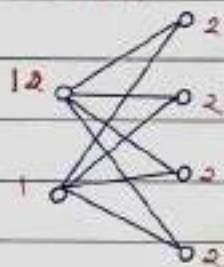
Bipartite Graph

* Every bipartite graph will always be 2-colorable (True)

* Every 2-colorable will always be bipartite graph.



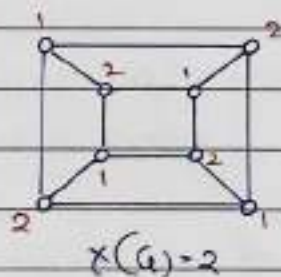
* $K_{2,4}$



* Bipartite graph does not contain odd length cycle

→ True (2-color)

→ Even length (2-color)



* Hypercube will be 2-colorable
So bipartite graph.

* Every 2-colorable will always be bipartite graph.

* 2 or more isolated vertices will also be bipartite graph with $x(G) = 1$.

→ Connected bipartite graph will be 2-colorable ($n \geq 2$) (True)

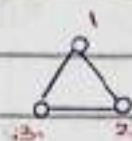
→ Bipartite graph will be 2-colorable (False)

eg: $x(G) = 1$

→ Every bipartite graph will be 2-colorable if connected + $n \geq 2$ (True)

$K_n (n \geq 1)$

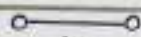
$\odot K_1 \chi(G) = 1$



$\chi(K_1) = 1$

$\chi(G) \leq n$

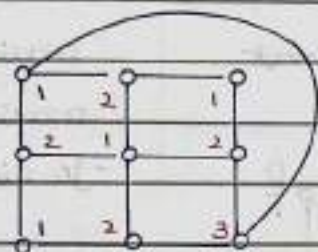
any graph



$\chi(G) = 2$

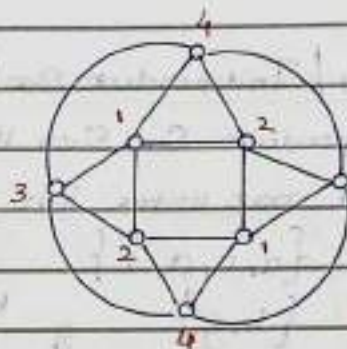
$\chi(K_n) = n$

①



$\chi(G) = 3$

②



Triangle Cycle

$\chi(G) = 4$

eg: Set of non-adjacent vertices

$1 \rightarrow \{a, c, d, f\}$

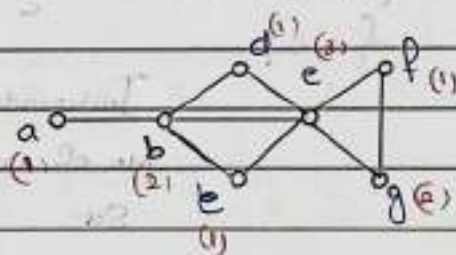
$2 \rightarrow \{b, g\}$

$3 \rightarrow \{e\}$

→ Chromatic no: 3

* writing Same color vertices in one set will give different Positions.

No. of positions = $\chi(G) = 3$



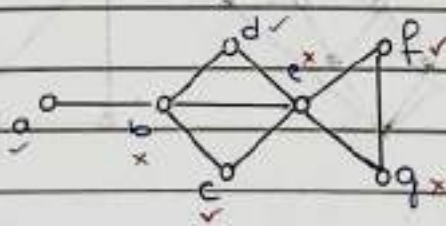
$\chi(G) = 3$

Independent Set:

$\{a, c, d, f\}$ ✓

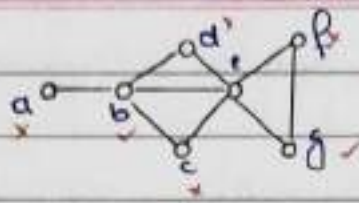
$\{a, c\}$ ✓

$\{a, c, d\}$ ✓



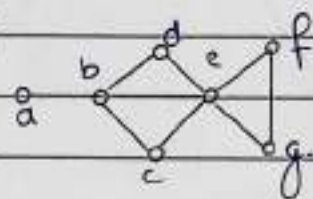
$\{a, c, d, f\}$ - 1 ✓

→ full - 10

$\{b\}$ - Is $\{b, g\}$ - Is ✓ $\{a\}$ - Is $\{a, e\}$ - Is ✓Maximal / Independent Set (MIS)

→ Independent Set such that we cannot add new vertices into this

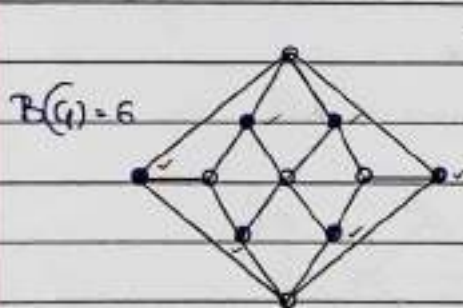
(MIS) $\left\{ \begin{array}{l} \{a, c, d, f\} \\ \{b, g\} \\ \{b, f\} \\ \{a, e\} \end{array} \right\}$



- * maximal $\neq$ maximum
- * maximal is not related to size but a property

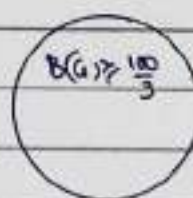
Independence No. ($B(G)$)

→ No. of vertices present in largest maximal Independent Set.

Determine the Chromatic numbers of the following graphs: $B(G)=2$

* 3 partitions = 3 independent Set (MIS)

MIS $\left\{ \begin{array}{l} \{ \dots \} \\ \{ \dots \} \\ \{ \dots \} \end{array} \right\}$



100 vertices

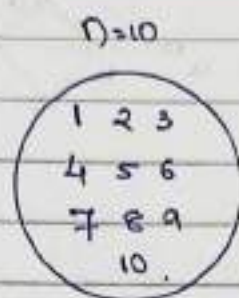
→ $\chi(G)=3$

3 partitions

$$B(G) \geq \frac{n}{\chi(G)}$$

$$B(G) = 4$$

$$\text{MIS} \left\{ \begin{array}{l} \{ \leftarrow 4 \rightarrow \} \\ \{ \leftarrow 3 \rightarrow \} \\ \{ \leftarrow 3 \rightarrow \} \end{array} \right.$$

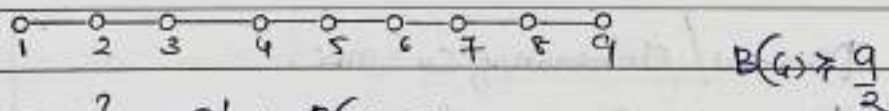
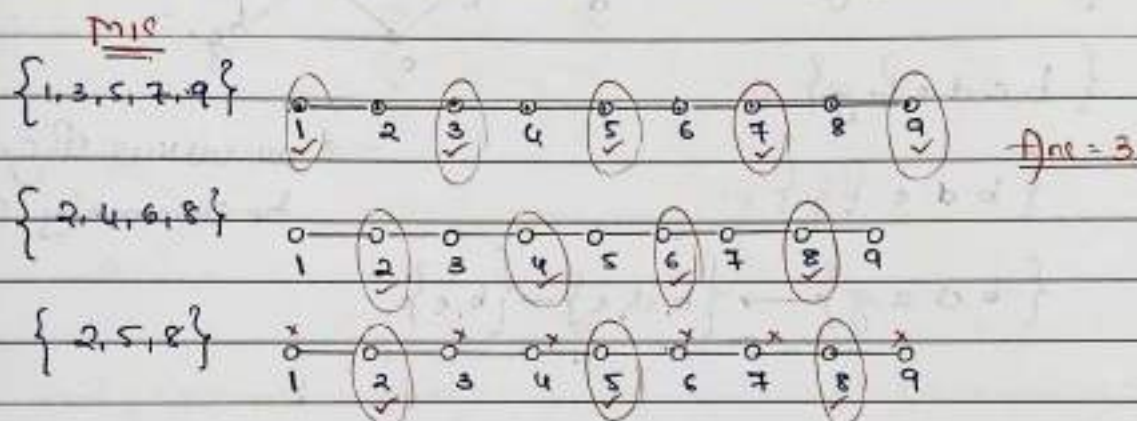


$$B(G) \geq \frac{10}{3} \quad \alpha(G) = \text{no. of paths}$$

$$B(G) \geq \frac{n}{\alpha(G)}$$

$$B(G) \cdot \alpha(G) \geq n$$

eg. What will be the smaller size of maximal independent set?



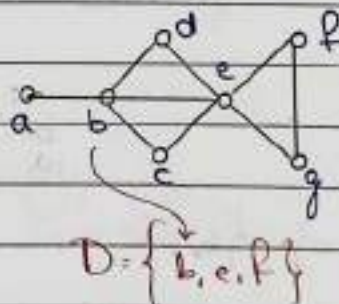
$$\left\{ \begin{array}{l} \{1, 3, 5, 7, 9\} - \text{C1 or } B(G) = 5 \\ \{2, 4, 6, 8\} - \text{C2 or } 4 \end{array} \right.$$

Dominating Set:

$$G = (V, E)$$

Dominating Set $D \subseteq V$

* If we take any vertex of v
either that vertex belongs to D
or its adjacent belongs to D

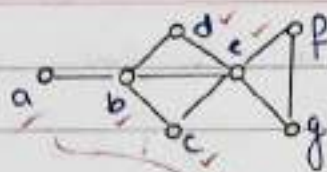


$$D_1 = \{b, e, f\} \text{ D.S.} \checkmark$$

$$D_2 = \{a, e\} \text{ D.S.} \checkmark$$

$$D_3 = \{a, e, f\} \text{ D.S.} \checkmark$$

$$\{a, c, d\} \text{ not D.S.} \times$$



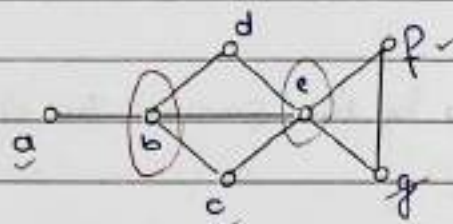
$$\{a, b, c, d, e, f, g\} \text{ D.S.} \checkmark$$

$$\{b, c, d, e, f, g\} \text{ D.S.} \checkmark$$

$$\{b, d, e, f, g\} \text{ D.S.} \checkmark$$

$$\{b, d, e, g\} \rightarrow \{b, d, e\} - \{b, e\}$$

minimal Dominating Set



→ all vertices of graph will be dominating set

Minimal / Dominating Set (MDS)

Dominating Set such that we cannot remove vertex from this

* minimal $\neq$ minimum

* minimal is not related to size but property.

maximal → Cannot add p to from set

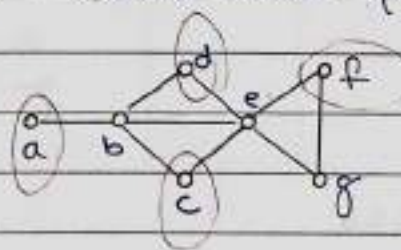
minimal → Cannot remove from set.

MDS:

$$\{b, e\} \checkmark \quad \{b, f\} \checkmark$$

$$\{a, e\} \checkmark \quad \{b, g\} \checkmark \quad \{a, c, d, f\} \checkmark$$

$$\hookrightarrow \text{Size} = 2 \therefore \alpha(G) = 2$$

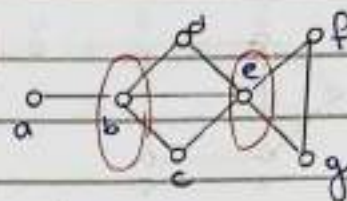


Dominations no ($\alpha(G)$)

→ No. of vertices present in Smallest minimal dominating set

$\beta(G) = 4$

$\alpha(G) \leq \beta(G)$



MIS	MOC
$\{a, c, d, f\}$	$\{a, c, d, f\}$
$\{b, f\}$	$\{b, f\}$
$\{b, g\}$	$\{b, g\}$
$\{a, e\}$	$\{a, e\}$
	$\{b, e\}$

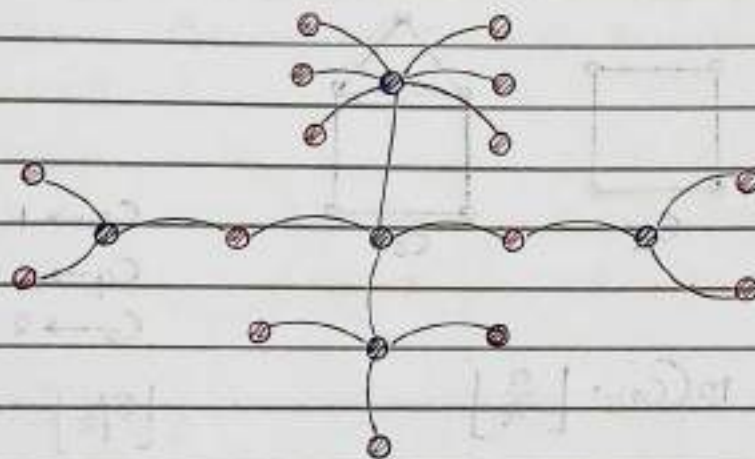
* Every MIS will always be moc ✓

* Every moc will be MIS (False)

eg: $\{b, e\}$

→ Domination ($\alpha(G) \leq$ Independence no ($\beta(G)$))

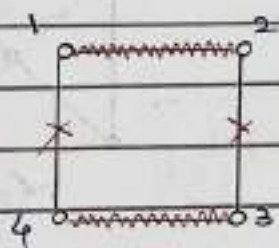
eg:



Independence no. $\beta(G) = 16$

Domination no. $\alpha(G) = 2$

Independent edge set: Set of non adjacent edges.
(matching set)



$\{1, 2\}$ - Matching set

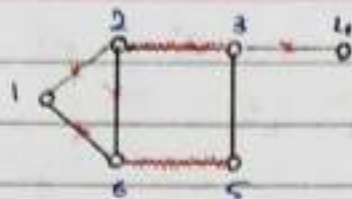
$\{1, 2, 4, 3\}$ - Matching set

maximal matching set.

Matching Set: Set of non-adjacent edges

Maximal matching Set: matching set such that we cannot add new edge into this

* Maximal is not related to size or property.


 $\{2,3\}$ - ms

 $\{2,3,6,5\}$ - maximal

matching set.

Matching No.

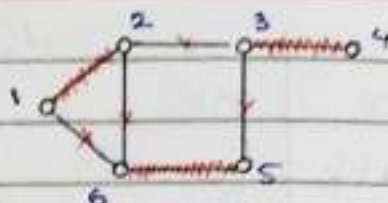
 $m(G)$

No. of edges

Present in larger

maximal matching

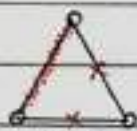
Set.


 $\{3,4\}$ - ms

 $\{3,4,6,5\}$ - ms

 $\{3,4,6,5,1,2\}$ - m.m.s.

Cycle Graphs


 C_3
 $m(C_3) = 1$

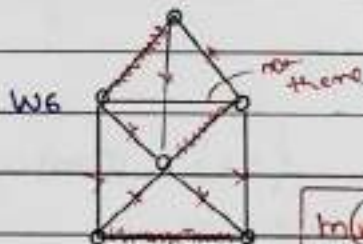
 C_4
 $m(C_4) = 2$

 C_5
 $m(C_5) = 2$
 $C_3 \rightarrow 1$
 $C_4 \rightarrow 2$
 $C_5 \rightarrow 2$

$$\Rightarrow m(C_n) = \lfloor \frac{n}{2} \rfloor$$

$$\lfloor \frac{5}{2} \rfloor = \lfloor 2.5 \rfloor$$

Wheel Graphs


 W_4
 $m(W_4) = 2$

 W_6
 $m(W_6) = 3$

$$m(W_n) = \lfloor \frac{n}{2} \rfloor$$


 W_5
 $m(W_5) = 2$

$$m(C_n) = m(K_n) = m(W_n) = \lfloor \frac{n}{2} \rfloor$$



$$m(K_{2,4}) = \min(2,4) = 2$$

$$m(K_{m,n}) = \min(m,n)$$

- Q1 Consider graph whose are represented as permutations of given set $\{1, 2, 3\}$, 2 vertices are connected if their adjacent elements are interchanged, $m(G) = ?$

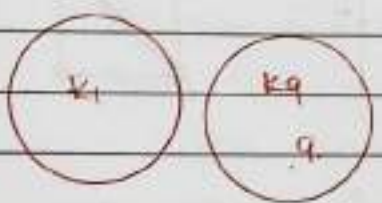
$$S = \{1, 2, 3\}$$

$$= \frac{3!}{1} = 6$$

$$\left. \begin{array}{l} 1 \ 2 \ 3 \\ 1 \ 3 \ 2 \\ 2 \ 1 \ 3 \\ 2 \ 3 \ 1 \\ 3 \ 1 \ 2 \\ 3 \ 2 \ 1 \end{array} \right\} C_6$$

$$m(G) = \lfloor \frac{6}{2} \rfloor = \underline{3}$$

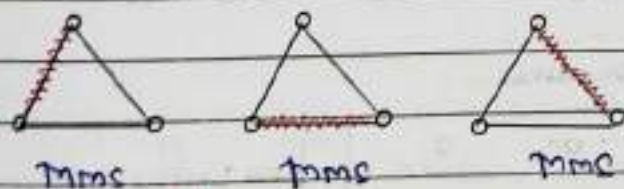
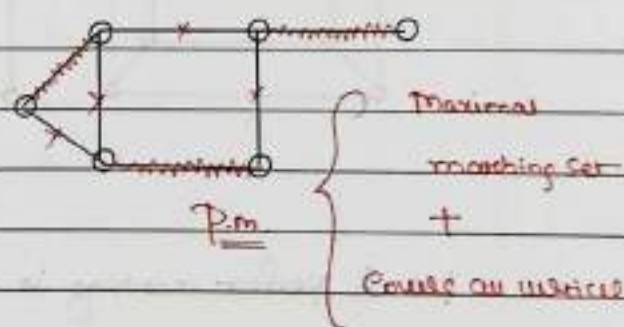
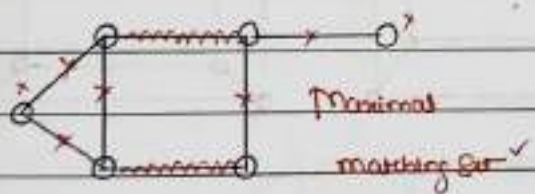
- Q2 Consider a disconnected graph of 10 vertices and maximum edges $m(G) + x(G) = ?$

$$\left\{ \begin{array}{l} 10 \text{ v} \\ k = 2 \end{array} \right.$$


$$x(K_9) = 9$$

$$m(K_9) = \lfloor \frac{9}{2} \rfloor = \underline{4}$$

Perfect Matching: maximal matching set + which saturates all the vertices.



* P.m will not exist

* If perfect matching exist then no. of vertices will be even.

* If (perfect matching exist) then (no. of vertices will be even)

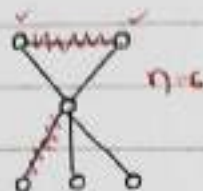
* If odd no. of vertices then P.m will not exist.

P $\rightarrow$ Q
Q $\rightarrow$ P

Contra $\neg P \rightarrow \neg Q$

Stmt 2 * If perfect matching exist then no. of vertices will be even.
 - vice versa is not true.

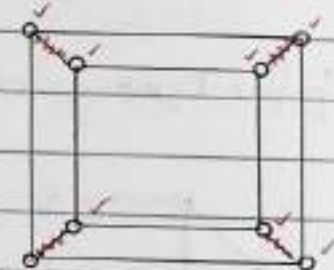
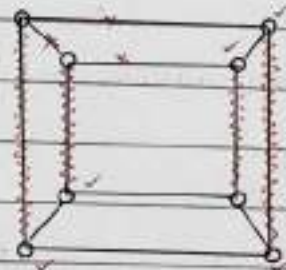
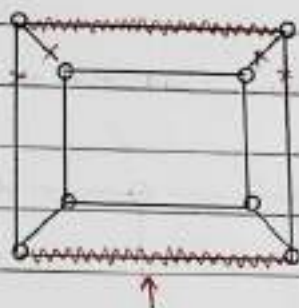
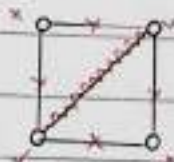
* If graph is even no. of vertices - then perfect matching exist.



Perfect match $\rightarrow$ maximal matching



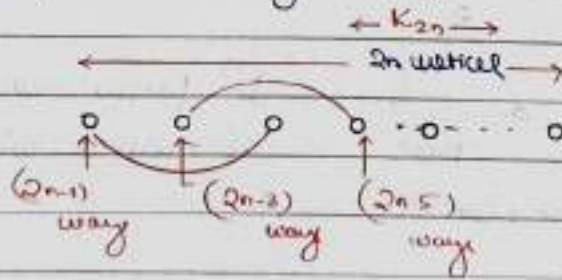
Maximal matching $\nrightarrow$ perfect matching



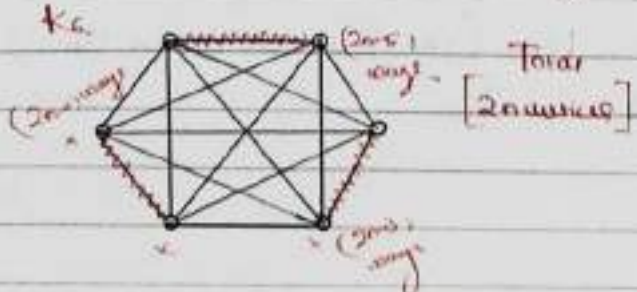
no. of perfect matchings = 3

* No. of perfect matching in K_{2n} .

* Total vertices = $2n$



eg:



$$\begin{aligned}
 \text{Total perfect matching} &= (2n-1) \times (2n-3) \times (2n-5) \dots \\
 &= \frac{2n}{2n} (2n-1) \frac{2n-2}{2n-2} (2n-3) \dots \\
 &= \frac{(2n)(2n-1)(2n-2)(2n-3) \dots}{2n(2n-2)(2n-4) \dots} \\
 &\Rightarrow \frac{2n!}{2^n \cdot n!}
 \end{aligned}$$

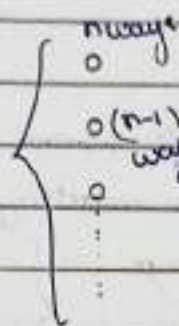
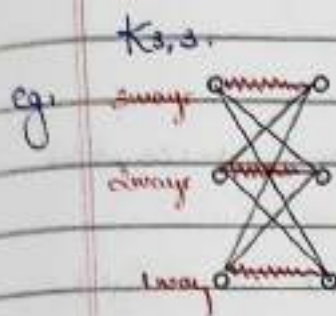
Q. What will be Pm in complete graph of 6 vertices?

Ans

$$2n = 6$$

$$n = 3$$

$$\frac{(2n)!}{2^n \cdot n!} = \frac{6!}{2^3 \cdot 3!} = \frac{720}{48} = 15$$



$$n \cdot (n-1) \cdot (n-2) \dots = n!$$

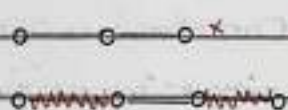
$$K_{n,n} \rightarrow n!$$

Tree:

* If Pm exist to tree, it will be unique.

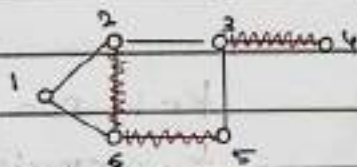
~~amino~~ ✓

* Total Pm in tree almost 1.

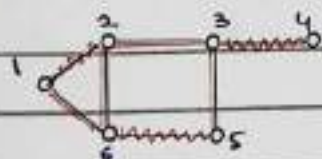


* Pm doesn't exist in all trees, but somehow if it exists there is no other representation so that is why it is unique.

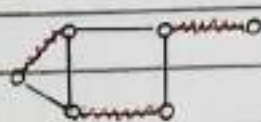
Covering Set (edge): Set of edge, such that all vertices must incident on atleast 1 edge.



Covering vertex set: Set of vertices, such that all edges, must incident on atleast 1 vertex.



$$C.S \{ 12, 16, 65, 23, 3/5, 34, 26 \}$$



$$C.S \{ 12, 16, 65, 23, 34, 26 \}$$

$$C.S \{ 12, 65, 34 \}$$

$$\{ 2, 3, 6 \} \rightarrow \text{Vertex Covering Set}$$

$\rightarrow$ Minimal Covering Set

Set of edges

↓
all vertices

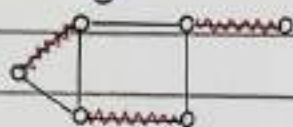
↓
incident

↓
atleast 1 edge

Minimal Covering Set: Covering set such that we cannot remove any edge from this.

Covering no. ($C(G)$): no. of edges present in smallest minimal covering set.

* Every P_n will be minimal covering set.

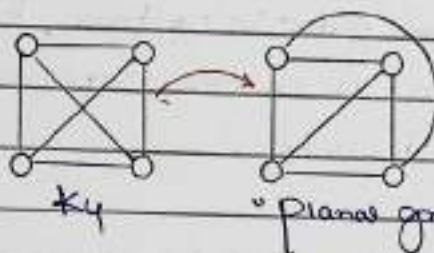


* Every P_n minimal covering set could be perfect matching. (False)



pendent will always be included into covering

Planar Graph.



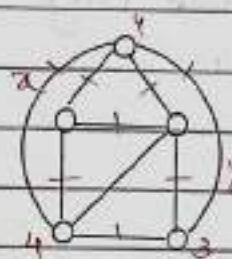
→ If we can draw a graph on a plane without intersection of its edges then that graph is called planar graph.
Otherwise it is non-planar graph.

eg: 1



K_5

$n=5, e=10$



Planar

$P=5$

$e=7$ (min)

$\{K_5 - e\}$

K_5 is non planar graph
Kurokawa's 1st Graph

K_5 is non-planar graph.

$n=1$ ○

$n=2$ ○○

$n=3$

$n=4$ K_4

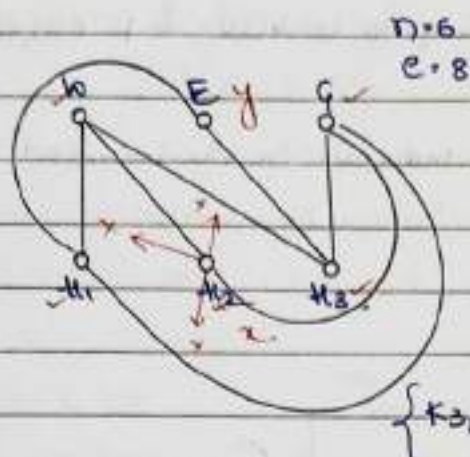
$n=5$ $\{K_5 - e\}$

$\{K_5 - e\}$
(planar)

K_5

→ non planar

eg 2

 $(K_{3,3})$

$$\begin{cases} n=6 \\ e=9 \end{cases}$$

$$\delta(u) = \frac{2e}{n} = \Delta(u) = 3$$

* Degree of each vertex will be 3:

* $K_{3,3}$ is non planar.

Non:

(Poor boy examples)

- * Both graphs are non-planar, removal of single edge from both the graphs will make Planar graph
- * Both graphs are Regular-Graph

Non Planar

K_5

$n=5$

$e=10$

$K_{3,3}$

$n=6$

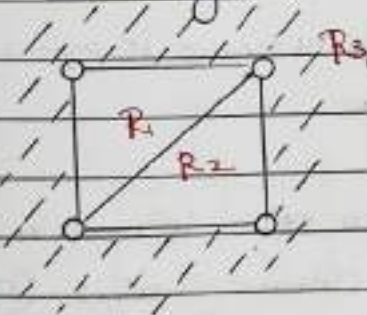
$e=9$

→ K_5 is non-planar graph with min. no. of vertices.

→ $K_{3,3}$ is non Planar with min no. of edges.

Draw a Planar graph on a Plane

embedding:



$$n - e + f = 2$$

$$n=4 \quad f=3 \\ e=5$$

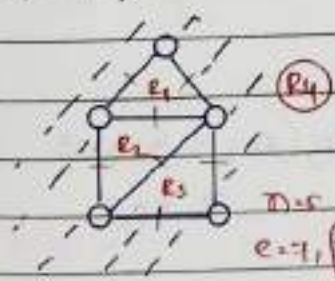
$$4 - 5 + 3 = 2$$

It creates or makes region/Face (r/f)

Closed
bounded
finite
eg: R_1, R_2

Open
bounded
infinite
eg: R_3

eg



(R4)

$n=5$

$e=7, f=4$

$n \rightarrow$ Total vertices
 $e \rightarrow$ no. of edges
 $f \rightarrow$ no. of face

Plane

Euler's Theorem

$$\rightarrow n - e + f = 2$$

Q1. Consider a planar graph having 10 vertices & 15 edges. What will be no. of faces?

$$n=10, e=15$$

$$n-e+f=2$$

$$10-15+f=2$$

$$f=7$$

Q2. What will be the bounded region?

$$\text{Total} = \text{bounded} + 1$$

$$\text{bounded} = 7-1 = 6$$

Case 1.

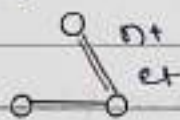


$$n-e+f=2$$

$$2-1+f=2$$

$$f=1$$

Case 2



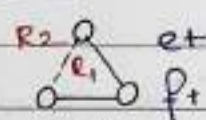
$$n=3, e=2$$

$$n-e+f=2$$

$$3-2+f=2$$

$$f=1$$

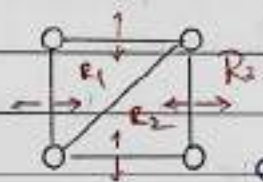
Case 3



$$n-e+f=2 \quad (n \geq 2)$$

$$3-3+f=2$$

$$f=2$$



Degree of regions = no of edges involved into this

$$e=5$$

$$\begin{cases} \deg(R_1)=3 \\ \deg(R_2)=3 \\ \deg(R_3)=4 \end{cases}$$

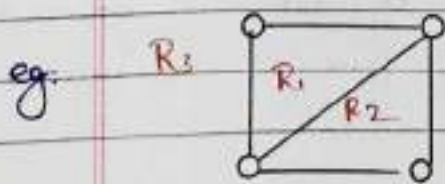
$$\deg(R_1) + \deg(R_2) + \deg(R_3) = 3+3+4$$

$$\sum d(R_i) = 10 = 2 \times 5$$

m of edge.

$$\boxed{\sum d(R_i) = 2e}$$

Sum of degree of all regions is equal to twice the no of edge.



$$\begin{cases} \deg(R_1)=3 \\ \deg(R_2)=3 \\ \deg(R_3)=4 \end{cases}$$

all region is made up by atleast 2 edge. ($n \geq 3$)

$$\deg(R_1) \geq 3$$

$$\deg(R_2) \geq 3$$

$$\deg(R_3) \geq 3$$

$$\deg(R_1) + \deg(R_2) + \deg(R_3) \geq 3+3+3$$

$$\geq 3 \times 3$$

no of region

$$\deg(v_1) + \deg(v_2) + \deg(v_3) \geq 3 \times 3 \quad \text{no. of faces}$$

$$\sum d(v_i) \geq 3f$$

$$2e \geq 3f$$

$$n - e + f = 2$$

$$f = 2 + e - n$$

$$n \geq 3$$

↑
all edges are made

by other edges

$$2e \geq 3(2 + e - n)$$

$$2e \geq 6 + 3e - 3n$$

$$3n - 6 \geq 3e - 2e$$

$$3n - 6 \geq e$$

$$e \leq 3n - 6$$

Theorem: If graph is planar then $e \leq 3n - 6$

* converse is not true

→ if $e \leq 3n - 6$ then G is planar (false)

eg: $K_{3,3}$

$$e = 9, n = 6$$

$$e \leq 3n - 6$$

$$9 \leq 3(6) - 6$$

$$9 \leq 12 \quad (\text{True})$$

Contrapositive:

if not $(e \leq 3n - 6)$ then G is

not planar

if $e > 3n - 6$ then G is non-planar.

$$P \rightarrow Q$$

Contrapositive

$$\neg P \rightarrow \neg Q$$

→ G is connected planar with 10 vertices. If the number of edges on each face is 3 then no. of edges in G .

$$e = 3n - 6$$

$$e = 3(10) - 6$$

$$e = 24$$



$$\deg(v_1) \geq 4$$

$$\deg(v_2) \geq 4$$

$$\deg(v_3) \geq 4$$

$$\deg(v_1) + \deg(v_2) + \deg(v_3) \geq 4 + 4 + 4$$

$$\sum d(v_i) \geq 4 \cdot 3 \quad \text{no. of regions/faces}$$

$$2e \geq 4f$$

Region is made up of atleast 4 edges

$$(n \geq 4)$$

$$\begin{aligned} \rightarrow & \quad \begin{aligned} 2e &\geq 4f \\ 2e &\geq 4(2+e-n) \\ 2e &\geq 8+4e-4n \\ 4n-8 &\geq 4e-2e \\ 4n-8 &\geq 2e \\ 2n-4 &\geq e \end{aligned} \quad \begin{cases} n-e+f=2 \\ f=2+e-n \end{cases} \end{aligned}$$

Theorem: If Graph is planar then $e \leq 2n-4$.

$$\begin{aligned} \rightarrow & \quad \begin{aligned} 2e &\geq 5f \\ 2e &\geq 5(2+e-n) \\ 2e &\geq 10+5e-5n \\ 5n-10 &\geq 5e-2e \\ 5n-10 &\geq 3e \\ 3e &\leq 5n-10 \\ e &\leq \left(\frac{5}{3}\right)n - \left(\frac{10}{3}\right) \end{aligned} \quad \begin{cases} n-e+f=2 \\ f=2+e-n \end{cases} \end{aligned}$$

Degree of each region is made up of atleast 5 edges.

Theorem: If Graph is planar then:
 $e \leq \left(\frac{5}{3}\right)n - \left(\frac{10}{3}\right)$

$\rightarrow$ If Graph is planar then $e \leq 3n-6$

$$\delta(G) \leq 2e/n \leq \Delta(G) \leq n-1$$

Connected,
Planar,
Simple

n

Connected
disconnected
Simple
Planar
non-planar

Connected
Planar
Simple

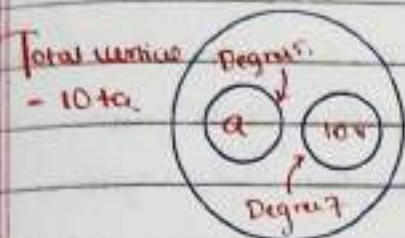
$$\delta(G) \leq \frac{2e}{n}$$

$$\delta(G) \leq \frac{2(3n-6)}{n} \leq \frac{6n-12}{n} \leq \frac{6n}{n} - \frac{12}{n} \leq 6 - \frac{12}{n}$$

$$\delta(G) \leq 6 - \frac{12}{n} \quad \left(\delta(G) \leq 5 \right) \quad \begin{cases} \text{Connected} \\ \text{Planar} \\ \text{Simple} \end{cases}$$

Theorem: If G is Connected Planar graph then $\delta(G) \leq 5$.

- Q1. A planar graph has vertices degree ≤ 7 , if there are 10 vertices of degree 7. Prove that atleast 22 vertices of degree ≤ 6 .



$$\sum d(v_i) = 2e$$

$$5a + 10 \cdot 7 = 2e$$

$$e = \frac{5a + 70}{2}$$

$$22 \leq a$$

$$e \leq 3n - 6$$

$$\frac{5a + 70}{2} \leq 3(10 + a) - 6$$

$$\frac{5a + 70}{2} \leq 30 + 3a - 6$$

$$35 - 30 + 6 \leq 3a - \frac{5a}{2}$$

$$11 \leq \frac{1}{2}a$$

- Q2. $\delta \rightarrow$ minimum degree ($\delta \geq 3$)

for all planar of n-vertices

- a) In any planar embedding, the no. of faces is atleast $\frac{n}{2} + 2$

less than $\frac{n}{2} + 2$

$$3n \leq 2e$$

$$3n \leq 2(n + f - 2)$$

$$3n \leq 2n + 2f - 4$$

$$3n - 2n + 4 \leq 2f$$

$$n + 4 \leq 2f$$

$$\frac{n}{2} + 2 \leq f$$

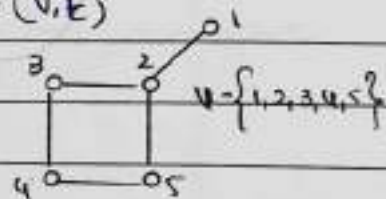
$$\begin{cases} n + f - 2 = e \\ n + f - 2 = e \end{cases}$$

Subgraph: (G)

- Edge disjoint Subgraph
- Vertex disjoint Subgraph
- induced Sub graph
- Spanning Sub graph (Spanning tree)

eg:

$$G = (V, E)$$



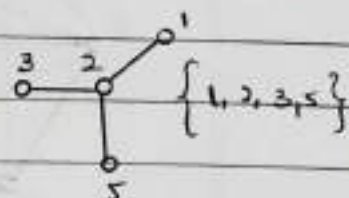
$$G_1 = (V_1, E_1) \text{ (Subgraph)}$$

$\rightarrow G_1$ is subgraph of G

$$\emptyset \neq V_1 \subseteq V \quad (V \subseteq V)$$

$$E_1 \subseteq E \quad (E \subseteq E)$$

* each edge E_1 is incident on G_1 is same as in G



Key Points:

1. Every Graph is subgraph of itself.

2. $G_2 \subseteq G_1 \subseteq G$, $G_2 \subseteq G$

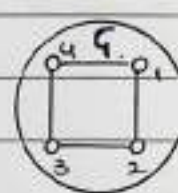
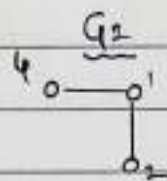
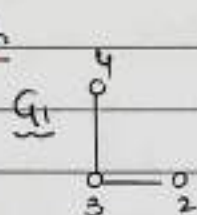
Subgraph of a subgraph of a graph is subgraph of a Graph

3. Every single vertex of a Graph is subgraph of a graph

4. Every single edge of a Graph is subgraph of a graph.

Edge disjoint Subgraph

→ Two graphs which does not have any common edge

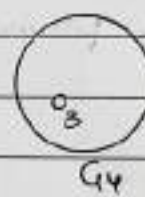
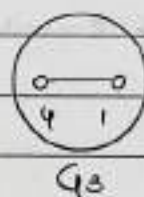


we may get common vertex.

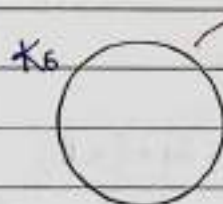
$$E = \{43, 32\} \quad E_2 = \{14, 12\}$$

Vertex disjoint Subgraph

→ Two graphs, which does not have any common vertex.

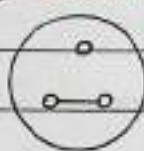
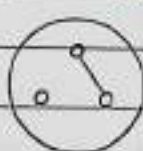
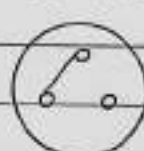
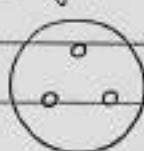


It will not have any common vertex so no pair of having common edge.

Q3. How many subgraph of K_6 satisfy $|V|=3$ 

Subgraph → 3 vertices

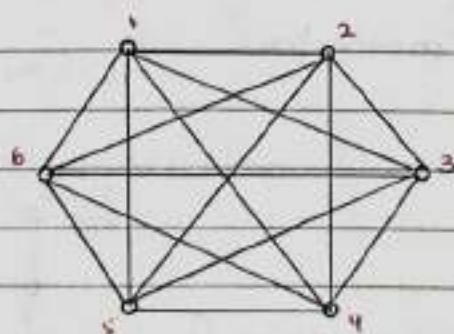
no. of subgraphs with 2 vertices



$$6C_3 \times 2^{\frac{3(3-1)}{2}}$$

$$\frac{3(3-1)}{2}$$

$$2$$



$$V = \{1, 2, 3, 4, 5, 6\}$$

In a class of 6 students, how many ways we can select 3 students

$$6C_3 \times 2^{\frac{3(3-1)}{2}}$$

Q4. How many Subgraphs are possible with n vertices?

→ 1 vertex

$$6C_1 \times 2^{\frac{1(1-1)}{2}}$$

2 vertices

$$6C_2 \times 2^{\frac{2(2-1)}{2}}$$

3 vertices

$$6C_3 \times 2^{\frac{3(3-1)}{2}}$$

4 vertices

$$6C_4 \times 2^{\frac{4(4-1)}{2}}$$

5 vertices

$$6C_5 \times 2^{\frac{5(5-1)}{2}}$$

6 vertices

$$6C_6 \times 2^{\frac{6(6-1)}{2}}$$

$$\Rightarrow \sum_{i=1}^n {}^nC_i \times 2^{\frac{i(i-1)}{2}} = \sum_{i=1}^n {}^nC_i \times 2^{\frac{i(i-1)}{2}}$$

Spanning Subgraph

G_1 is Spanning Subgraph of G

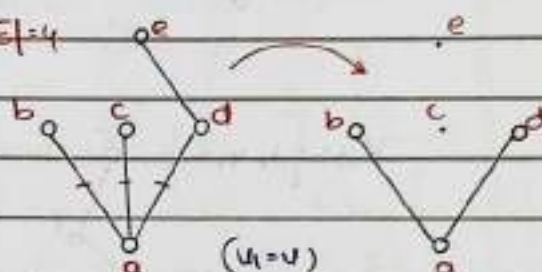
$$G_1 = (V_1, E_1) \quad G = (V, E)$$

1. Subgraph

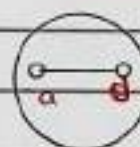
2. $V_1 = V$

$$(E_1 \subseteq E)$$

$|E|=4$



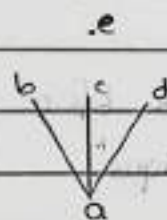
2 Spanning Subgraphs are possible



not Spanning Subgraph



$|E|=4$

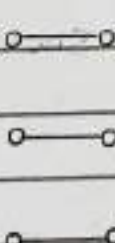


* all vertices must be present, edge may or may not.

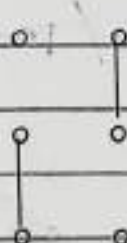
$|E|=5$



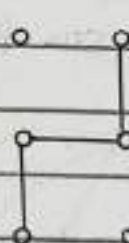
$V_1 = V$

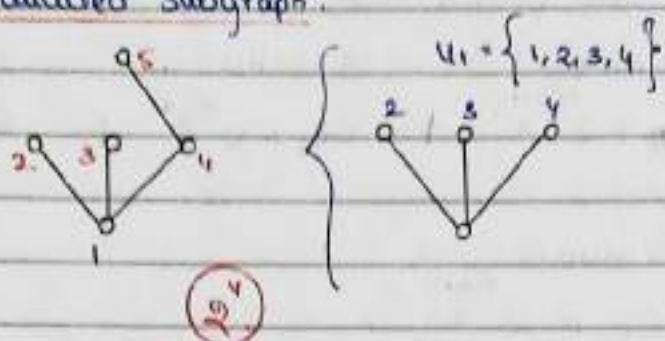


$V_1 = V$

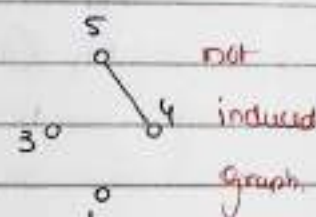
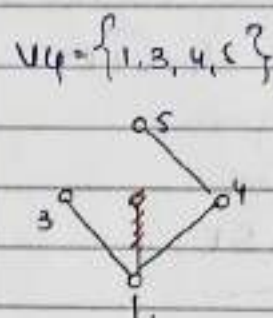
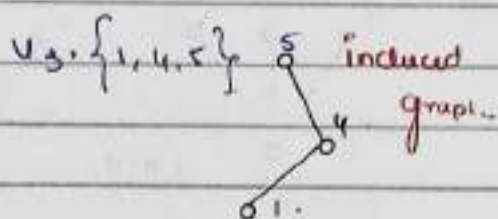
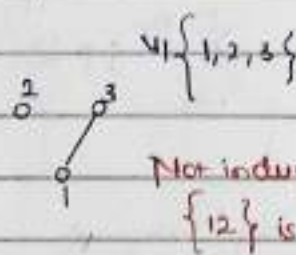
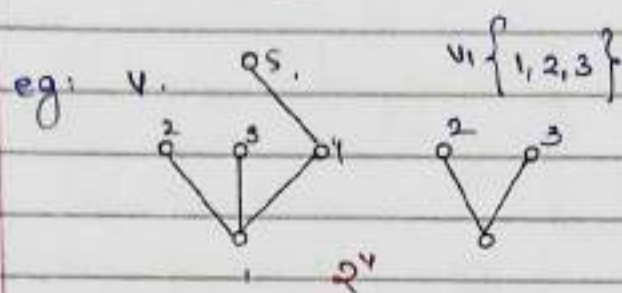


$V_1 = V$

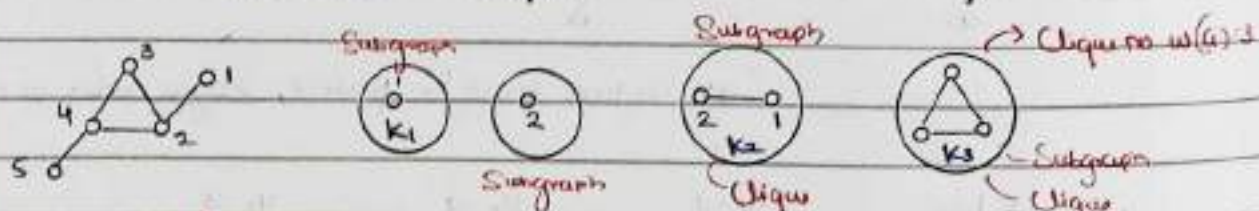


Induced Subgraph:

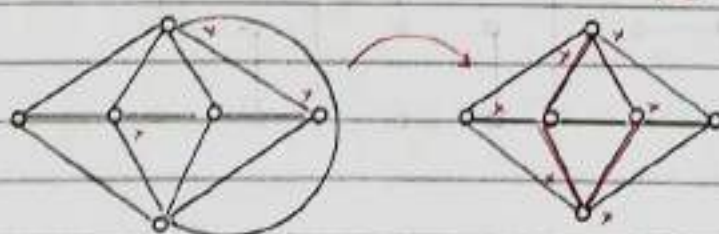
If we take any vertex set, all edge related to those vertices must be present.

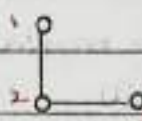
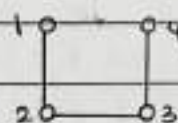
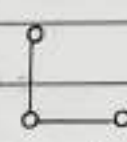
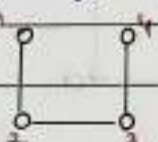
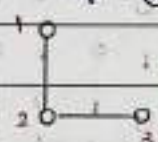
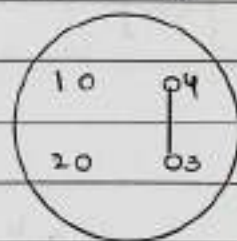
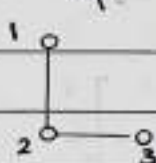
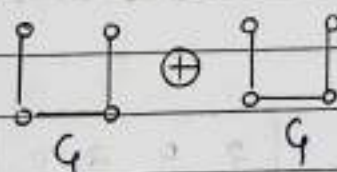


Clique: The Subgraph of a graph is complete graph then it is called "Clique".



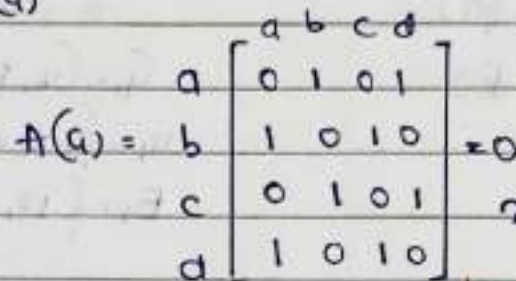
Maximum size of Clique / Minimum size of independent set.



Graph Operations:1. Union ($\cup$)2. Intersection ($\cap$)3. Ring Sum ($\oplus$) G_1 G_2  $G_3 = (V_3, E_3)$ $G_1 = (V_1, E_1)$ $G_2 = (V_2, E_2)$ $V_1 = \{1, 2, 3\}$ $\cup V_2 = \{1, 2, 3, 4\} = V_3$ $E_1 = \{12, 23\}$ $\cup E_2 = \{12, 23, 34\} = E_3$  $G_3 = (V_3, E_3)$ Intersection: ($\cap$) $G_4 = (V_4, E_4)$ $V_4 = V_1 \cap V_2$ $E_4 = E_1 \cap E_2$ $G_1 = (V_1, E_1)$ $G_2 = (V_2, E_2)$  $V_1 = \{1, 2, 3\} \cap \{1, 2, 3, 4\} = V_4 \{1, 2, 3\}$ $E_1 = \{12, 23\} \cap \{12, 23, 34\} = E_4 \{12, 23\}$ Ring Sum ($\oplus$) $G_1 = (V_1, E_1)$ $G_2 = (V_2, E_2)$ $G_6 = (V_6, E_6)$ $V_6 = V_1 \cup V_2$ $E_6 = (E_1 \cup E_2) - (E_1 \cap E_2)$ G_1 G_2  $\rightarrow$ edge in G_1 or G_2 but not in both $* G \oplus G = \text{null graph}$ $\rightarrow G \oplus G = \text{null graph}$ G  G

null graph (set of isolated vertices)

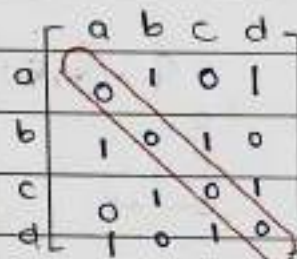
Adjacency matrix / Connection matrix
 $A(G)$



rijd } connection
between
water
and

Tab. 26a

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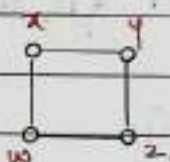


→ no self loop

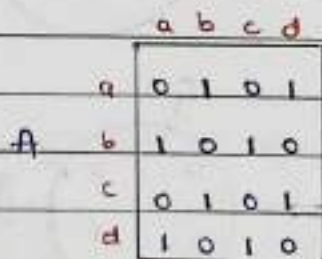
→ diagonal elements will be zero

→ no. of is present in row
or column will give degree

A triangle is drawn on lined paper. The top vertex is labeled 'a', the bottom-left vertex is labeled 'b', and the rightmost vertex is labeled 'c'.



$$A(G) = \begin{bmatrix} A(G_1) & \\ & A(G_2) \end{bmatrix}$$



$$A^2(i, i) = \text{degree of vertex } i$$

$$0 \times 0 + 1 \times 1 + 0 \times 0 + 1 \times 1 = 2$$

$$A^2 = A \times A$$

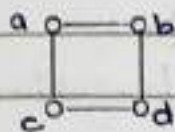
○	○	○	○
○	○	○	○
○	○	○	○
○	○	○	○

○	○	○	○
○	○	○	○
○	○	○	○
○	○	○	○

2A

	a.	b.	c.	d.
a.	2	0	2	0
b.	0	2	0	2
c.	2	0	2	0
d.	0	2	0	2

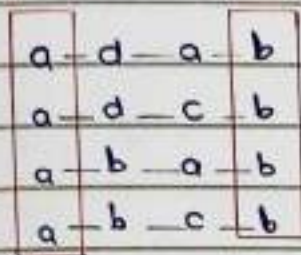
$a \rightarrow b$ (length 3) walk.



$$A^2 = \begin{bmatrix} 2 & 0 & 2 & 0 \\ 0 & 2 & 0 & 2 \\ 2 & 0 & 2 & 0 \\ 0 & 2 & 0 & 2 \end{bmatrix}$$

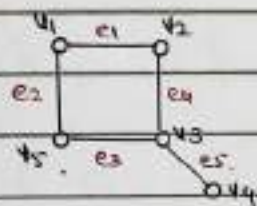
$A^3(a, b) = \begin{cases} 4 \text{ walks of length 3} \\ \text{from } a, b \end{cases}$

$A^n(i, j) = \text{no. of walks from } i \text{ to } j \text{ of length } n.$



$$A^3 = \begin{bmatrix} 2 & 0 & 2 & 0 \\ 0 & 2 & 0 & 2 \\ 2 & 0 & 2 & 0 \\ 0 & 2 & 0 & 2 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}$$

$$A^3 = \begin{matrix} & a & b & c & d \\ \begin{matrix} a \\ b \\ c \\ d \end{matrix} & \begin{bmatrix} 0 & 4 & 0 & 4 \\ 4 & 0 & 4 & 0 \\ 0 & 4 & 0 & 4 \\ 4 & 0 & 4 & 0 \end{bmatrix} \end{matrix}$$


Incidence matrix $M(G)$

$$M(G) = \begin{matrix} & \begin{matrix} e_1 & e_2 & e_3 & e_4 & e_5 \end{matrix} \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \end{matrix} & \begin{bmatrix} 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 \end{bmatrix} \end{matrix}$$

$\text{rank}(M(G)) = n - 1$

- Every column will have only 2 ones.
- Every row will give degree of a vertex.

MATHEMATICAL LOGICConnectives:

- Conjunction
- Disjunction
- Single implication
- Double implication

Modifier:

- not/negation

Conjunction ($\wedge$) (And/But)

S ₁	S ₂	C
T	T	T
T	F	F
F	T	F
F	F	F

$$(F \wedge _) = F$$

$$(F \wedge _ \wedge _) = F$$

OR : $\rightarrow$ Inclusive OR ($\vee$) $\rightarrow$ Exclusive OR ($\oplus$)

P	q	$P \vee q$	$(T \vee _) = T$
T	T	T	
T	F	T	$(T \vee _ \vee _) = T$
F	T	T	
F	F	F	

Conditional Statement ($\rightarrow$)

$$P \rightarrow q$$

if P then q

if P, q

* q if P

q when P

q whenever P

* q unless $\neg$ P

P implies q

* P only if q

eg: if you win the match then i will

P

give pizza

q.

 $F \rightarrow T$

win(T)	pizza(T)	T
win(T)	pizza(F)	F
win(F)	pizza(T)	T
win(F)	pizza(F)	T

$$P \rightarrow q$$

→ if perfect matching exist then no. of vertices would be even:

Theorem

$P.m(T)$	$Even(T)$	T	$\left\{ \begin{array}{l} T \rightarrow T = T \\ T \rightarrow F = F \\ F \rightarrow T = T \\ F \rightarrow F = T \end{array} \right.$
$P.m(T)$	$Even(F)$	F	
$P.m(F)$	$Even(T)$	$True$	
$P.m.$	$Even.$	$True$	

$$P \rightarrow q$$

$$P.m \rightarrow E.m.$$

$$P \rightarrow q \neq q \rightarrow p$$

$$P \rightarrow q \neq \neg p \rightarrow \neg q$$

(inverse)

$$P \rightarrow q = \neg q \rightarrow \neg p$$

Inverse

$$\neg p \rightarrow \neg q$$

$$\neg(P.m) \rightarrow \neg(E.m)$$

Converse

$$q \rightarrow p$$

$$E.m \rightarrow P.m.$$

Contrapositive

$$\neg q \rightarrow \neg p$$

$$\neg(E.m) \rightarrow \neg(P.m)$$

→ if G is planar then $e \leq 3n-6$

$$P \rightarrow q = \neg q \rightarrow \neg p$$

$\neg(e \leq 3n-6)$ then G is not planar

if $e > 3n-6$ then G is non planar

D	A		B		C
p	p	q	$p \rightarrow q$	$\neg p \rightarrow \neg q$	$\neg q \rightarrow \neg p$
T	T	T	T	F	F
T	T	F	F	F	T
F	F	T	T	T	F
F	F	F	T	T	T

* Expression A is logically equivalent to expression B.

$$A \equiv B$$

or.

2. A, B are having same behaviour.

Double Implication ($\leftrightarrow$)

P	q	$P \leftrightarrow q$
T	T	T
F	F	T
T	F	F
F	T	F

* E.g. $\leftrightarrow$ degree of an sentence as even

 $A \equiv B$

1. Same column
2. Same behaviour

A	B	$A \leftrightarrow B$
T	T	T
F	F	T
T	T	T
F	F	T

→ tautology (1=2=3)

1. $A \leftrightarrow B$ tautology
 $A \equiv B$

2. A, B are having same column

3. They are having same behaviour

	T
	T
	T
	T

→ tautology/valid.

- all valid are
Satisfiable

- All contingency
are Satisfiable.

	F
	F
	F
	F

→ Contradiction

	T
--	---

Satisfiable

→ at least
1 true.

	T
	F

→ Contingency

Not tautology

Not contradiction

→ at least

[illegible]

$P \wedge P \equiv P$	$P \wedge T \equiv P$	$P \vee T \equiv P$
$P \vee P \equiv P$	$P \vee F \equiv P$	$P \wedge F \equiv F$

$P \vee Q = Q \vee P$	$P \wedge (Q \wedge R) = (P \wedge Q) \wedge R$
$P \wedge Q = Q \wedge P$	$P \vee (Q \vee R) = (P \vee Q) \vee R$

Logic	Set	Digital
$\wedge$	$\cap$	$\cdot$
$\vee$	$\cup$	$+$
T	$\mathcal{U}$	1
F	$\emptyset$	0

$$\rightarrow P \wedge (Q \vee R) = (P \wedge Q) \vee (P \wedge R)$$

$$\rightarrow P \vee (q \wedge r) = (P \vee q) \wedge (P \vee r)$$

Absorption:

$$a \vee (a \wedge b) = a$$

$$a \wedge (a \vee b) = a$$

$$\underline{a = 1}$$

$$T \vee (T \wedge \underline{\quad}) = T$$

$$\underline{a = T}$$

$$T \wedge (T \vee \underline{\quad}) = T$$

$$\underline{a = F}$$

$$F \vee (F \wedge \underline{\quad}) = F$$

$$\underline{a = F}$$

$$F \wedge (F \vee \underline{\quad}) = F$$

$$p \rightarrow q = \neg p \vee q$$

$$p \rightarrow q = \neg q \rightarrow \neg p$$

De Morgan's Rule:

$$\neg(a \vee b) = \neg a \wedge \neg b$$

$$\neg(a \wedge b) = \neg a \vee \neg b$$

$$(p \rightarrow q) \wedge (p \rightarrow R) = p \rightarrow (q \wedge R)$$

$$(\neg p \vee q) \wedge (\neg p \vee R)$$

$$\neg p \vee (q \wedge R)$$

$$p \rightarrow (q \wedge R)$$

$$(p \rightarrow R) \wedge (Q \rightarrow R) = (p \vee Q) \rightarrow R$$

$$(\neg p \vee R) \wedge (\neg Q \vee R)$$

$$(\neg p \wedge \neg Q) \vee R$$

$$\neg(p \vee Q) \vee R$$

$$(p \vee Q) \rightarrow R$$

Demorgan's

Rule

$$\left\{ \begin{array}{l} (p \rightarrow q) \wedge (p \rightarrow R) = p \rightarrow (q \wedge R) \\ (p \rightarrow R) \wedge (Q \rightarrow R) = (p \vee Q) \rightarrow R \end{array} \right.$$

$$\left\{ \begin{array}{l} (p \rightarrow Q) \vee (p \rightarrow R) = p \rightarrow (Q \vee R) \\ (p \rightarrow R) \vee (Q \rightarrow R) = (p \wedge Q) \rightarrow R \end{array} \right.$$

$$(p \rightarrow q) \wedge (\neg q \vee p)$$

$$p \leftrightarrow q = (p \rightarrow q) \wedge (q \rightarrow p)$$

$$= (\neg p \vee q) \wedge (q \rightarrow p)$$

$$(\neg p \vee q) \wedge (\neg q \vee p)$$

$$(p \wedge q) \vee (\neg p \wedge \neg q)$$

$$\begin{aligned}
 & (\tau p \vee q) \wedge (\tau q \vee p) \\
 & A \wedge (\tau q \vee p) \\
 & (A \wedge \tau q) \vee (A \wedge p) \\
 & (\tau q \wedge A) \vee (p \wedge A) \\
 & (\tau q \wedge (\tau p \vee q)) \vee (p \wedge (\tau p \vee q)) \\
 & (\tau q \wedge \tau p) \vee (p \wedge q)
 \end{aligned}$$

$$\begin{aligned}
 x: & \tau q \wedge (\tau p \vee q) \\
 & (\tau q \wedge \tau p) \vee (\tau q \wedge q) \\
 & (\tau q \wedge \tau p) \vee F \\
 & = \tau q \wedge \tau p
 \end{aligned}$$

$$\begin{aligned}
 y: & p \wedge (\tau p \vee q) \\
 & (p \wedge \tau p) \vee (p \wedge q) \\
 & F \vee (p \wedge q) = p \wedge q
 \end{aligned}$$

Q1. The simplest form of $(p \wedge (\neg r \vee q \vee \neg q)) \vee ((\tau v t \vee \neg r) \wedge \neg q)$ is

Ans

$$\begin{aligned}
 & p \wedge (\tau p \vee \tau) \quad \vee \quad (\tau \vee \tau p \vee \tau) \wedge \tau q \\
 & \quad \downarrow \quad \quad \quad \downarrow \\
 & p \quad \quad \quad \tau \wedge \tau q = \tau q \\
 & \Rightarrow \underline{p \vee \tau q}
 \end{aligned}$$

Q2. The simplest form of $(p \vee (p \wedge q) \vee (p \wedge q \wedge \neg r)) \wedge ((p \wedge \neg t) \vee t)$

Ans

$$\begin{aligned}
 & p \vee (p \wedge q \wedge \tau r) \\
 & \quad \downarrow \\
 & p \wedge t \\
 & \Rightarrow \underline{p \wedge t}
 \end{aligned}$$

Q3. P and Q are two propositions. Which of the following logical expressions are equivalent?

Ans I. $p \vee \neg Q \rightarrow I = II$

II. $\neg(\neg p \wedge Q)$

III. $(p \wedge Q) \vee (p \wedge \neg Q) \vee (\neg p \wedge \neg Q)$

IV. $(p \wedge Q) \vee (p \wedge \neg Q) \vee (\neg p \wedge Q)$

I. $p \vee \tau q$

II. $\tau(\tau p \wedge Q) = p \vee \tau Q$

$(p \wedge Q) \vee (p \wedge \tau Q) \vee (\tau p \wedge \tau Q)$

$p \wedge (Q \vee \tau Q) \vee (\tau p \wedge \tau Q)$

$(p \wedge \tau) \vee p \vee (\tau p \wedge \tau q)$

$p \vee (\tau p \wedge \tau q)$

$(p \vee \tau p) \wedge$

$(p \vee \tau q)$

$\tau \vee (p \vee \tau q)$

$(p \vee \tau q)$

Q4. Which of the following is not equivalent to $p \leftrightarrow q$?

Ans $(\neg p \wedge q) \vee (p \wedge \neg q)$

Premises $\rightarrow$ Conclusion (i) Premise
True Check $\therefore$ Conclusion

eg: Premises — Conclusion Or Premises
True ∴ Conclusion

$P \vee Q$	$Q \vee P$
$\neg Q$	$\neg Q$
$\therefore P$	$\therefore P$

Tyda Chinese

L	R	Lvr	TL
T	T	T	F
T	F	T	F
F	T	T	T
F	F	F	T

P	Q	$P \rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

True False

$$[(P \rightarrow Q) \wedge P] \rightarrow Q$$

P	Q	P → Q	$(P \rightarrow Q) \wedge P$	$((P \rightarrow Q) \wedge P) \rightarrow Q$
T	T	T	T	T
T	F	F	F	T
F	T	T	F	T
F	F	T	F	T

P₁: if p.m exist then no of wickets will be even

P₂: no of wicket even

Q: P.m will exist

$$\begin{array}{c} \text{T} \\ ((P \rightarrow Q) \wedge Q) \rightarrow P \end{array}$$

P → Q

Q

∴ P

P	Q	P → Q
T	T	T
T	F	F
F	T	T
F	F	T

$$\begin{array}{c} P \vee Q \\ \neg P \quad \text{or} \quad \neg Q \\ \hline \therefore Q \end{array} \quad \begin{array}{c} P \vee Q \\ \neg P \quad \text{or} \quad \neg Q \\ \hline \therefore P \end{array} \quad \begin{array}{c} \neg P \vee Q \\ P \\ \hline \therefore Q \end{array} = \left\{ \begin{array}{c} P \rightarrow Q \\ P \\ \hline Q \end{array} \quad \begin{array}{c} P \rightarrow Q \\ Q \\ \hline \therefore P \end{array} \right. \quad (\text{fallacy})$$

P₁: if it rains then cricket match will not be played

P₂: Cricket match was played

∴ There was no rain

$$\begin{array}{c} P \rightarrow \neg P \equiv P \rightarrow \neg R \\ P \\ \hline \neg R \end{array} \quad \begin{array}{c} P \rightarrow \neg R \\ P \\ \hline \neg R \end{array} \quad (\text{valid})$$

P₁: if it rains Cricket match will not be played

P₂: it did not rain

∴ Cricket match was played

$$\begin{array}{c} P \rightarrow \neg P \equiv P \rightarrow \neg R \\ \neg R \\ \hline P \end{array} \quad \begin{array}{c} P \rightarrow \neg R \\ \neg R \\ \hline P \end{array} \quad (\text{invalid})$$

$$(1) \quad \begin{array}{l} P \rightarrow Q \text{ or } ((P \rightarrow Q) \wedge P) \rightarrow Q \\ \underline{P} \\ \therefore Q \end{array} \quad \text{(Modus ponens)}$$

Example:

P	Q
Q	X
∴ P	

$$(2) \quad \begin{array}{l} P \wedge Q \equiv \neg Q \rightarrow \neg P \quad ((P \rightarrow Q) \wedge \neg Q) \rightarrow \neg P \\ \underline{\neg Q} \quad \underline{\neg Q} \\ \therefore \neg P \quad \neg P \end{array} \quad \text{(Modus tollens)}$$

$$(3) \quad \begin{array}{l} P \rightarrow Q \quad ((P \rightarrow Q) \wedge (Q \rightarrow R)) \rightarrow (P \rightarrow R) \\ \underline{Q \rightarrow R} \\ P \rightarrow R \end{array} \quad \text{(Hypothetical Syllogism)}$$

$$(4) \quad \begin{array}{l} P \vee Q \quad ((P \vee Q) \wedge \neg P) \rightarrow Q \\ \underline{\neg P} \\ \therefore Q \end{array} \quad \text{(Disjunctive Syllogism)}$$

$$(5) \quad \begin{array}{l} \frac{P \wedge Q}{P} \quad P \wedge Q \rightarrow P \\ \text{or} \\ P \wedge Q \rightarrow Q \end{array} \quad \text{(Simplification)}$$

$$(6) \quad \begin{array}{l} \underline{P} \quad P \rightarrow P \vee Q \\ \therefore P \vee Q \end{array} \quad \text{(Addition)}$$

$$(7) \quad \begin{array}{l} P \vee Q \quad ((P \vee Q) \wedge (\neg P \vee \neg Q)) \\ \underline{\neg P \vee \neg Q} \quad \rightarrow Q \vee R \\ Q \vee R \end{array} \quad \text{(Resolution)}$$

$$C \wedge (A \vee B) \wedge (\neg B \vee C) \rightarrow C$$

$$1: A$$

$$2: \underline{A \rightarrow B} \quad \text{(modus ponens)} \\ B$$

$$3: \underline{\neg B \vee C} \\ C$$

$$C \wedge (A \rightarrow B) \wedge (\neg B \vee C) \wedge (C \rightarrow X) \rightarrow X$$

$$1: A$$

$$2: \underline{A \rightarrow B} \quad \text{(M.P.)} \\ B$$

$$3: \underline{\neg B \vee C} \quad \text{(D.S.)} \\ C$$

$$4: \underline{C \rightarrow X} \quad \text{(M.P.)} \\ X$$

Q1. The statement formula $\{(A \rightarrow C) \wedge (B \rightarrow D) \wedge (C \rightarrow \neg D)\} \rightarrow (\neg A \vee \neg B)$ is

$$A \rightarrow C = \neg A \vee C$$

$$B \rightarrow D = \neg B \vee D \quad \neg A \vee C$$

$$\underline{C \rightarrow \neg D} = \neg C \vee \neg D \quad \neg B \vee D$$

$$\neg A \vee \neg B$$

$$\neg A \vee \neg B$$

Q2. Establish the validity of the following arguments

$$(a) \quad [P \wedge (P \rightarrow Q) \wedge (\neg Q \vee R)] \rightarrow R$$

$$(b) \quad P \rightarrow Q \equiv \neg P \vee Q$$

$$\underline{\neg P} \quad \underline{\neg Q} \quad \neg P \wedge \neg Q$$

$$\underline{\neg R} \quad \underline{\neg R} \quad \neg R$$

$$\therefore \neg(P \vee R)$$

$$\neg R$$

$$\neg(P \vee R)$$

$$P \rightarrow Q \equiv \neg P \vee Q$$

$$P \rightarrow Q \equiv \neg Q \rightarrow \neg P$$

$$(c) \begin{array}{l} p \rightarrow q = \neg q \rightarrow \neg p \\ r \rightarrow \neg q \quad \neg q \\ \hline r \\ \hline \therefore \neg p \end{array}$$

$$(d) \begin{array}{l} p \wedge q \\ p \rightarrow (r \wedge q) \\ r \rightarrow (s \vee t) \\ \hline \neg s \\ \hline \therefore \neg t \end{array}$$

$$\begin{array}{l} p \wedge q \\ \downarrow \\ p \\ p \rightarrow (r \wedge q) \\ \hline r \wedge q \rightarrow r \\ r \rightarrow (s \vee t) \\ \hline s \vee t \end{array}$$

$$(e) \begin{array}{l} 1. p \rightarrow (q \rightarrow r) \\ 2. p \vee s \\ 3. t \rightarrow q \\ 4. \neg s \\ \hline \therefore \neg r \rightarrow \neg t \end{array}$$

$$\begin{array}{l} 2. p \vee s \\ 4. \neg s \\ \hline p \end{array}$$

$$1. p \rightarrow (q \rightarrow r)$$

$$q \rightarrow r \quad \neg t \rightarrow r \equiv \neg r \rightarrow \neg t$$

$$t \rightarrow q$$

$$\begin{array}{l} \neg s \\ \hline t \end{array}$$

$$Q3 \quad S1: \{ (\neg p \rightarrow (q \rightarrow \neg w)) \wedge (\neg s \rightarrow q) \wedge \neg t \wedge (\neg p \vee t) \} \rightarrow (w \rightarrow s)$$

$$S2: \{ (q \rightarrow t) \wedge (s \rightarrow r) \wedge (\neg q \rightarrow s) \} \rightarrow (\neg t \rightarrow r)$$

$$S2: q \rightarrow t \equiv \neg q \vee t$$

$$s \rightarrow r \equiv \neg s \vee r$$

$$\begin{array}{l} \neg q \rightarrow s \equiv q \vee s \\ \neg t \rightarrow r \end{array} \quad \begin{array}{l} \neg q \vee t \\ q \vee r \end{array} \quad t \vee r$$

Predicate Logic:

$P(x)$: x is even number (Open statement)

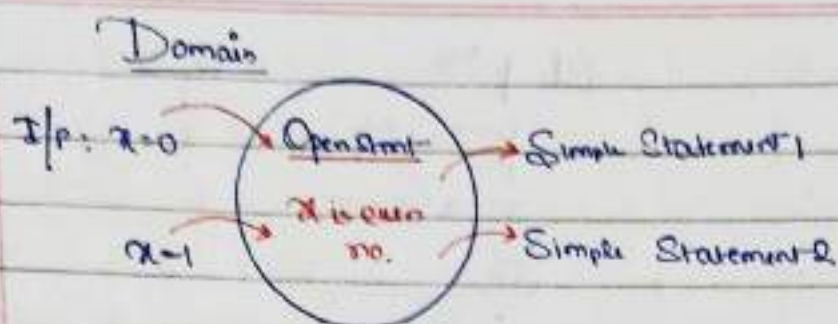
$P(0)$: 0 is even number (True) $\rightarrow$ Simple

$P(1)$: 1 is even number (False) $\rightarrow$ Simple

$\hookrightarrow$ predicate variable

Open Statement: It is a type of statement where we cannot directly define truth value. Once we provide input, it changes to Simple Statement.

Domain / Domain of discourse / Universe: possible choice for open statement.



Quantifying → Truth value.

All (universal quantifier) ($\forall x$)

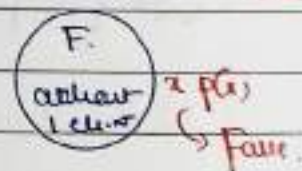
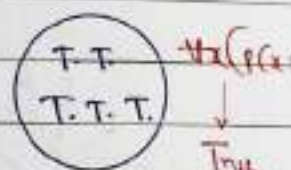
→ Tool used to find out the truth value in terms of quantifying (All)

$\forall x P(x)$

for all values of x , such that it satisfies $P(x)$

for all of x , such that $P(x)$

for every value of x , such that $P(x)$



eg:

$D: \{1, 2, 3\}$

$P(x): x^2 \leq 9$

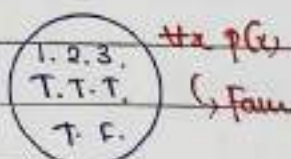
$\forall x P(x)$
or

$\forall x (x^2 \leq 9)$

→ True

$D: \{1, 2, 3, 4\}$

$P(x): x^2 \leq 9$



$D: 2$

$P(x): x^2 \geq 0$

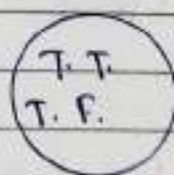
$\forall x (x^2 \geq 0)$

→ True



$\forall x P(x) \rightarrow$ True

- When it's true for all elements in domain.



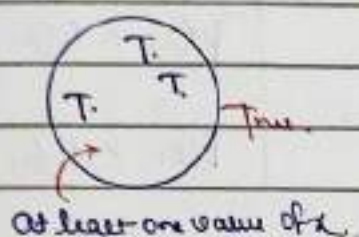
$\forall x P(x) \rightarrow$ False

- at least 1 element is false.

Existential Quantifier ($\exists x$): Tool which is used to find out truth in terms of quantity.

$\exists x P(x)$

- there exist x , which satisfies $P(x)$
- there exist x , such that $P(x)$
- Some of x , such that $P(x)$
- atleast 1 value of x , such that $P(x)$.



$\exists x P(x) \rightarrow \text{True}$



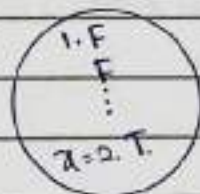
$\exists x P(x) \rightarrow \text{False}$

eg: $D: \{1, 2, 3, \dots, 10\}$

$$P(x) = x^2 = 4$$

$\exists x P(x)$
or

$\exists x (x^2 = 4)$
(True)

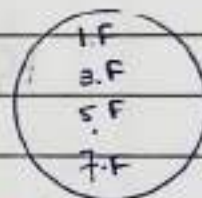


$D: \{1, 3, 5, 7\}$

$$P(x) = x^2 = 4$$

$\exists x (x^2 = 4)$

(False)



$D: \mathbb{Z}$

$$P(x) = x^2 \geq 0$$

$\forall x P(x)$

$\forall x (x^2 \geq 0)$
(True)

$D: \mathbb{Z}$

$$P(x) = x^2 \geq 0$$

$\exists x P(x)$

$\exists x (x^2 \geq 0)$

$$\forall x (x^2 \geq 0) \rightarrow \exists x (x^2 \geq 0)$$

True True.

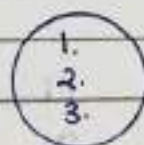
$$\forall x P(x) \rightarrow \exists x P(x)$$

True True

Negation of Quantifier

$$\neg \forall x P(x) = \exists x \neg P(x)$$

$$\neg \exists x P(x) = \forall x \neg P(x)$$



$D: \{1, 2, 3\}$

$$\begin{aligned} \neg \forall x P(x) &= \neg \{P(1) \wedge P(2) \wedge P(3)\} \\ &= \neg P(1) \vee \neg P(2) \vee \neg P(3) \\ &= \exists x \neg P(x) \end{aligned}$$

$$D: \{1, 2, 3\}$$

$$P(x): x+1=4 \quad (x=3)$$

$$x=1 \quad 1+1=4 \quad (F)$$

$$x=2 \quad 2+1=3 \quad (F)$$

$$x=3 \quad 3+1=4 \quad (T)$$

$$D: \{1, 2, 3\}$$

$$Q(x): 2x+1=7 \quad (x=3)$$

$$x=1 \quad 2(1)+1=3 \quad (F)$$

$$x=2 \quad 2(2)+1=5 \quad (F)$$

$$x=3 \quad 2(3)+1=7 \quad (T)$$

$$\exists x [P(x) \wedge Q(x)]$$

$$D: \{1, 2, 3\}$$

$$\exists x [P(x) \wedge Q(x)] \rightarrow \exists x P(x) \wedge \exists x Q(x)$$

$P(1) \wedge Q(1)$	F	$P(1)$	F	$Q(1)$
$P(2) \wedge Q(2)$	F	$P(2)$	F	$Q(2)$
$P(3) \wedge Q(3)$	T	$P(3)$	T	$Q(3)$

True $\wedge$ True

True

$$\exists x P(x) \wedge \exists x Q(x) \rightarrow \exists x [P(x) \wedge Q(x)]$$

T	F
$\vee$	$\vee$
F	T

True True

$T \wedge F$	F
$F \wedge T$	F

False

eg

$$\forall x [P(x) \wedge Q(x)] = \forall x P(x) \wedge \forall x Q(x)$$

$$D: \{1, 2\}$$

$$P(1) \wedge Q(1)$$

$$P(2) \wedge Q(2)$$

$$\left(\begin{matrix} P_1 \\ \wedge \\ P_2 \end{matrix} \right) \left(\begin{matrix} P_2 \\ \wedge \\ Q_2 \end{matrix} \right)$$

- Case 1: $\overbrace{\forall x [P(x) \vee Q(x)]}^{\text{True}} \rightarrow \overbrace{\forall x P(x) \vee \forall x Q(x)}^{\text{False}}$

$$\begin{array}{|c|c|c|}
 \hline
 T & F & T \\
 \hline
 \wedge & & \\
 \hline
 F & T & T \\
 \hline
 \end{array}$$

$$\overbrace{\forall x P(x) \vee \forall x Q(x)}^{\text{True}} \rightarrow \overbrace{\forall x [P(x) \vee Q(x)]}^{\text{True}}$$

$$\begin{array}{|c|c|c|} \hline T & & \\ \hline \wedge & \vee & \wedge \\ \hline T & & \\ \hline \end{array}$$

$$\begin{array}{|c|c|} \hline T & \vee \\ \hline \wedge & \\ \hline T & \vee \\ \hline \end{array}$$

$$T \vee F$$

$$T$$

$p(x): x \leq 3$ $q(x): x+1$ is odd.

Ans.

$$\neg p(3) \vee (\neg T) = F$$

$$p(x) : x \leq 3$$

$p(3) : 3 \leq 3 \text{ (True)} \Rightarrow p(3) \wedge q(4)$

Q3. $P(x) : x^2 - 7x + 10 = 0$ $x(x) : x < 0$

$$q(x): x^2 - 2x - 3 = 0$$

Ans:

$$\forall x [P(x) \rightarrow \neg R(x)]$$

$$x=1 \quad \boxed{P(1) \rightarrow \neg R(1)} \quad T$$

T

$$x=2 \quad \boxed{P(2) \rightarrow \neg R(2)} \quad T$$

T

T

$$x=3 \quad \boxed{P(3) \rightarrow \neg R(3)} \quad T$$

Q2. $P(x): x^2 - 7x + 10 = 0$

$Q(x): x^2 - 2x - 3 = 0$

$R(x): x < 0$

$$\forall x (P(x) \rightarrow \neg R(x))$$

$$x=1 \quad P(1) \rightarrow \neg R(1) \quad T$$

$$x=2 \quad P(2) \rightarrow \neg R(2) \quad T$$

$$x=3 \quad P(3) \rightarrow \neg R(3) \quad T$$

$$x=4 \quad P(4) \rightarrow \neg R(4) \quad T$$

$$x=5 \quad P(5) \rightarrow \neg R(5) \quad T$$

H.W

Q4. $P(x): x^2 - 8x + 15 = 0$

$Q(x): x \text{ is odd}$

$R(x): x > 0$

Negation of Quantifier

Expression: $\forall x [P(x) \rightarrow Q(x)]$

negate it:

$$\neg \forall x [P(x) \rightarrow Q(x)]$$

$$\neg \forall x [\neg P(x) \vee Q(x)]$$

$$\exists x \neg [\neg P(x) \vee Q(x)]$$

$$\neg \forall x P(x) = \exists x \neg P(x)$$

$$\neg \exists x P(x) = \forall x \neg P(x)$$

$$\rightarrow \exists x [\neg (\neg P(x)) \wedge \neg Q(x)]$$

$$\exists x [P(x) \wedge \neg Q(x)]$$

Q5. Negate and Simplify each of the following:

(a) $\exists x [P(x) \vee Q(x)]$

$$\neg \exists x [P(x) \vee Q(x)]$$

$$\forall x \neg [P(x) \vee Q(x)] \Rightarrow \forall x [\neg P(x) \wedge \neg Q(x)]$$

$$(b) \exists x [p(x) \rightarrow q(x)]$$

$$\exists x [\neg p(x) \vee q(x)]$$

$$(d) \exists x [(p(x) \vee q(x)) \rightarrow r(x)]$$

$$\neg \exists x [p(x) \vee q(x) \rightarrow \neg r(x)]$$

$$\neg \exists x [\neg (p(x) \vee q(x)) \vee R(x)]$$

$$\forall x [(p(x) \vee q(x)) \wedge \neg R(x)]$$

English Statement $\rightarrow$ Logical Statement

eg: All mothers are female

All of x , x is mother x is female

$\forall x$ $MT(x)$ $F(x)$

$$\forall x [MT(x) \rightarrow F(x)]$$

Case 1: $\forall x [MT(x) \wedge F(x)]$

Case 2: $\forall x [MT(x) \rightarrow F(x)]$

False

False	$\wedge$	F
F	$\wedge$	T
T	$\wedge$	T

T

F	$\rightarrow$	T
F	$\rightarrow$	F
F	$\rightarrow$	T

eg: Some cars are black

Some of x , x is car, x is black

$\exists x$ $c(x)$ $b(x)$

1. $\exists x [c(x) \rightarrow b(x)]$

$\exists x [c(x) \wedge b(x)]$

$\exists x [c(x) \leftrightarrow b(x)]$

T

T	$\rightarrow$	F
T	$\rightarrow$	T
F	$\rightarrow$	T

T

T	$\wedge$	F
T	$\wedge$	T
	$\wedge$	

T

T	$\leftrightarrow$	F
T	$\leftrightarrow$	T
	$\leftrightarrow$	

F

- eg: 1. All of my friends are perfect. $\forall x [F(x) \rightarrow P(x)]$
2. Not all of my friends are perfect.
 $\neg \forall x [F(x) \rightarrow P(x)]$
 $\neg \forall x [\neg F(x) \vee P(x)]$
 $\exists x [F(x) \wedge \neg P(x)]$
3. None of my friends are perfect. $\forall x [F(x) \rightarrow \neg P(x)]$
4. All of my friends are not perfect. $\forall x [P(x) \rightarrow \neg P(x)]$

- eg: 2. All graphs are connected. $\forall x [G(x) \rightarrow C(x)]$
- Not all graphs are connected. $\neg [\forall x [G(x) \rightarrow C(x)]]$

- No graphs are connected. $\forall x [G(x) \rightarrow \neg C(x)]$
- All graphs are not connected. $\forall x [G(x) \rightarrow \neg C(x)]$

Note:

1. not all $\rightarrow \neg \forall x [\rightarrow]$

2. no / none $\rightarrow \forall x [\rightarrow]$

$D: \{1, 2, 3\}$

$P(x, y): x + y \leq 9$

$\forall x \forall y (x + y \leq 9)$

for all values of x , all values of y
 for every value of x , every value of y

for all values of x , all values of y
 for $x = 1$ to 3 $x = 1, y = 1, 2, 3$
 for $y = 1$ to 3 $x = 2, y = 1, 2, 3$
 $x + y \leq 9$ $x = 3, y = 1, 2, 3$

$D: \{1, 2, 3\}$

$x = 1, y = 1$

$(x + y \leq 9) \quad (1 + 1 \leq 9) (T)$

$x = 1, y = 2 \quad 1 + 2 \leq 9 (T)$

$x = 1, y = 3 \quad 1 + 3 \leq 9 (T)$

$x = 2, y = 1 \quad 2 + 1 \leq 9 (T)$

$$x \times y \leq 9$$

$x \quad y$



$$x=1 \quad y=1 \quad 1 \times 1 \leq 9 (T)$$

$$\forall x \forall y \rightarrow \text{True}$$

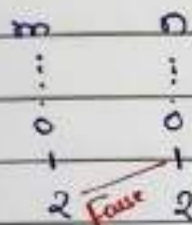
when all edges are True

$$\forall x \forall y \rightarrow \text{False}$$

at least 1 edge is false.

$$D: 2$$

$$\forall m \forall n (m \cdot n = 0)$$



$$m=2, n=1$$

$$m \cdot n = 0$$

$$2 \cdot 1 \neq 0$$

$$D: 2$$

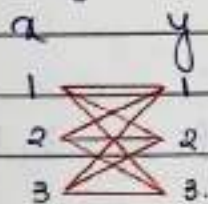
$$\forall x \forall y (x \in z)$$



↘ false

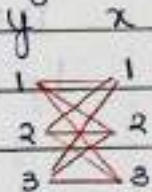
$$D: \{1, 2, 3\}$$

$$\forall x \forall y (x \times y \leq 9)$$



$$D: \{1, 2, 3\}$$

$$\forall y \forall x (x \times y \leq 9)$$



$$\forall x \forall y = \forall y \forall x$$

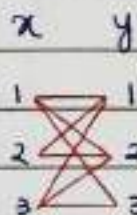
$$\forall x \forall y \leftrightarrow \forall y \forall x$$

$$D: \{1, 2, 3\}$$

$$\exists x \exists y (x \times y = 4)$$

there exist x , there exist y .
at least 1 value of x , at least 1 value of y .
for some value of x , some value of y .

$$\exists x \exists y (x \times y = 4)$$



$$x=1 \quad y=1$$

$$1 \times 1 \neq 4$$

$$x=1 \quad y=2$$

$$1 \times 2 \neq 4$$

$$x=1 \quad y=3$$

$$1 \times 3 \neq 4$$

$$\rightarrow \exists x \exists y \rightarrow \text{True}$$

* at least 1 edge is True

$$\rightarrow \exists x \exists y \rightarrow \text{False}$$

* when all edges are false.

$$D: \mathbb{Z}$$

$$\exists x \exists y (x \in y)$$

x	y
...	...
2	2
2	3
3	...

True

True

$$D: \mathbb{Z}$$

$$\exists x \exists y (x + y = 10)$$

x	y
...	...
1	9
2	8
...	...

True

True

$$x=1 \quad y=9$$

$$1+9=10 (T)$$

$$x=2 \quad y=8$$

$$2+8=10 (T)$$

$$D: \mathbb{Z}$$

$$\exists x \exists y (x+y=10 \wedge 2x+y=15)$$

x	y
...	...
5	5

True

$$x+y=10$$

$$2x+y=15$$

$$-x = -5$$

$$x=5$$

$$y=5$$

$$D: \{1, 2, 3\}$$

$$\exists x \exists y (x \times y = 4)$$

x	y
1	1
2	2
3	3

$$D: \{1, 2, 3\}$$

$$\exists y \exists x (x \times y = 4)$$

y	x
1	1
2	2
3	3

$$\exists x \exists y = \exists y \exists x$$

$$\forall x \forall y = \forall y \forall x$$

$$\exists x \exists y = \exists y \exists x$$

$$\forall x \forall y \rightarrow \exists x \exists y$$

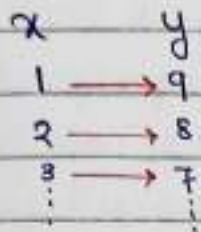
$$\forall y \forall x \rightarrow \exists x \exists y$$

$$\forall y \forall x \rightarrow \exists y \exists x$$

$$\forall x \forall y \rightarrow \exists y \exists x$$

Domain is fixed

$D: \mathbb{Z}$ (True)
 $\forall x \exists y (x+y=10)$



$$x+y=10$$

$$x=1$$

$$1+y=10$$

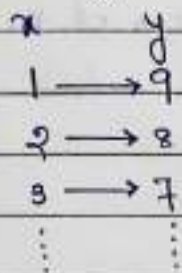
$$y=9$$

$$x=2$$

$$y=8$$

for all value of x , there exist y
 for every value of x , there exist y

$D: \mathbb{Z}$
 $\forall x \exists y (x+y=10) \not\rightarrow \exists y \forall x (x+y=10)$ (false)



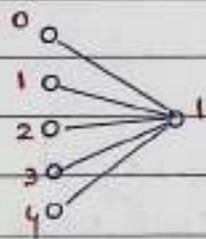
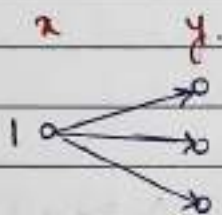
there exist y , and x
 Forcing y to be
 Constant.

$$x+y=10$$

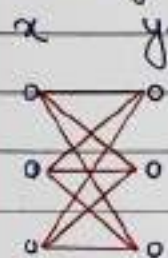
$$y=10-x$$

$$y=10-$$

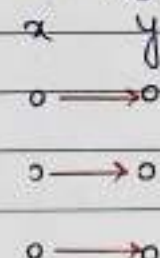
$$\exists y \forall x \left(\frac{x}{y} \in \mathbb{Z} \right) \rightarrow \forall x \exists y \left(\frac{x}{y} \in \mathbb{Z} \right)$$



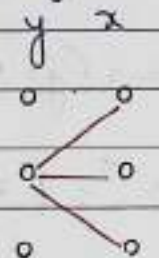
1.
 $\forall x \forall y$



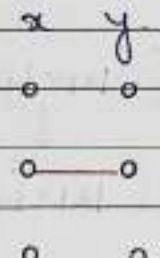
2.
 $\forall x \exists y$



3.
 $\exists y \forall x$



4.
 $\exists x \exists y$



O.S find
 Domain: find.

$$1. \forall x \forall y = \forall y \forall x \quad 5.$$

$$2. \boxed{\exists y \forall x} = \boxed{\exists x \forall y} \quad 6.$$

$$3. \forall x \exists y \quad 7. \forall y \exists x$$

$$4. \exists x \exists y \quad 8. \exists y \exists x$$

$$1 \rightarrow \text{all}$$

$$5 \rightarrow \text{all}$$

$$\text{all} \rightarrow 4, 8$$

$$2 \rightarrow 3$$

no relation between 2nd & 8

Q1. Consider the Open Statement $p(x, y) : y - x = y + x^2$ where the universe for each of the variables x, y comprises all integers. Determine the truth value for each of the following statements.

$$(a). p(0, 0) \text{ (T)}$$

$$(b). p(0, 1) \text{ (T)}$$

$$(c). \exists y p(1, y) \text{ (F)}$$

$$(d). \exists y \forall x p(x, y) \text{ (F)}$$

$$(e). p(1, 1) \text{ (F)}$$

$$(f). \forall y p(x, y) \text{ (T)}$$

$$(g). \forall x \exists y p(x, y) \text{ (F)}$$

$$(h). \forall y \exists x p(x, y) \text{ (F)}$$

$$p(x, y) : y - x = y + x^2$$

$$x + x = 0$$

$$x(x+1) = 0$$

$$x = 0 \quad x = -1$$

$$Q2. \forall x \forall y [(x < y) \rightarrow \exists z (x < z < y)]$$

$$\neg \forall x \forall y [\neg (x < y) \vee \exists z (x < z < y)] \equiv (x < y \wedge \neg \exists z (x < z < y))$$

$$\exists x \exists y [\neg (x < y) \wedge \neg \exists z (x < z < y)]$$

$$\forall z (x \geq z \text{ or } z \geq y)$$

$$Q3. \neg \forall x \forall y [(|x| = |y|) \rightarrow (y = \pm x)]$$

$$\exists x \exists y (|x| = |y| \wedge (y \neq \pm x))$$

Q4. $[\forall x, x \rightarrow (\exists y, B \rightarrow (\forall u, \exists v, x))]$
 Ans. $[\exists x, x \rightarrow (\forall y, B \wedge (\exists u, \forall v, x))]$

Q5. $p(x, y)$: x is divisor of y .
 $S_1: \forall x \exists y p(x, y)$ (F)
 $S_2: \forall y \exists x p(x, y)$ (T)
 $S_3: \exists x \forall y p(x, y)$ (F)
 $S_4: \exists y \forall x p(x, y)$ (T)

D: Z

$$p(x, y) = \frac{y}{x}$$

all elements are $\div$ by element called as 1.

$$S_1: \forall x \exists y \left(\frac{y}{x} \in Z \right)$$

$$x \quad y$$

$$0$$

$$1 \rightarrow 2$$

$$2 \rightarrow 4$$

$$\vdots$$

Q6. D: Z

S4:

$$S_1: \forall n \exists m (n+m=5) \quad \exists m \forall n (m+n=0) \text{ (False)}$$

$$S_2: \exists n \forall m (n+m=m) \quad m \quad n \quad m+n=0$$

$$S_3: \forall m \exists n (m+n=1) \quad n=2$$

$$S_4: \exists m \forall n (m+n=0) \quad m+n=0$$

$$S_1: \forall n \exists m (n+m=5)$$

$$n \quad m \quad (True)$$

$$1 \rightarrow 4$$

$$2 \rightarrow 3$$

$$4 \rightarrow 1$$

$$0 \rightarrow 5$$

$$0 \rightarrow 5$$

(True).

$$\forall n \exists m (n^2 \leq m)$$

$$n \quad m$$

$$1 \rightarrow 2$$

$$2 \rightarrow 5$$

$$S_2: \exists n \forall m (n+m=m)$$

$$n \quad m \quad (True)$$

$$1 \rightarrow 2$$

$$1 \rightarrow 3$$

$$1 \rightarrow 4$$

$$1 \rightarrow 5$$

$$1 \rightarrow 5$$

$$1 \rightarrow 5$$

$$1 \rightarrow 5$$

$$1 \rightarrow 5$$

$$1 \rightarrow 5$$

$$1 \rightarrow 5$$

$$1 \rightarrow 5$$

$$1 \rightarrow 5$$

$$1 \rightarrow 5$$

$$1 \rightarrow 5$$

$$1 \rightarrow 5$$

$$1 \rightarrow 5$$

$$1 \rightarrow 5$$

$$1 \rightarrow 5$$

$$1 \rightarrow 5$$

$$1 \rightarrow 5$$

$$n=3$$

$$m=3$$

$$S_3: \exists m \forall n (m \cdot n = 1)$$

$$2 \rightarrow x$$

$$(False)$$

$$m=2$$

$$2 \cdot n = 1$$

$$n = \frac{1}{2}$$

arrow doesn't exist.

(True)

$$\exists n \forall m (n \leq m^2)$$

- there exist element which is less than square of any int.

$$-1 \rightarrow 1$$

$$-1 \rightarrow 4$$

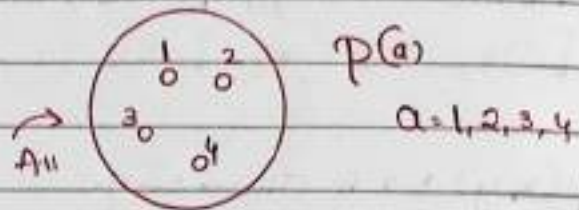
$$-1 \rightarrow 9$$

$$-1 \rightarrow 16$$

$$-1 \rightarrow 25$$

$$-1 \rightarrow 36$$

$$-1 \rightarrow 49$$

Quantifier with I.R:Universal Specifications:

$\forall x P(x) \rightarrow P(a)$ (a is for all elements).

$$p(x): x^2 \leq 9$$

$$\forall x P(x)$$

$$D: \{1, 2, 3\}$$

$$\forall x P(x) \rightarrow P(a) \text{ a is for all.}$$

$$\forall x (x^2 \leq 9) \rightarrow p(1) (1^2 \leq 9)$$

$$\rightarrow p(2) (2^2 \leq 9)$$

$$\rightarrow p(3) (3^2 \leq 9)$$

Existential Specifications:

$$\exists x P(x) \rightarrow p(a)$$

[a is specific]

$$D: \{1, 2, 3\}$$

$$p(x): x^2 = 4$$

$$\exists x P(x)$$

$$\exists x (x^2 = 4) \rightarrow p(1), p(2), p(3)$$

Universal Generalization:

$$p(a) \rightarrow \forall x P(x)$$

a is for all.

Existential Generalization:

$$p(a) \rightarrow \exists x P(x)$$

a is fixed

eg: All students in class will go to IIT $\forall x [C(x) \rightarrow IIT(x)]$

Some students in class are not solving problems $\exists x [C(x) \wedge \neg Q(x)]$

Some students who are not solving questions will go to IIT. $\exists x [\neg Q(x) \wedge IIT(x)]$

$$[\forall x [C(x) \rightarrow IIT(x)] \wedge \exists x [C(x) \wedge \neg Q(x)]] \rightarrow \exists x [\neg Q(x) \wedge IIT(x)]$$

$$1. \forall x [C(x) \rightarrow IIT(x)] \text{ Premise (T)}$$

$$3. \exists x [C(x) \wedge \neg Q(x)] \text{ Given (T)}$$

(a = x)

$$2. C(a) \rightarrow IIT(a) \text{ [u/a is for all students]}$$

$$4. C(a) \wedge \neg Q(a) \text{ [a is fixed]}$$

$$5. (Ca) [4(A), a=5]$$

$$(a=5)$$

$$6. \text{IT}(a) [a=5]$$

$$4A. (a) [4, \text{Simplification Rule}]$$

$$a=5$$

$$4B. \neg Q(a) [4, \text{Simplification}]$$

$$7. \neg Q(a) \wedge \text{IT}(a) [a=5]$$

$$8. \exists x [\neg Q(x) \wedge \text{IT}(x)]$$

$$Q_1. (\forall x P(x) \wedge \forall x [P(x) \rightarrow Q(x)]) \rightarrow \forall x Q(x)$$

$$1. \forall x [P(x) \rightarrow Q(x)] [\text{Given/T}]$$

$$3. \forall x P(x) [\text{Given}]$$

$$2. P(a) \rightarrow Q(a) [u.s / a \text{ is for all}]$$

$$4. P(a) [a \text{ is for all}]$$

$$5. P(a)$$

$$6. Q(a) [a \text{ is all/m.p}]$$

$$7. \forall x Q(x) [6, u.g.]$$

$$Q_2. (\forall x [P(x) \rightarrow Q(x)] \wedge \exists x P(x)) \rightarrow \exists x Q(x)$$

$$(1) \forall x (P(x) \rightarrow Q(x))$$

$$(2) \exists x P(x)$$

$$(1A) P(a) \rightarrow Q(a) [a = \text{all} / u.s.]$$

$$(3) P(a) [a = \text{fixed}]$$

$$(4) \neg Q(a) ([a \text{ is fixed}] / 3)$$

$$(4A) Q(a) [a \text{ is fixed}]$$

$$(4B) \exists x Q(x) [\text{Existence generalisation}]$$

$$\exists x [P(x) \wedge Q(x)] \rightarrow \exists x P(x) \wedge \exists x Q(x)$$

$$1. \exists x [P(x) \wedge Q(x)]$$

$$2. P(a) \wedge Q(a) [a \text{ is fixed } a=3]$$

$$3. P(a) (2, \text{Simpl})$$

$$4. Q(a) (2, \text{Simpl})$$

$$\downarrow$$

$$\exists x P(x) [E.g.]$$

$$\downarrow$$

$$\exists x Q(x) [E.g.]$$

$$\exists x P(x) \wedge \exists x Q(x)$$

$$\exists x P(x) \wedge \exists x Q(x) \rightarrow \exists x [P(x) \wedge Q(x)]$$

$$1. \exists x P(x)$$

$$3. \exists x Q(x)$$

$$2. P(a) [a \text{ is fixed}]$$

$$4. Q(b) [b \text{ is fixed}]$$

$$\exists x P(x) \wedge \exists x Q(x) \not\rightarrow \exists x [P(x) \wedge Q(x)]$$

$$\boxed{T \wedge F} \quad F$$

$$\vee$$

$$\boxed{F \wedge T} \quad F$$

$$[\exists x [P(x) \rightarrow Q(x)] \wedge \exists x P(x)] \rightarrow \exists x Q(x)$$

$$1. \exists x [P(x) \rightarrow Q(x)]$$

$$3. \exists x P(x)$$

$$2. P(a) \rightarrow Q(a) \quad [a \text{ is fixed}]$$

$$(a=5)$$

$$4. P(a) \quad [a \text{ is fixed}]$$

$$(a=3)$$

$$Q. \quad \forall x [P(x) \vee Q(x)] \longrightarrow P(a) \vee Q(a) \quad [a = \text{all}]$$

$$\exists x \neg P(x) \longrightarrow \neg P(a) \quad [a = \text{fixed}]$$

$$\forall x [\neg Q(x) \vee R(x)] \longrightarrow Q(a) \quad [a = \text{fixed}]$$

$$\forall x [S(x) \rightarrow \neg R(x)]$$

$$\exists x \neg S(x)$$

$$\neg Q(a) \vee R(a) \quad [a = \text{all}]$$

$$R(a) \quad [\text{fixed}]$$

$$\neg S(a) \vee \neg R(a) \quad [a = \text{all}]$$

$$\neg S(a) \quad (\text{fixed})$$

No mothers are male

$$\forall x [MT(x) \rightarrow \neg mT(x)]$$

Some males are politicians

$$\exists x [mT(x) \wedge PL(x)]$$

Some politicians are not mothers

$$\exists x [PL(x) \wedge \neg mT(x)]$$

SET THEORY

Set: Collection of objects.

$$A = \{1, 2, 3\}$$

- Repetition is not allowed

- Order is not important

$$1 \in A$$

$$4 \notin A$$

element $\rightarrow$ Set

A:

$$\text{eg: } \{e_1, \{e_2, \{e_3, \{1, \{2\}\}\}\}\}$$

Element $\in$ Set

$$1 \in A (T)$$

$$\{1\} \in A (T)$$

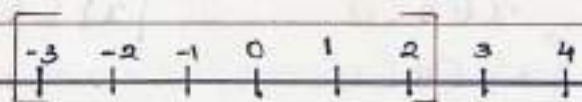
Representation of Set:

① Roster: $A = \{1, 2, 3\}$

② Set builder: $\{x \mid \text{condition}\}$ eg: $\{x \mid 1 \leq x \leq 3\}$

Closed interval:

$$[a, b] = \{x \mid a \leq x \leq b\}$$



$$[-3, 2] = \{-3, -2, -1, 0, 1, 2\}$$

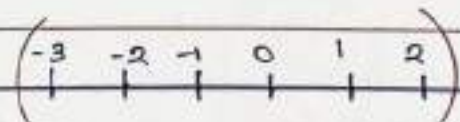
$$A = \{1, 2, 3\}$$

$$A = \{x \mid 1 \leq x \leq 3\}$$

$$A[1, 3] = \{1, 2, 3\}$$

Open interval:

$$(a, b) = \{x \mid a < x < b\}$$



$$= \{-1, 0, 1, 2\}$$

Subset: ($\subseteq$)

$$A \subseteq B$$

$$A = \{1, 2, 3\} \quad B = \{1, 2, 3, 4\}$$

if $x \in A$ then $x \in B$

$$A \subseteq B \quad \forall x (x \in A \rightarrow x \in B)$$

$$\underline{\underline{F \rightarrow T}}$$

Any Set:

$$\begin{cases} \emptyset \subseteq A \\ A \subseteq A \end{cases}$$

$$\forall x (x \in \emptyset \rightarrow x \in A)$$

Empty Set / null Set :

$A = \{ \}$ $|A| = 0$ if Cardinality is zero then Empty set
 $A = \{ \{ \} \}$ $|A| = 1$
 $\hookrightarrow$ not Empty Set.

Proper Subset ($\subset$)

$\forall x (x \in A \rightarrow x \in B \wedge \exists x (x \in B \wedge x \notin A))$

$A = \{1, 2, 3\}$ $B = \{1, 2, 3, 4\}$

Subset :

$A \subseteq B$

$|A| \leq |B|$

Proper Subset

$A \subset B$

$|A| < |B|$

eg. $A = \{ \overset{e_1}{x}, \overset{e_2}{\{x\}}, \overset{e_3}{\{\{x\}\}} \}$

element $\in$ Set

Set $\subseteq$ Set

element $\in$ Set

$x \in A$ (T)

$\{x\} \subseteq A$ (T)

$\{x\} \in A$ (T)

Q.1 $A = \{1, \{1\}, \{2\}\}$

a. $1 \in A$ (T)

d. $\{\{1\}\} \subseteq A$ (T)

g. $\{\{2\}\} \subseteq A$ (T)

b. $\{1\} \in A$ (T)

e. $\{2\} \in A$ (T)

h. $\{\{2\}\} \subseteq A$ (T)

c. $\{1\} \subseteq A$ (F)

f. $\{2\} \subseteq A$ (F)

Except if all are false.

Q.2 a. $\phi \in \phi$ (False)

e. $\phi \subset \{\phi\}$ (True)

b. $\phi \subset \phi$ (False)

f. $\phi \subseteq \{\phi\}$ (True)

c. $\phi \subseteq \phi$ (True)

$\phi \subseteq$ anything except ϕ

d. $\phi \in \{\phi\}$ (True)

$\phi \subseteq$ anything

$$\rightarrow A = \{1, 2, 3\}$$

$$\phi \quad \{1, 2\}$$

$$\{1\} \quad \{2, 3\}$$

$$\{2\} \quad \{3, 1\}$$

$$\{3\} \quad \{1, 2, 3\}$$

$$\text{Power set} : \mathcal{P}(A) / 2^A$$

$\hookrightarrow$ Collection of Subsets.

$$\mathcal{P}(A) = \{\phi, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{2, 3\}, \{3, 1\}, \{1, 2, 3\}\}$$

Total no. of elements in $\mathcal{P}(A) = 2^A$.

$$\{\phi \in \mathcal{P}(A)\}$$

$$\{\phi \subseteq \mathcal{P}(A)\}$$

Operations in Set:

1. Set Difference ($-$)

2. Symmetric Difference (Δ / $\oplus$)

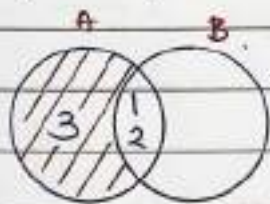
eg. $A = \{1, 2, 3\} \quad B = \{1, 2\}$

$$A - B = \{3\} \quad B - A = \phi$$

Set Difference ($-$)

$$A = \{1, 2, 3\} \quad B = \{1, 2\} \quad A - B = A - (A \cap B)$$

$$= \{3\}$$

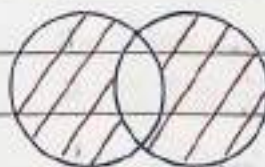


$$B - A = \{\} = \phi$$

Symmetric Difference (Δ / $\oplus$)

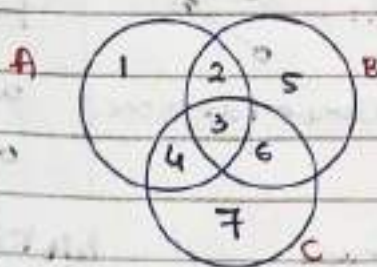
$$A \Delta B = (A - B) \cup (B - A)$$

$$= (A \cup B) - (A \cap B)$$

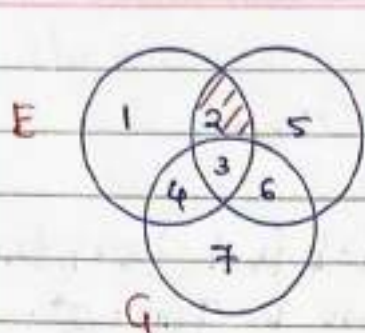


$$A \Delta B = B \Delta A$$

$$A \Delta (B \Delta C) = (A \Delta B) \Delta C$$



$$(A - B) - C = 1$$



$$x = (E \cap F) - (F \cap G)$$

$$y = (E - (E \cap G)) - (E - F)$$

$$\{E \cap F\} - (F \cap G) = \{2\} \quad x = y$$

$$E - (E \cap G) - (E - F) = \{2\}$$

$$A_n = [-2n, 3n]$$

$$n=3$$

$$A_3 = [-6, 9]$$

$$A_4 = [-8, 12]$$

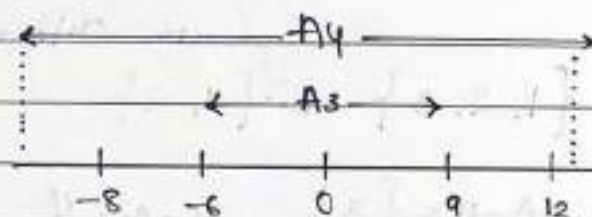
$$\bigcup_{i=1}^7 A_i = A_7$$

$$\bigcap_{i=1}^7 A_i = A_1$$

$$A_3 - A_4 = \emptyset$$

$$A_4 - A_3$$

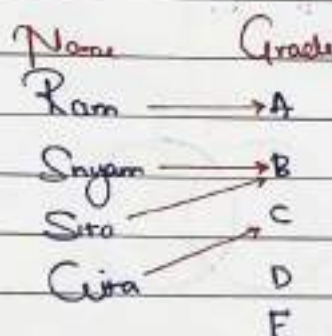
$$= [-8, 6] \cup [9, 12]$$



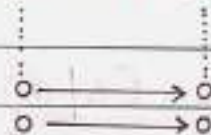
$$\bigcup_{i=1}^7 A_i = A_1 \cup A_2 \cup A_3 \dots A_7 = [-14, 21]$$

$$A_1 = [-2, 3]$$

Function/Assignment/Mapping



f: Set 1 $\rightarrow$ Set 2



Set 1 is mapped to Set 2

Domain

Co-domain

Rule 1

$$a \rightarrow a \checkmark$$

Some
all not

paired, valid.

Rule 2

$$a \rightarrow a \checkmark$$

$$a \rightarrow a$$

left side

Should always point to right side

Rule 3

$$a \rightarrow a \checkmark$$

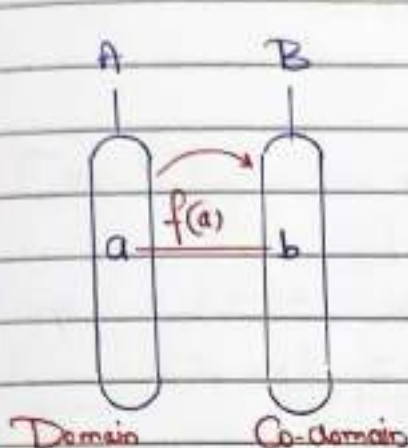
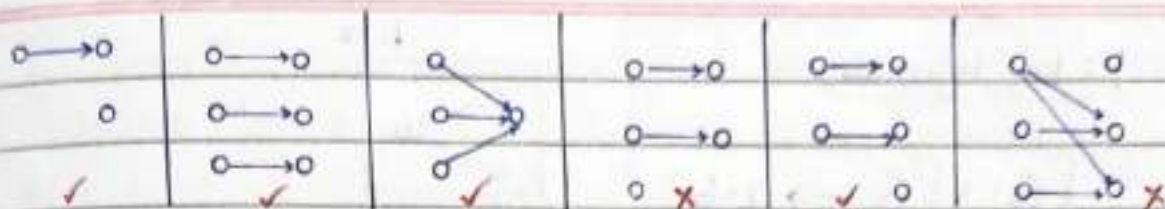
two elements point to one.

Rule 4

$$a \rightarrow a \times$$

$$a \rightarrow a$$

left should not point multiple



$$f: A \rightarrow B$$

$$f(a) = b$$

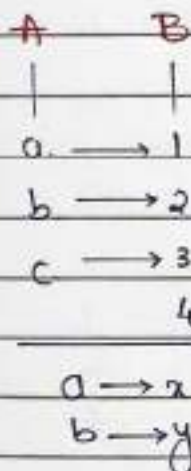
b is image of a

a is called preimage of b .

Range = Set of image

$$\text{Range} \subseteq \text{Codomain}$$

eg:



$$f: A \rightarrow B$$

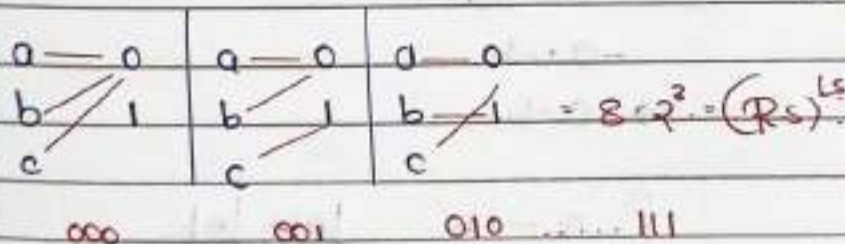
$$\text{Range} = \{1, 2, 3\}$$

$$\text{Co-domain} = \{1, 2, 3, 4\}$$

$$\text{Range} = \{x, y\} \quad \text{Co-domain} = \{x, y\}$$

$$\begin{aligned} a &\rightarrow x \\ b &\rightarrow y \end{aligned}$$

eg: $f: A \rightarrow B$ $|A| = 3$ $|B| = 2$ Total no. of functions
= Total no. of different arrow representations.

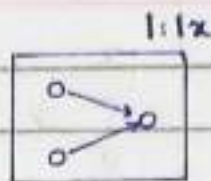


$$= 8 \cdot 2^3 = (2^3)^{2^3}$$

Q. $f: x \rightarrow y$ Total no. of functions = 97
 $(2^3)^{2^3} = y^x \rightarrow |x| = 1 \wedge |y| = 97$
 $(97)^1 = 97$

1:1 (Injection)

$$(f(a) = f(b) \rightarrow a = b)$$



eg.

$$f: \mathbb{Z} \rightarrow \mathbb{Z}$$

$$f(x) = x^2$$

$$0 \rightarrow 0$$

$$1 \rightarrow 1$$

$$a = 2 \quad b = 2$$

$$f(a) = 4 \quad f(b) = 4$$

$$a = 2 \rightarrow 4$$

$$b = 2 \rightarrow 4$$

$$f(a) = f(b) \rightarrow a = b$$

$$\underline{4 = 4} \rightarrow \underline{2 = 2}$$

True

False.

eg.

$$f: \mathbb{Z} \rightarrow \mathbb{Z}$$

$$f(x) = x+1$$

$$f(x) = x+1$$

$$1 \rightarrow 2$$

$$f(x) = x+1$$

$$2 \rightarrow 3$$

$$f(x) = x+1$$

$$3 \rightarrow 4$$

$$f(x) = x+1$$

$$a \quad 0 \rightarrow 0 \quad f(a)$$

$$f(w) = f(p)$$

$$w+1 = p+1$$

$$b \quad 0 \rightarrow 0 \quad f(b)$$

$$w = p$$

$$f: \mathbb{R} \rightarrow \mathbb{R} (1/2)$$

$$f(x) = \frac{4x}{2x-1} (1:1) \checkmark$$

$$2x-1$$

$$f(a) = f(b)$$

$$\underline{4a} = \underline{4b}$$

$$\underline{2a-1} = \underline{2b-1}$$

$$a(2b-1) = b(2a-1)$$

$$2ab - a = 2ab - b$$

$$-a = -b$$

$$\Rightarrow \underline{a = b}$$

$$f: A \rightarrow B$$

$$|A| = 5 \quad |B| = 3$$

$$0 \rightarrow 0$$

$$0 \rightarrow 0$$

$$0 \rightarrow 0$$

$$0 \rightarrow 0$$

Not 1:1

Functions.

$$f: A \rightarrow B$$

$$|A| = 3 \quad |B| = 5$$

$$|A| \leq |B|$$

$$|L.S| \leq |R.S|$$

$$0 \rightarrow 0$$

$$0 \rightarrow 0$$

$$0 \rightarrow 0$$

$$0 \rightarrow 0$$

ways

$$= \frac{5!}{2!}$$

$$= \frac{5!}{2!}$$

$$= \frac{5!}{2!}$$

$$= \frac{5!}{2!}$$

$$= \frac{5!}{2!} \cdot 5P_3 = R.S.$$

$$= \frac{5!}{2!} \cdot 5P_3 = P.L.S$$

$$f: A \rightarrow B$$

$$|A| = 3 \quad |B| = 7$$

$$\# 1:1 \rightarrow \text{Rs. } P_{1:1} = {}^7P_3$$

$$\# \text{ Functions} = (\text{Rs.})^{Ls} = 7^3$$

What is the probability of 1:1 functions?

$$\rightarrow \frac{{}^7P_3}{7^3}$$

Onto (Right side must be full)

Range	Codomain	Range	Codomain	Range	Codomain
$\begin{array}{c} \circ \rightarrow \circ \\ \circ \end{array}$	$\begin{array}{c} \circ \\ \circ \end{array}$	$\begin{array}{c} \circ \rightarrow \circ \\ \circ \rightarrow \circ \end{array}$	$\begin{array}{c} \circ \\ \circ \end{array}$	$\begin{array}{c} \circ \rightarrow \circ \\ \circ \rightarrow \circ \end{array}$	$\begin{array}{c} \circ \\ \circ \end{array}$
Function $\checkmark$					
1:1 $\checkmark$		1:1 $\checkmark$		1:1 $\times$	
Onto $\times$		Onto $\times$		Onto $\checkmark$	

eg.

$$f: \mathbb{Z} \rightarrow \mathbb{Z}$$

$$f(x) = x+1$$

$$x = a \quad a \in \mathbb{Z}$$

$$1 \rightarrow 2 \quad a-1 \in \mathbb{Z}$$

$$2 \rightarrow 3$$

$$3 \rightarrow 4$$

$$f(x) = x+1$$

$$y = x+1$$

$$y-1 = x$$

eg2:

$$f: \mathbb{Z} \rightarrow \mathbb{Z}$$

$$f(x) = x^2$$

$$1 \rightarrow 1$$

$$2 \rightarrow 4$$

$$y = 2$$

$$f(x) = x^2$$

$$2 = x^2$$

$$\sqrt{2} = x$$

$$\sqrt{2} \in \mathbb{R}$$

Not Onto!

$$1. f(x) = x+7$$

$$4. f(x) = 2x-3 \rightarrow \text{not onto}$$

$$2. f(x) = -x+5$$

$$5. f(x) = x^2$$

$$3. f(x) = x^2+1$$

$$6. f(x) = x^3$$

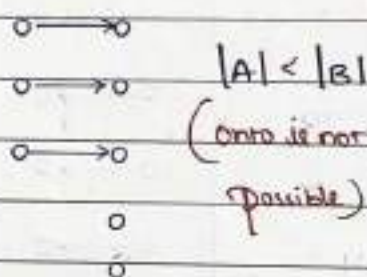
$$f: A \rightarrow B$$

$$|A| = m \quad |B| = n$$

$$\text{Total Onto} = \sum_{i=0}^n (-1)^i \cdot n C_i \cdot (n-i)^m$$

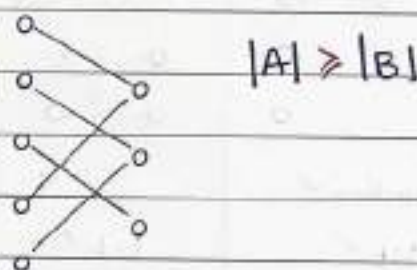
$$f: A \rightarrow B$$

$$|A| = 3 \quad |B| = 5$$



$$f: A \rightarrow B$$

$$|A| = 5 \quad |B| = 3$$

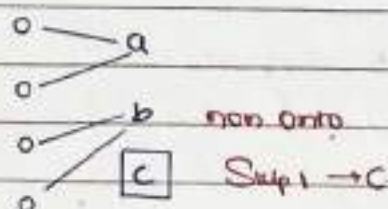
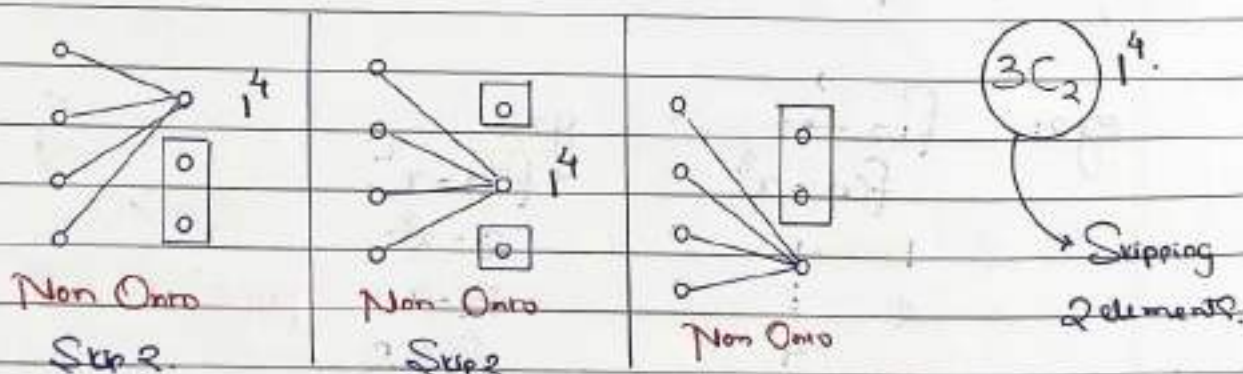


$$\rightarrow |A| = 3 \quad |B| = 2 \quad \text{Total onto functions} = \text{Total} - \text{Total non-onto}$$

$$= 2^3 - 2$$

$$= 6 \text{ Functions (Onto)}$$

$$\text{eg: } f: A \rightarrow B \quad |A| = 4 \quad |B| = 3 \quad \text{Total Onto} = \text{Total} - \text{Total non-onto}$$



$$\text{Total functions if we are not printing on } C \rightarrow 2^4$$

(P.S.) L.S.

$f: A \rightarrow B$ Total onto = Total - Total non onto

$$|A| = 4 = m$$

$$|B| = 3 = n$$

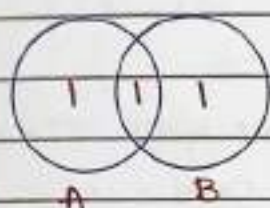
$$= 3^4 - 3C_1 \cdot 2^4 + 3C_2 \cdot 1^4$$

↓
Skipping 1

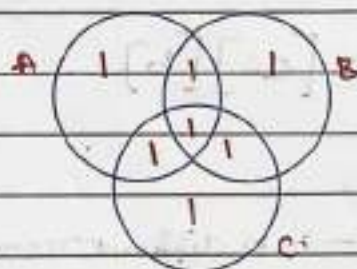
↓
Skipping 2 elements

$$\sum_{i=0}^n (-1)^i \cdot nC_i \cdot (n-i)^m = 3C_0 (3-0)^4 - 3C_1 (3-1)^4 + 3C_2 (3-2)^4$$

$$= nC_0 (n-0)^m - nC_1 (n-1)^m + nC_2 (n-2)^m - nC_3 (n-3)^m$$



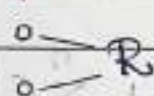
$$\Rightarrow A \cup B = A + B - A \cap B$$



$$\Rightarrow A + B + C - A \cap B - B \cap C - A \cap C + A \cap B \cap C$$

Q1. How many ways we can arrange 7 different guest to 4 different rooms, such that none of the rooms must be empty?

A



$f: A \rightarrow B$

$$|A| = 7 \quad |B| = 4$$

$$m = 7 \quad n = 4$$

$$nC_0 (n-0)^m - nC_1 (n-1)^m + nC_2 (n-2)^m -$$



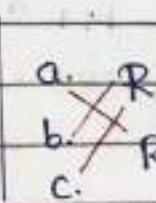
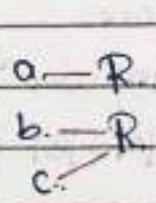
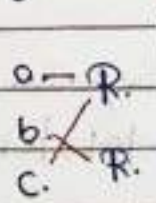
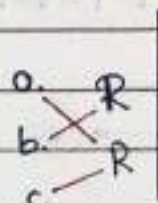
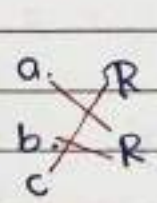
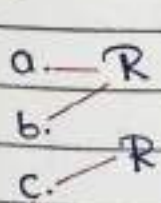
$$= 4C_0 (4-0)^7 - 4C_1 (4-1)^7 + 4C_2 (4-2)^7 - 4C_3 (4-3)^7 + 4C_4 (4-4)^7$$

$$Ans = \underline{8400}$$

$$Total\ onto = 2^3 - 2 = 6$$

Q2. $f: A \rightarrow B$

3 different guest $\rightarrow$ 2 same room



$$(ab)(c)$$

$$(ac)(b)$$

$$(a)(bc)$$

3 different guest $\rightarrow$ 2 different rooms $Onto = 6$
 3 different guest $\rightarrow$ 2 identical rooms $\frac{6}{2} = \frac{Onto}{2!}$

Onto

$[ab]_1, [c]_2 \rightarrow [ab][c]$
 $[ab]_2, [c]_1$

3 guest

2 identical rooms

$[bc]_1, [a]_2 \rightarrow [bc][a]$
 $[bc]_2, [a]_1$

$[ac]_1, [b]_2 \rightarrow [ac][b]$
 $[ac]_2, [b]_1$

3 different guest $\rightarrow$ 2 identical $\frac{Onto}{2!}$

$\rightarrow$ m different guest $\rightarrow$ n different room (none of the rooms should be empty)

$$Onto \rightarrow \sum_{i=0}^n (-1)^i \cdot nC_i \cdot (n-i)^m$$

$\rightarrow$ m different guests $\rightarrow$ n identical rooms

Stirling second kind no. $S(m, n) = \frac{Onto}{n!} = \frac{1}{n!} \sum_{i=0}^n (-1)^i \cdot nC_i \cdot (n-i)^m$

diff.

identical.

1:1 (Injective)

Onto (Surjective)

1:1 Correspondance1:1 $\wedge$ onto

1:1

o—o

o—o

o

 $|A| \leq |B|$ Onto

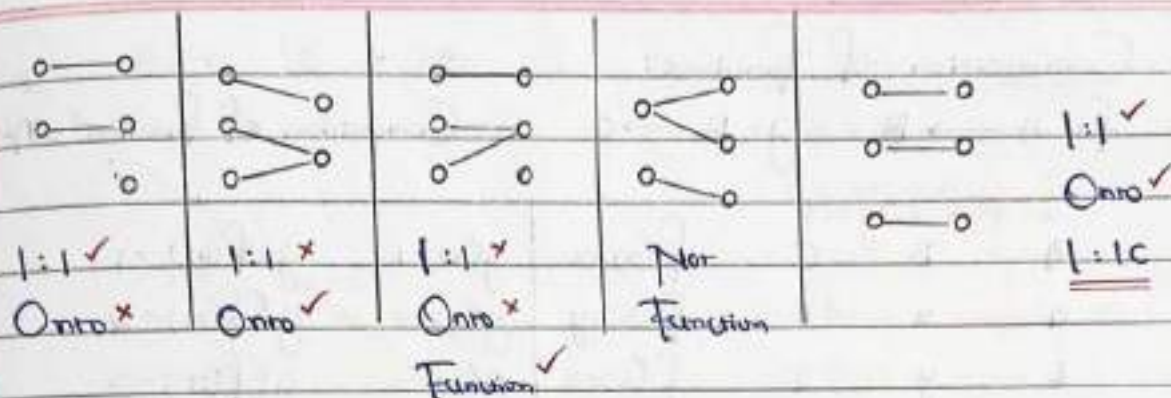
o—o

o—o

o—o

 $|A| \geq |B|$ Bijection = injective $\wedge$ surjective

$$= |A| \leq |B| \wedge |A| \geq |B| \rightarrow (A \sim B)$$

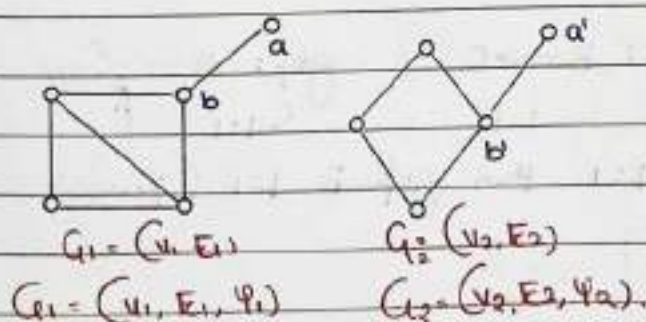


$f(x) = x+1$ $f: 2 \rightarrow 2$ $f(a) = f(b) \rightarrow a=b$ $a+1 = b+1 \rightarrow a=b$ <u>1:1</u>	$f(x) = x+1$ $y = x+1$ $y-1 = x$ <u>Onto</u>	$1 \rightarrow 2$ $2 \rightarrow 3$ $3 \rightarrow 4$ $0 \rightarrow 0$ $0 \rightarrow 0$
--	---	---

eg: $\{1, 2, 3\}$

ϕ	000	FFF
$\{1\}$	001	FFr
$\{2\}$	010	
$\{3\}$	011	
$\{1, 2\}$	100	
$\{1, 3\}$	101	
$\{2, 3\}$	110	
$\{1, 2, 3\}$	111	TTT

$|x| = 30$
 $f: x \rightarrow y$ (1:1C)
 $|x| = |y| = 30$



$$f: G_1 \rightarrow G_2$$

$$f: v_1 \rightarrow v_2 \quad |v_1| = |v_2|$$

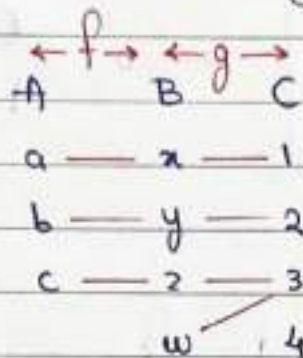
$$f: a \rightarrow a' \quad f: E_1 \rightarrow E_2$$

$$f: b \rightarrow b'$$

$f: A \rightarrow B \quad |A| = |B| = n \quad \text{Total 1:1C Functions} = n!$
(1:1C)

Composition of functions:

$$f: A \rightarrow B \quad g: B \rightarrow C$$

Composition of function: $A \rightarrow C$ 

$$f(a) = x$$

$$f(b) = y$$

$$f(c) = z$$

$$g(x) = 1$$

$$g(y) = 2$$

$$g(z) = 3$$

$$g(w) = 3$$

$$g(f(a)) = 1$$

$$g(f(b)) = 2$$

$$g(f(c)) = 3$$

$$g(f(c)) \downarrow \\ \text{gof}$$

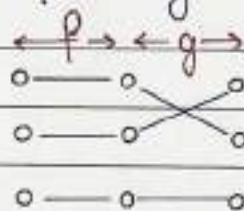
$$f: A \rightarrow B \quad g: B \rightarrow C \\ \text{gof}: A \rightarrow C$$

eg: $f: \mathbb{Z} \rightarrow \mathbb{Z} \quad g: \mathbb{Z} \rightarrow \mathbb{Z}$
 $f(x) = 2x+3 \quad g(x) = x^2$
 $g(a) = a^2$
 $\text{gof} = g(f(x)) = g(2x+3)$
 $\text{gof} = (2x+3)^2$

$$\text{fog} = f(g(x)) \\ = f(x^2) \\ = 2x^2+3$$

$$\text{gof} \neq \text{fog}$$

1. If f and g are 1:1 then gof is 1:1 (True)

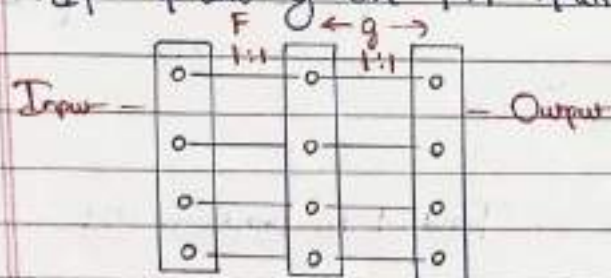


$$\text{gof} \rightarrow 1:1$$

$$f: A \rightarrow B \quad g: B \rightarrow C \quad \text{gof}: A \rightarrow C$$

$\hookrightarrow 1:1 \quad \hookrightarrow 1:1 \quad \hookrightarrow 1:1$

→ If f and g are 1:1 then gof is 1:1 (True)



$$f: \mathbb{Z} \rightarrow \mathbb{Z}$$

$$f(x) = x+1$$

(1:1)

$$g: \mathbb{Z} \rightarrow \mathbb{Z}$$

$$g(x) = x+2$$

(1:1)

$$g \circ f = g(f(x))$$

$$= g(x+1)$$

$$= (x+1)+2 = x+3$$

$$\begin{array}{ccc} x+1 & & x+2 \\ 1 \rightarrow 2 & \rightarrow & 4 \\ 2 \rightarrow 3 & \rightarrow & 5 \\ 3 \rightarrow 4 & \rightarrow & 6 \end{array}$$

$$1 \rightarrow 4$$

$$2 \rightarrow 5$$

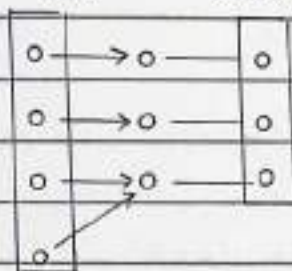
$$3 \rightarrow 6$$

→ If f and g are onto then $g \circ f$ is onto (True)

$$\leftarrow f \rightarrow \leftarrow g \rightarrow$$

onto

onto



→ If $g \circ f$ is onto then f is onto (False)

$$a \rightarrow o_1 \rightarrow ox$$

$$b \rightarrow o_2 \rightarrow oy$$

$$z \xrightarrow{f} o_3$$

$$g \circ f: \mathbb{R} \rightarrow \mathbb{R}$$

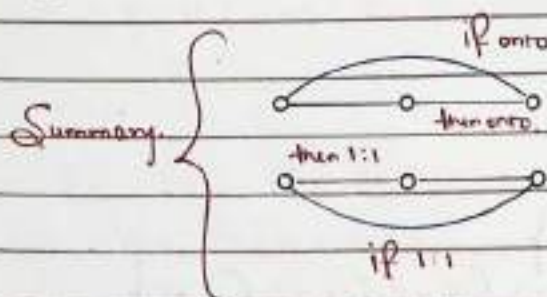
$$a \rightarrow x$$

$$b \rightarrow y$$

→ If $g \circ f$ is onto then g is onto (True)

→ If $g \circ f$ is 1:1 then f is 1:1 (True)

→ If $g \circ f$ is 1:1 then g is 1:1 (False)



Inverse function

$$f: A \rightarrow B$$

$$a \rightarrow x \quad f(a) = x$$

$$b \rightarrow y \quad f(b) = y$$

$$c \rightarrow z \quad f(c) = z$$

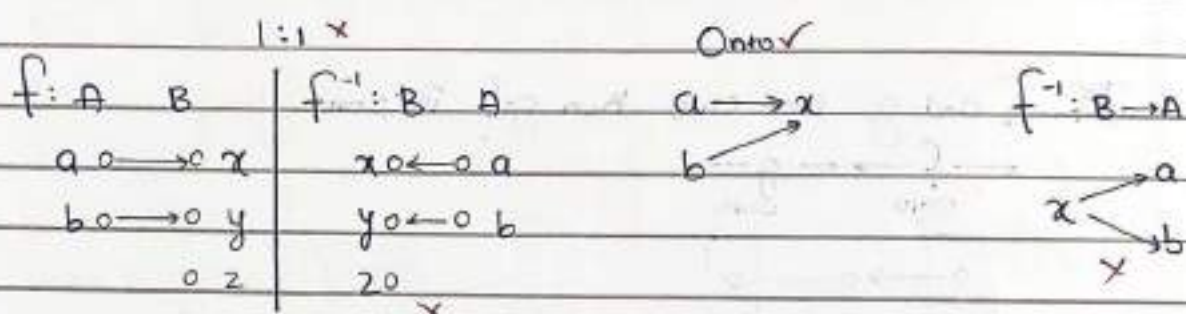
Invertible

$$f^{-1}: B \rightarrow A$$

$$a \leftarrow x \quad f^{-1}(x) = a$$

$$b \leftarrow y \quad f^{-1}(y) = b$$

$$c \leftarrow z \quad f^{-1}(z) = c$$



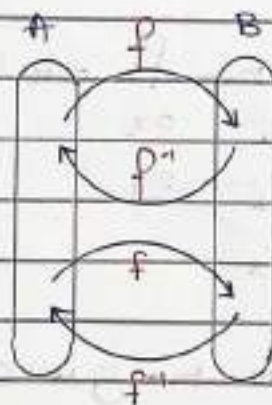
$$i_A: A \rightarrow A$$

Identity function

$$f: A \rightarrow B \quad (1:1)$$

$$f \circ f^{-1} = i_B$$

$$f^{-1} \circ f = i_A$$



$$i_B: B \rightarrow B$$

$$i_A: A \rightarrow A$$

Relations

$$P_1 R P_2$$

$$\{ e_1 R e_2$$

$$s > 3$$

$$Q_1 R Q_2$$

$$C_1 C Q_2$$

$$\text{Set } P \text{ Set } 2$$

$$A \subseteq B$$

eg. $A = \{1, 2, 3\}$ $B = \{2, 3\}$
 $A \times B = \{(1, 2), (1, 3), (2, 2), (2, 3), (3, 2), (3, 3)\}$

After passing through a condition:

Result: $\{(2,2) (2,3)\}$ - Relation

Result: $\{(1,2) (2,3) (4,5)\}$ - Relation $(a < b)$

eg: $A = \{1, 2\}$ $B = \{1\}$
 $A \times B = \{(1,1), (2,1)\}$



$$R_1 = \{\} = \phi$$

$$R_2 = \{(1,1)\}$$

$$R_3 = \{(2,1)\}$$

$$R_4 = \{(1,1) (2,1)\}$$

$A \times B$

$2^{A \times B}$

2

Relation: Subset of Cross product of A, B .

$$|A| = m \quad |B| = n$$

$$|A \times B| = m \cdot n$$

Total Relations = Total no. of subsets

$$= \underline{2^{m \cdot n}}$$

$$|A| = n, |A \times A| = n^2 \quad \text{Total Relations} = \underline{2^{n^2}}$$

↳ When only one set is given.

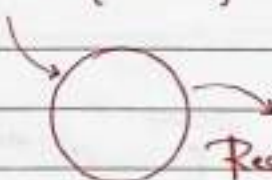
Symmetric Relation

$$\forall a, b [(a, b) \in R \rightarrow (b, a) \in R]$$

$$\text{if } (a, b) \in R \text{ then } (b, a) \in R$$

$$A = \{\}$$

$$A \times A = \{\dots\}$$

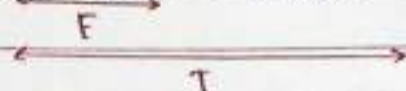


Result:

Symmetric

$$R_1 = \{\}$$

$$(a, b) \in R \rightarrow (b, a) \in R$$



$$R_2 = \{(1,2) (2,1)\} \text{ Symmetric.}$$

$$(a, b) \in R \rightarrow (b, a) \in R$$

$$(1,2) \in R \rightarrow (2,1) \in R$$

T

T

$$R_3 = \{(1,2)\}$$

$$(a, b) \in R \rightarrow (b, a) \in R.$$

$$(1,2) \in R \rightarrow (2,1) \in R$$

T

F

$$a=1, b=2$$

Not Symmetric

$$R_4 = \{(1,1)\} \text{ Symmetric.}$$

$$(a,b) \in R \rightarrow (b,a) \in R$$

$$\underbrace{(1,1)}_T \in R \rightarrow \underbrace{(1,1)}_T \in R \quad a=1 \quad b=1.$$



$$R_1 = \{(2,3) \times, (2,2) \checkmark\}$$

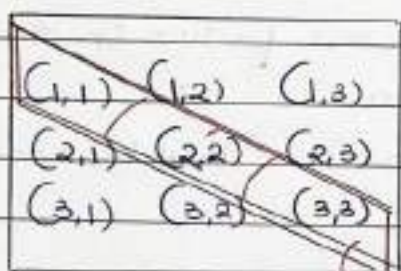
$$R_2 = \{(1,1) \checkmark, (2,2) \checkmark, (3,3) \checkmark, (1,3) \times\}$$

$$R_3 = \{(1,1) \checkmark, (2,2) \checkmark, (3,3) \checkmark\}$$

$$A = \{1, 2, 3\} \quad |A| = n$$

$$n^2 = A \times A =$$

$$\rightarrow 2 \times 2$$



$$A \times A$$

$$\{11 \quad 22 \quad 33 \quad (12) \quad (21) \quad (13) \quad (31) \quad (23) \quad (32)\}$$

$$2 \text{ choice}$$

$$2^n$$

$$\text{Total elements} = n^2$$

$$\text{Total diagonal} = n$$

$$\text{non-diagonal} = n^2 - n$$

$$\{ \underbrace{(11) \quad (22) \quad (33)}_n \mid \underbrace{(12) \quad (21)}_{n^2-n} \mid \underbrace{(13) \quad (31)}_{n^2-n} \mid \underbrace{(23) \quad (32)}_{n^2-n} \}$$

$$2 \text{ row}$$

$$w_1$$

$$w_2$$

$$0$$

$$0$$

$$1$$

$$1$$

$$1$$

$$1$$

$$1$$

$$1$$

$$1$$

$$2 \text{ row}$$

$$w_1$$

$$w_2$$

$$0$$

$$0$$

$$1$$

$$1$$

$$1$$

$$1$$

$$1$$

$$1$$

$$\{(1,1) \quad (1,2)\}$$

$$\{ \phi \quad 00 \}$$

$$\{(1,1)\} \quad 01$$

$$\{(1,2)\} \quad 10$$

$$\{(1,1) \quad (1,2)\} \quad 11$$

$$\{(1,1) \quad (1,2)\} \quad 11$$

$$\{(1,1) \quad (1,2)\} \quad 11$$

$$\{(1,1) \quad (1,2)\} \quad 11$$

$$\{(1,1) \quad (1,2)\} \quad 11$$

Reflexive: $A \rightarrow n$ on empty set.

$$A = \{1, 2, 3\}$$

$$\forall a \in A (a, a) \in R$$

$$A = \{1, 2, 3\}$$

$$\forall a \in A (a, a) \in R$$

$$R_1 = \{(1,1) (2,2)\} \text{ not Reflexive}$$

$$A = \{1, 2, 3\} \quad R_2 = \{(1,1)\} \quad \times$$

$$\forall a \in A (a, a) \in R$$

$$|A| = n$$

$$|A \times A| = n^2$$

(1,1)	(1,2)	(1,3)
(2,1)	(2,2)	(2,3)
(3,1)	(3,2)	(3,3)

1 choice
Power 1

$$\text{Total elements} = n^2$$

$$\text{Diagonal elements} = n$$

$$\text{non-diagonal elements} = n^2 - n$$

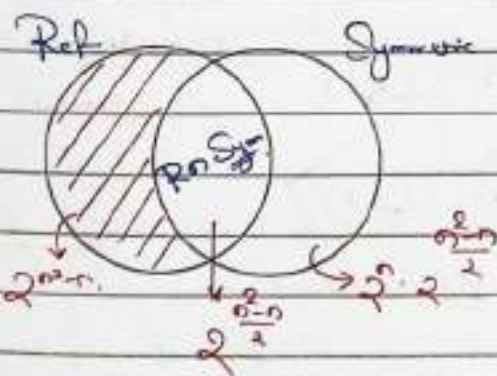
2 choice for non-diagonal

$$\#R = 2^{n^2 - n}$$

$$A \times A = \{(1,1) (2,2) (3,3) (1,2) (2,1) (2,3) (3,2) (1,3) (3,1)\}$$

$$R_1 = \{ \boxed{11 \ 22 \ 33} (1,2) \}$$

$$R_2 = \{ \boxed{11 \ 22 \ 33} (2,1) \}$$



$$|A| = n$$

$$\#R = 2^{n^2 - n}$$

$$\#S = 2^n \cdot 2^{\frac{n^2 - n}{2}}$$

diagonal must be present.

$$\text{Ref} \cap \text{Sym} = 1^n \cdot 2^{\frac{n^2 - n}{2}} = 2^{\frac{n^2 - n}{2}}$$

Irreflexive : (have some element) $A \rightarrow$ non empty set

$$A = \{1, 2, 3\}$$

$$\forall a \in A, (a, a) \notin R$$

$$R_1 = \{(1, 2)\} \text{ IRR } \checkmark$$

$$R_2 = \{(2, 3)\} \text{ IRR } \checkmark$$

$$R_3 = \{(1, 1)\} \text{ IRR } \times$$

$$R_4 \times$$

$$R_5 = \{(1, 2), (2, 3), (2, 2)\} \text{ IRR } \times$$

$$R_6 \times$$

$$R_7 = \{\} \checkmark \text{ IRR } \checkmark$$

$$R_8 = \{(1, 2)\} \checkmark$$

$$R_9 = \{(1, 2), (2, 1), (2, 3), (3, 2)\} \checkmark$$

$$R_{10} = \{(2, 1)\} \checkmark$$

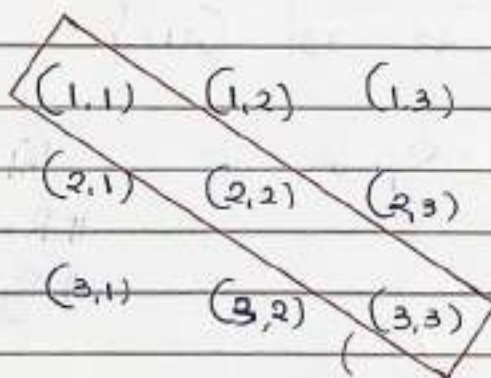
$$R_{11} = \{(1, 3), (3, 2)\} \text{ IRR } \times$$

$$R_{12} = \{(1, 1)\} \times$$

$$\#R = 2^{n^2 - n} \left\{ \begin{array}{l} \boxed{11 \ 22 \ 33} (1, 2) \\ \boxed{11 \ 22 \ 33} (1, 2) (2, 1) \end{array} \right.$$

$$\#IRR = 2^{n^2 - n} \left\{ \begin{array}{l} \boxed{\text{absent}} (1, 2) \\ \boxed{\text{absent}} (1, 2) (2, 1) \end{array} \right.$$

$$A = \{1, 2, 3\}$$



$$\text{Total} = n^2$$

$$\text{Diagonal} = n$$

$$\text{non-diagonal} = n^2 - n$$

$$(2 \text{ choice} \rightarrow \text{absent})$$

$$\rightarrow \text{present}$$

$$2^{n^2 - n}$$

 $\rightarrow$ If A is non empty set.

$$R = \{\}$$

Reflexive $\times$ Irreflexive $\checkmark$

1 choice absent.

Anti Symmetric : — Does not allow Pepping
— Allow same element

$$\forall a \forall b [(a,b) \in R \wedge (b,a) \in R \rightarrow a=b]$$

$$R_1 = \{ \} \rightarrow \text{Anti} \checkmark$$

$$(a,b) \in R \wedge (b,a) \in R \rightarrow a=b$$

$$\frac{F \wedge F}{F}$$

$$\frac{F}{\text{True}}$$

$$R_2 = \{ (1,1) \} \rightarrow \text{Anti} \checkmark$$

$$(a,b) \in R \wedge (b,a) \in R \rightarrow a=b$$

$$\frac{(1,1) \in R \wedge (1,1) \in R \rightarrow 1=1}{T \wedge T \rightarrow \text{True}}$$

$$a=1, b=1$$

$$R_3 = \{ (1,2) (2,1) \} \rightarrow \text{Anti} \times$$

$$(a,b) \in R \wedge (b,a) \in R \rightarrow a=b$$

$$\frac{(1,2) \in R \wedge (2,1) \in R \rightarrow 1=2}{T \wedge T \rightarrow \text{False}}$$

$$\frac{T}{\text{False}}$$

$$R_4 = \{ (1,2) \}$$

$$(a,b) \in R \wedge (b,a) \in R \rightarrow a=b$$

$$\frac{(1,2) \in R \wedge F}{F}$$

$$\frac{F}{\text{True}}$$

$$R_5 = \{ \} \checkmark$$

$$R_6 = \{ (2,2) (1,1) \} \checkmark$$

$$R_7 = \{ (2,2) (1,1) (2,2) (3,3) (1,1) \} \checkmark$$

$$R_8 = \{ (2,2) \} \checkmark$$

$$R_9 = \{ (1,3) \} \checkmark$$

$$R_{10} = \{ (1,1) (2,2) (1,2) (2,1) \} \times$$

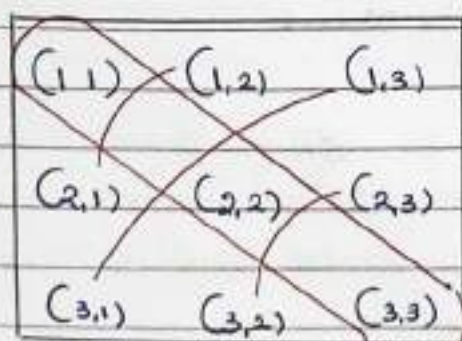
Not anti

$$A = \{ 1, 2, 3 \}$$

$$A \times A = \{ (1,1) (2,2) (3,3) \mid (1,2) (2,1) \mid (1,3) (3,1) \mid (2,3) (3,2) \}$$

n n^2 $n^2 - n$ $\text{boxes} = \frac{n^2 - n}{2}$

$2^n \times 3^{\frac{n^2 - n}{2}}$



$$\text{Total} = n^2 \quad \text{Diagonal} = n$$

$$\text{Non-Diagonal} = n^2 - n$$

$$\text{boxes} = \frac{n^2 - n}{2}$$

3 choice

2 choice 2^n $\{ 2^n \times 3^{\frac{n^2 - n}{2}} \}$

$$\rightarrow A = \{0, 1, 2, 3\}$$

(i) What is Smaller size of Anti Symmetric $\rightarrow 0$

(ii) larger size of Anti Symmetric $\rightarrow \left\{n + \frac{n-1}{2}\right\}$

(iii) Total no. of Anti Symmetric: $2^{\frac{n(n-1)}{2}}$

(iv) Symmetric Smaller size $\rightarrow 0$

(v) Symmetric larger size $\rightarrow n^2$

(vi) Reflexive Smaller $\rightarrow 0$

(vii) Reflexive Smaller $\rightarrow n$

(viii) Reflexive larger $\rightarrow n^2 - n$

(ix) Reflexive larger $\rightarrow n^2$

Asymmetric - Does not allow same element

- Does not allow flipping.

$$\forall a \neq b \quad (a, b) \in R \rightarrow (b, a) \notin R$$

$$\begin{array}{l} R_1 = \{ \} \text{ Asymmetric } \checkmark \\ (a, b) \in R \rightarrow (b, a) \notin R \\ \text{F} \rightarrow \text{True.} \end{array}$$

$$\begin{array}{l} R_2 = \{(1, 1)\} \text{ Asy } \checkmark \\ (1, 1) \in R \rightarrow (1, 1) \notin R \\ \text{T} \quad \text{F} \\ \text{False.} \end{array}$$

$$R_3 = \{(1, 2)\} \text{ Asy } \checkmark$$

$(a, b) \in R \rightarrow (b, a) \notin R$ - if (a, b) belongs to R then (b, a) should not belong to R .

$$\begin{array}{l} (1, 2) \in R \rightarrow (2, 1) \notin R \\ \text{T} \quad \text{T} \end{array}$$

- if (a, b) is present then (b, a) must be absent.

(1, 1)	(1, 2)	(1, 3)
(2, 1)	(2, 2)	(2, 3)
(3, 1)	(3, 2)	(3, 3)

$$\rightarrow 3^{\frac{3 \times 2}{2}}$$

1 Choice
Absent.

Symmetric	AntiSymmetric	Asymmetric
* allow same element.	* allow same element.	* allow same element.
* demands flipping $\frac{n^2-n}{2}$ $2^2 - 2$	* no flipping $\frac{n^2-n}{2}$ $2^2 - 3$	* no flipping $\frac{n^2-n}{2}$ $3^2 - 2$
$(a,b) \in R \rightarrow (b,a) \in R$	$(a,b) \in R \wedge (b,a) \in R \rightarrow a=b$	$(a,b) \in R \rightarrow (b,a) \notin R$

Q. $A = \{1, 2, 3, 4\}$ $R = \{(1,1) (2,2) (2,2) (2,3) (2,4) (3,1) (3,2) (3,3) (3,4)\}$ Reflexive $\times$ (4,4) is absentAntiSymmetric $\times$ Transitive $\times$ hate sameAsymmetric $\times$ Symmetric $\times$

	Ref	Tr	Sym	Anti	Asy.
Ref.	$\times$	0	1. $\frac{n^2-n}{2}$ $2^2 - 2$	1. $\frac{n^2-n}{2}$ $2^2 - 3$	0
Tr		$\times$	1. $\frac{n^2-n}{2}$ $2^2 - 2$	1. $\frac{n^2-n}{2}$ $2^2 - 3$	1. $\frac{n^2-n}{2}$ $3^2 - 2$
Sym.			$\times$	2	1
Anti				$\times$	3. $\frac{n^2-n}{2}$ $3^2 - 2$
Asy.					$\times$

 $A \rightarrow$ non empty set. $A = \{1, 2, 3\}$ $R_1 \rightarrow$ reflexive relation on A $R_1 = \{(1,1) (2,2) (3,3) (1,1) (2,2) (3,3) \dots\}$ $R_2 \rightarrow$ reflexive relation on A $R_2 = \{(1,1) (2,2) (3,3) (1,1) \dots\}$ $R_1 \cup R_2 = \{ \} \}$ Both $R_1 \cap R_2 = \{ \} \}$ Reflexive

	Union (U)	Intersection (∩)	$R_1 = \{(1,2)\}$	$R_2 = \{(2,1)\}$
R_1, R_2 reflexive.	✓	✓	Not ✓	Not ✓
R_1, R_2 transitive.	✓	✓		
R_1, R_2 Symmetric	✓	✓	$R_1 \cup R_2 = \{(1,2), (2,1)\}$	
Anti Symmetric	✗	✓	Not Anti	
Asymmetric	✗	✓	$R_1 \cap R_2 = \{\}$	

$$R_1 = \{a, b \mid a + b \leq 3\}$$

Anti ✗

Asy. ✗

Set: 2

2x2

Reflexive ✗

$$aRa$$

$$a + a \leq 3$$

$$2a \leq 3$$

Symmetric:

$$(a,b) \in R \rightarrow (b,a) \in R$$

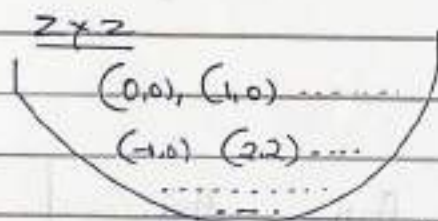
$$aRb \rightarrow bRa$$

$$\frac{a+b \leq 3}{T} \rightarrow \frac{b+a \leq 3}{T}$$



$$a+b \leq 3$$

$$R_{\text{reflexive}}: (0,0), (1,1)$$



$$a+b \leq 3 \rightarrow \{(1,2), (2,1), (0,3), (3,0), \dots\}$$

$$R_2 = \{(a,b) \mid a = b+1\}$$

Reflexive: ✗

$$aRa$$

$$a = a+1$$

Symmetric:

$$aRb \rightarrow bRa$$

$$a = b+1 \rightarrow b = a+1$$

$$(2,1) \rightarrow 1 \neq 2+1$$

Anti Symmetric ✓

Asymmetric ✓

$$G = \{G_1, G_2, G_3, \dots\}$$

$$G \times G$$

$$(G_1, G_2), (G_1, G_3), \dots$$

$$(G_1, G_3), \dots$$

$$R = \{(G_1, G_2) \mid G_1 \text{ is isomorphic to } G_2\}$$

$$R = G_1 R G_2$$

$$\text{Symmetric: } G_1 R G_2 \rightarrow G_2 R G_1$$

Anti ✓

Asy ✓

Ref ✓

$$A = \{1, 2, 3\}$$

$$P(A) = \{\phi, \{1\}, \{2\}, \dots, \{1, 2, 3\}\}$$

$$P(A) \times P(A) = \{(\phi, \phi), (\phi, \{1\}), (\{1\}, \{1\}), (\{1\}, \{2\}), \dots, (\{1, 2, 3\}, \{1, 2, 3\})\}$$

$$P: \{(A, B) \mid A \subseteq B\}$$

$$Ref: A \subseteq A \quad A \subseteq A$$

$$Sym: A \subseteq B \rightarrow B \subseteq A \quad (\text{no flipping})$$

$$A \subseteq B \rightarrow B \subseteq A$$

$$\{1\} \subseteq \{1, 2\} \rightarrow \{1, 2\} \subseteq \{1\}$$

Anti ✓

Asymmetric ✗

Transitive

$$\{(a, b) \in R \wedge (b, c) \in R \rightarrow (a, c) \in R\}$$

$$R_1 = \{ \}$$

$$\underbrace{\underbrace{(a, b) \in R}_F \wedge \underbrace{(b, c) \in R}_F}_{F} \rightarrow (a, c) \in R$$

$$R_2 = \{(1, 1)\}$$

$$(a, b) \in R \wedge (b, c) \in R \rightarrow (a, c) \in R$$

$$\underbrace{\underbrace{(1, 1) \in R}_T \wedge F}_T$$

F

$$R_3 = \{(1, 2)\}$$

$$(a, b) \in R \wedge (b, c) \in R \rightarrow (a, c) \in R$$

$$\underbrace{(1, 2) \in R}_F \wedge F$$

$$a=1, b=2$$

F

True

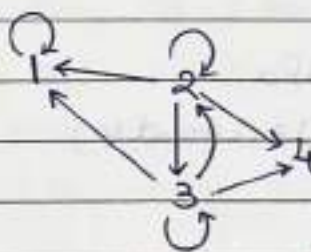
$$(a, b) \in R \wedge (b, c) \in R \rightarrow (a, c) \in R$$

$$(2, 1) \in R \wedge (1, 2) \in R \rightarrow (2, 2) \in R$$

$$R_4 = \{(1, 2), (2, 1), (1, 1)\} \quad \text{Not Transitive}$$

$$(a, b) \in R \wedge (b, c) \in R \rightarrow (a, c) \in R$$

$$\underbrace{(1, 2) \in R}_T \wedge \underbrace{(2, 1) \in R}_T \rightarrow \underbrace{(1, 1) \in R}_T$$



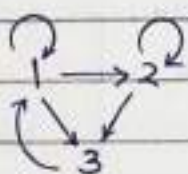
$$\begin{matrix} 3 \\ \downarrow \\ (3, 1) (3, 2) (3, 3) \\ (3, 4) \end{matrix}$$

$$\rightarrow \{ (1, 1) (2, 1) (3, 2) (2, 2) (2, 4) (3, 1) (3, 2) (3, 3) (3, 4) \}$$

$$\begin{matrix} 1 & 2 \\ \downarrow & \downarrow \\ (1, 1) & (2, 2) (2, 3) (2, 4) (2, 1) \end{matrix}$$

$$\rightarrow A = \{1, 2, 3\}$$

$$R = \{ (1, 2) (2, 3) (1, 3) (1, 1) (2, 2) (3, 1) \}$$



$$R' = \begin{matrix} 1 \rightarrow \{ (1, 1) (1, 2) (1, 3) \\ 2 \rightarrow (2, 2) (2, 1) (2, 3) \\ 3 \rightarrow (3, 1) (3, 2) (3, 3) \end{matrix}$$

if $R = R'$ then R is transitive

Equivalence Relation ::

Reflexive
Symmetric
Transitive

 (R_{SR})

Then

Equivalence R.

$$A = \{1, 2, 3\}$$

$$R_1 = \{ \underbrace{(1,1), (2,2), (3,3)}_{R_{SR}}, \boxed{(1,2), (2,1)} \}$$

 R_{SR}

$$R_1 = \{ (a,b) \mid a+b = \text{even} \}$$

Re: aRa

$$a+a = \text{even}$$

$$2a = \text{even}$$

Symmetric:

$$aRb \rightarrow bRa$$

$$a+b = \text{even} \rightarrow b+a = \text{even}$$

if $a+b = \text{even}$ then $b+a = \text{even}$

$$(0,4) \in R \rightarrow (4,0) \in R$$

$$0+4 = \text{even} \rightarrow 4+0 = \text{even}$$

Transitive ✓

$$aRb \wedge bRc \rightarrow aRc$$

$$a+b = \text{even} \wedge b+c = \text{even} \rightarrow a+c = \text{even}$$

$$(0,4) \in R \wedge (4,6) \in R$$

$$0+4 = \text{even} \wedge 4+6 = \text{even} \rightarrow 0+6 = \text{even}$$

$$1+3 = \text{even} \wedge 3+5 = \text{even} \rightarrow 1+5 = \text{even}$$

$$R_2 = \{ (a,b) \mid a \equiv b \pmod{4} \}$$

R: aRa ✓

$$a \equiv a \pmod{4}$$

$$aRb \wedge bRc \rightarrow aRc$$

$$a \equiv b \pmod{4} \wedge b \equiv c \pmod{4}$$

$$\rightarrow a \equiv c \pmod{4}$$

Sym:

$$aRb \rightarrow bRa$$

$$a \equiv b \pmod{4} \rightarrow b \equiv a \pmod{4}$$

$$(0,4) \in R \rightarrow (4,0) \in R$$

 a, b are having same remainder wrt 4.

$$(0,4) \in R \quad 0 \equiv 4 \pmod{4}$$

$$(1,5) \in R \quad 1 \equiv 5 \pmod{4}$$

$$(0,4) \in R \wedge (4,8) \in R \rightarrow (0,8) \in R$$

$$\underbrace{0=4 \pmod{4}}_{\rightarrow 0} \wedge \underbrace{4=8 \pmod{4}}_{\rightarrow 0} \rightarrow \underbrace{0=8 \pmod{4}}_{\rightarrow 0}$$

→ Equivalence Relation Create Partition On Set.

Set: A

A × A



Check (EQR)

Result: { ... }

eg: $A = \{1, 2, 3, 4\}$
 $R = \{(1,2), (3,4)\}$

How to Create a Partition?

$$\{(a,b) \in R \text{ or } aRb\}$$

a, b will go in one Partition.

$$(1,2) \in R$$

1, 2 will go in one partition.

Set: A

1, 2	→ P ₁
3, 4	→ P ₂

eg: Set: Z
 $Z \times Z$ partition.

$$R = \{(a,b) \mid a+b = \text{even}\}$$

(Rsr)

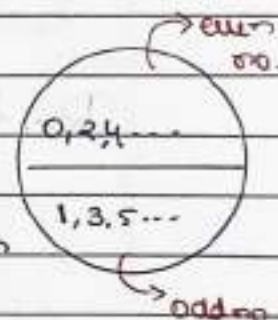
EQR ✓

$$(1,3) \in R \quad 1+3 = \text{even}$$

$$(0,0) \in R \quad 0+0 = \text{even}$$

$$(0,1) \in R \quad 0+1 \neq \text{even}$$

$$(0,2) \in R \quad 0+2 = \text{even}$$



$$R = \{(a,b) \mid a=b \pmod{4}\}$$

(Rsr) EQR ✓

$$(0,4) \in R \quad 0=4 \pmod{4}$$

$$(1,5) \in R \quad 1=5 \pmod{4}$$

0, 4
1, 5
2, 6
3, 7

$$R = \{(s_1, s_2) \mid s_1, s_2 \text{ will go in same state}\}$$

$$R = S R S$$

$$S_1 = S_1 R S_2 \rightarrow S_2 R S_1$$

$$T: S_1 R S_2 \wedge S_2 R S_3$$

$$01 R 01 \wedge 01 R 011 \rightarrow 01 R 011$$

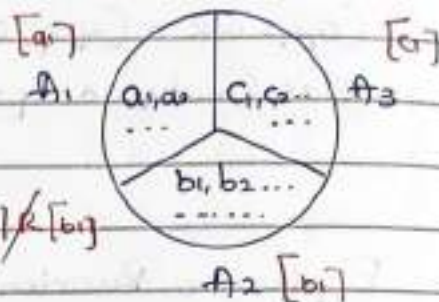


Set A

EQR Create partition on Set A

(equivalence class)

(A_1, A_2, A_3)



$$1) A_1 \cup A_2 \cup A_3 = A$$

$$2) A_1 \cap A_2 \cap A_3 = \phi$$

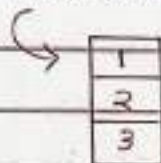
$[]_R$: Representation of a class



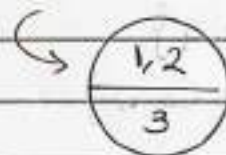
Set : A = {1, 2, 3}

Total equivalence = Total no. of partitions

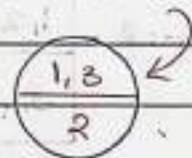
$$R_1: \{(1,1) (2,2) (3,3)\}$$



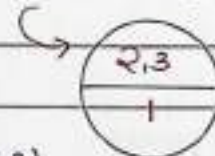
$$R_2: \{(1,1) (2,2) (3,3) (1,2) (2,1)\}$$



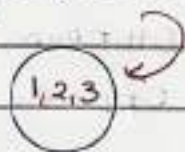
$$R_3: \{(1,1) (2,2) (3,3) (1,3) (3,1)\}$$



$$R_4: \{(1,1) (2,2) (3,3) (2,3) (3,2)\}$$



$$R_5: \{(1,1) (2,2) (3,3) (1,2) (2,1) (2,3) (3,2) (3,1) (1,3)\}$$



$\therefore$ Total no. of partitions = 5

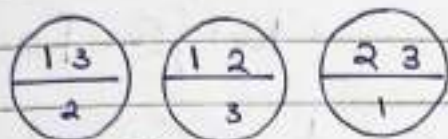
Total Equivalence relation = Total Partitions

$$= S(3,1) + S(3,2) + S(3,3)$$

$$= \sum_{i=1}^B S(3,i)$$

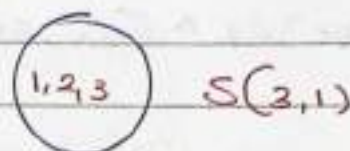
$$B(n) = \sum_{i=1}^n S(m,n)$$

How many ways we can adjust 2 guest 2 identical rooms?

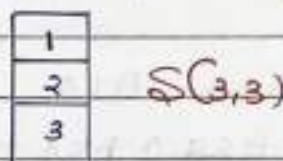


$S(3,2)$
diff. $\rightarrow$ identical

How many ways we can adjust 3 guest to 1 room.



3 guest and 3 identical rooms.



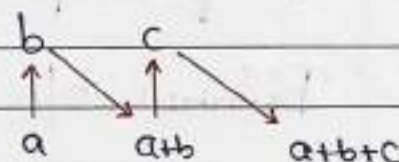
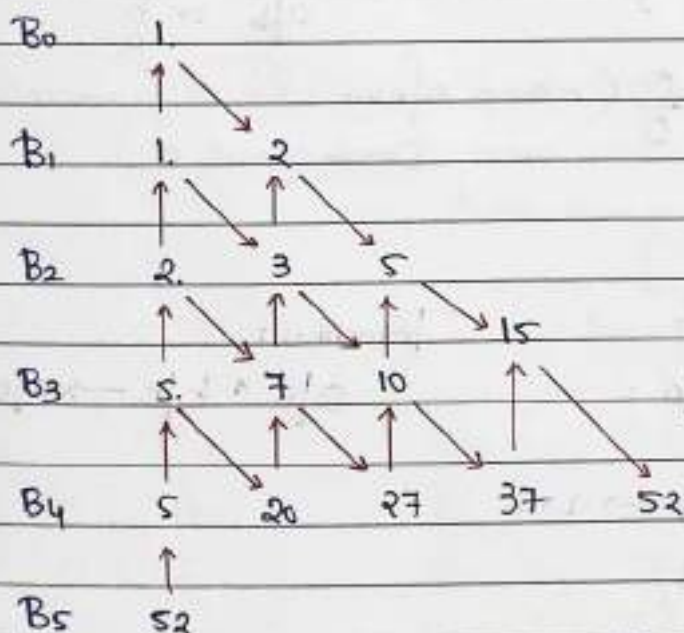
$$S(3,1) + S(3,2) + S(3,3)$$

$$\Rightarrow \sum_{i=1}^3 S(3,i)$$

$$B(n) = \sum_{i=1}^n S(n,i)$$

↓
Bell Number

Triangle



$$R: \{(a,b) \mid a \leq b\}$$

Partial Order Relation:

$$R: a R a \quad a \leq a(T)$$

(Ref) { Reflexive
Antisymmetric
Transitive

$$\text{Anti: } a R b \wedge b R a \rightarrow a = b$$

$$a \leq b \wedge b \leq a \rightarrow a = b$$

$$2 \leq 3 \wedge 3 \leq 2$$

$$T \wedge F \rightarrow T$$

$$I. a \leq b \wedge b \leq c \rightarrow a \leq c$$

$$2 \leq 3 \wedge 3 \leq 5 \rightarrow 2 \leq 5$$

$$(2,3) \in R \wedge (3,5) \in R \rightarrow (2,5) \in R$$

$$R_1: \{(a,b) \mid a \leq b\} \text{ Rat.}$$

$$R_2: \{(a,b) \mid a < b\} \text{ Rat.}$$

$$R: \{(a,b) \mid a \leq b\} \text{ (Rat.)}$$

$$R: \{(a,b) \mid \gcd(a,b) = 1\}$$

$$R: A \leq A \quad A \leq A$$

$$\text{Anti: } A \leq B \wedge B \leq A \rightarrow A = B$$

$$T: A \leq B \wedge B \leq A \rightarrow A \leq C$$

$$\times R: \gcd(2,2) = 2 \neq 1$$

$$\times \text{Anti: Relation allows flipping}$$

So not anti.

$$\times \text{Transitive:}$$

$$\gcd(3,5) \wedge \gcd(5,9) \rightarrow \gcd(3,9)$$

$$= 1 \quad = 1 \quad = 3 \neq 1$$

$$R: \{(a,b) \mid a \text{ divides } b\}$$

$$R: \{(a,b) \mid a \mid b\}$$

$$a \text{ divides } b$$



$$a \mid b \text{ or } \frac{b}{a}$$

$$\text{Reflexive: } 0 \mid 0 \quad \frac{0}{0} \text{ (it is not reflexive when Domain allows '0')}$$

$$D: \mathbb{Z}^+$$

$$R: \{(a,b) \mid a \mid b\}$$

$$R: a \leq a \quad a \mid a$$

$$\text{Transitive:}$$

$$a \mid b \wedge b \mid c \rightarrow a \mid c.$$

$$\text{Anti: } a \mid b \wedge b \mid a \rightarrow a = b$$

$$3 \mid 6 \wedge 6 \mid 3$$

F

$$\text{Set: } \mathbb{Z}$$

$$\text{Relation: } \{(a,b) \mid a \leq b\}$$

$$\text{Set: } \mathbb{Z}^+$$

$$\text{Relation: } \{(a,b) \mid a \mid b\}$$

Set: $\{a, b, \dots\}$

aRb or bRa

a, b are called
Comparable.

Relation: 1

Set: $2^1 \{1, 2, 3, 4, \dots\}$

Only partial elements are Comparable

$a=3$ $b=6$

aRb or bRa

3, 6 are called

$3R6$ or $6R3$

Comparable

$3/6$ or $6R3$

T

Domain: $2^2 \{-2, -1, 0, +1, 2, \dots\}$

$R: \{(a, b) \mid a \leq b\}$

↓
Total Order Relation
(Linear order
relations)

$a=3$ $b=5$

aRb or bRa

$a \leq b$ or $b \leq a$

$3 \leq 5$ or $5 \leq 3$

T

$a=100$ $b=-1$

$a \leq b$ or $b \leq a$

$100 \leq -1$ or $-1 \leq 100$

T

Relations $\rightarrow$ $R \cap T \rightarrow$ Partial order relations



All elements are Comparable
(YES)



Total Order Relations

Por	Por	Por	Por
$R_1 \cup R_2$	R_2	$R_1 \cap R_2$	R_2
$R \cup R$	R ✓	$R \cap R$	R ✓
Anti $\cup$ Anti	Anti ✗	Anti $\cap$ Anti	Anti ✓
Tr. $\cup$ T	T ✗	T $\cap$ T	T ✓

Equivalence Relations

$= R \cap T$

Partial Order Relations

$= R \cup T$

$A: \{1, 2, 3\}$

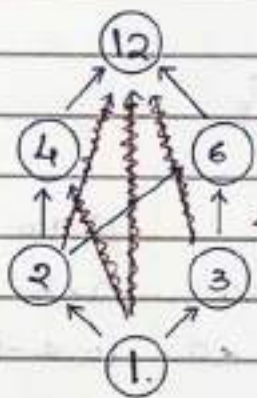
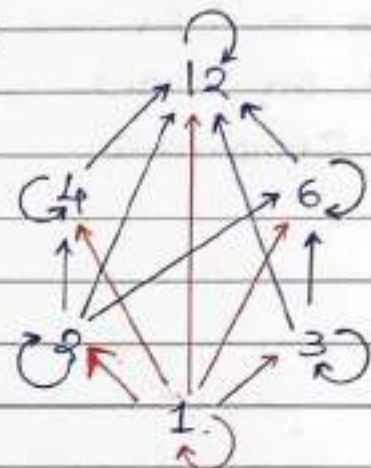
Por

$R_1: \{(1,1), (2,2), (3,3), (1,2)\}$

Por

$R_2: \{(1,1), (2,2), (3,3), (2,1)\}$

Set: $\{1, 2, 3, 4, 6, 12\}$ Partial Order Relation
 $R: \{(1,1) (1,2) (1,3) (1,4) (1,6) (1,12) \}$
 $(2,2) (2,4) (2,6) (2,12)$
 $(3,3) (3,6) (3,12)$
 $(4,4) (4,12)$
 $(6,6) (6,12)$
 $(12,12) \}$

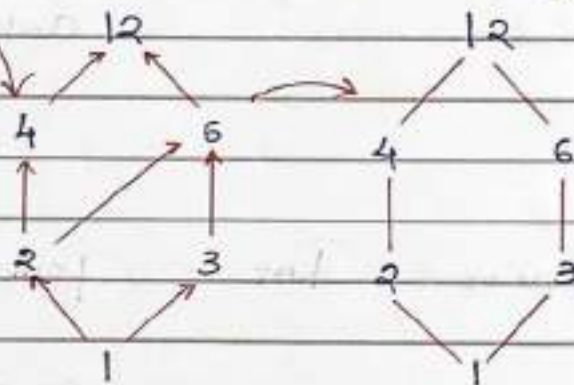


Simplification:

Remove Self loop

Remove transitive

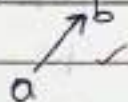
Hasse Diagram



aRb
 or
 $(a,b) \in R$

b
 $\uparrow$
 a

a , will be at lower level
 b , will be at higher level.

Set: $\mathbb{Z}^+$ Relations: $\leq$ Per

(Set, Relations) (Power)

Partial
Order
Set

Total Order Relation

 $(\mathbb{Z}, \leq)$

Total Order Set

(TOSET)

Set: $\{1, 2, 3, 4, 6, 12\}$

$R: \{(1,1) (1,2) (1,3) (1,4) (1,6) (1,12)$

$(2,2) (2,4) (2,6) (2,12)$

$(3,3) (3,6) (3,12)$

$(4,4) (4,12)$

$(6,6) (6,12)$

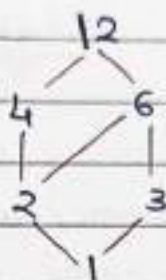
$(12,12) \}$

(Set, Relation)

$(\{1, 2, 3, 4, 6, 12\}, |)$

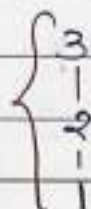
Divisor of 12

$(D_{12}, |)$



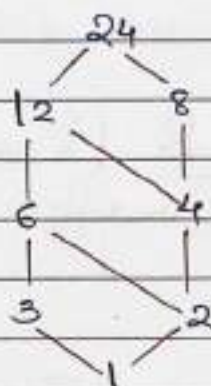
(Set, Relation)

$(\{1, 2, 3\}, \leq)$



$(D_{24}, |)$

$(\{1, 2, 3, 4, 6, 8, 12, 24\}, |)$



#edges = 10

$(D_{36}, |)$

$(\{1, 2, 3, 4, 6, 9, 12, 18, 36\}, |)$



$(P(A), \subseteq)$ $A = \{1, 2, 3\}$

$123 (111)$

$(110) 12$

$13 (101)$

$23 (011)$

$(100) 1$

$2 (100)$

$3 (001)$

$\phi (000)$

$\{ \text{Hyper Cube} \}$

Power set: $P(A)/2^A$

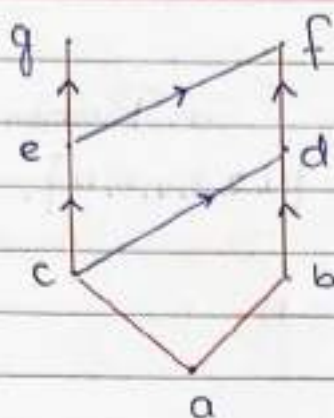
$A = \{1, 2, 3, 4\}$

$2^n \cdot n = 2^4$

$(P(A), \subseteq)$

$n=4$

$e = 2 \cdot n = 4 \cdot 2 = 32$

 (A, R) posetGreater Element (GE) x is called Greater Element of (A, R) all elements $\leq x \in R$ Check for g: g is not GE $abcde \leq g$ $a \leq g(?)$ $b \leq g$ (False)

Note:

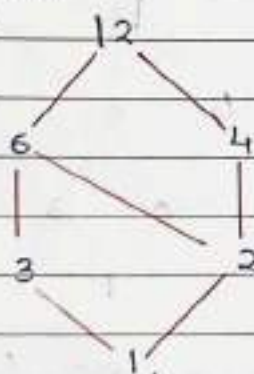
 $(a, b) \in R$

or

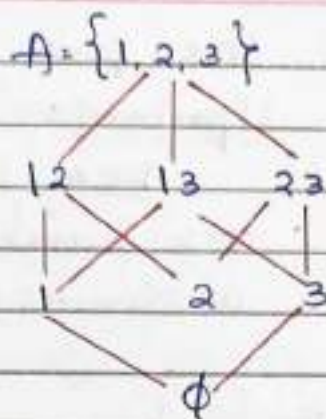


or

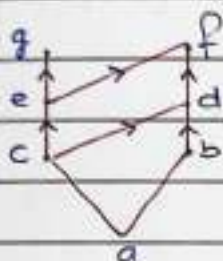
 $a R b$ $[a \leq b]$ $a R b$ $a \leq b$ a will be at lower level b will be at higher levelCheck for f: f is also not GE
 $abcde \leq f$

$$\begin{cases} a \leq f(?) & c \leq f(?) & e \leq f(?) \\ b \leq f(?) & d \leq f(?) & f \leq f(?) \end{cases}$$
 $g \leq f$ (False)
 $(D_{12}, |)$ Greater Elementall elements $\leq x \in A$ (12 is G.E)Check for 12: $1, 2, 3, 4, 5, 6, 12 \leq 12$ (True)Theorem: If greater element exist then it will be uniqueAssume x_1 and x_2 are G.Eall elements $\leq x_1 \wedge$ all elements $\leq x_2$ $x_2 \leq x_1 \wedge x_1 \leq x_2 \rightarrow x_1 = x_2$ $\therefore$ assumption is wrong
 $\therefore x_1$ and x_2 are unique

eg: $A = \{1, 2, 3\}$
 $(2^A, \subseteq)$
 or
 $(P(A), \subseteq)$



$G.E: \{1, 2, 3\}$
 $G.E: 2A$



(A, R) poset

Least Element (Minimum element)

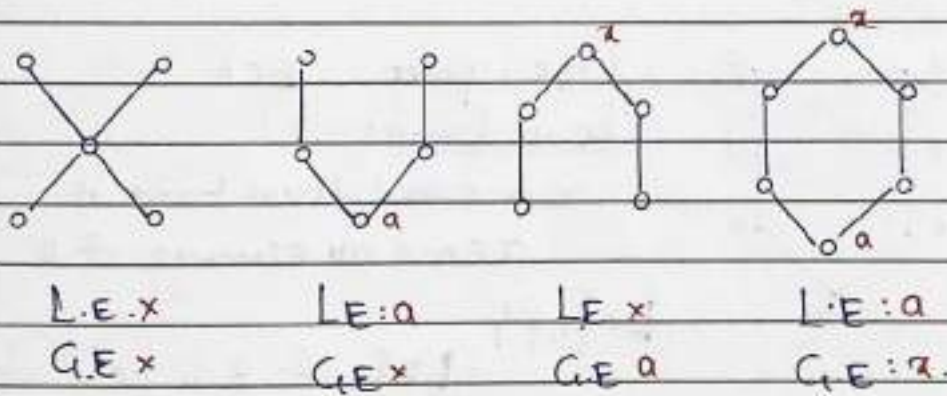
x is called least element if:

$x \in A \leq$ all elements of A .

Check for 'a':

as $a \leq b, c, d, e, f$ $\therefore a$ is least element.

* If least element exist it will be unique.



$(\{1, 2, 3, 4, 5\}, \leq)$

$(\mathbb{Q}^+, \leq)$

$(\mathbb{Z}, \leq)$

5 (G.E)

4

3

2

1 (L.E)

∞

4

3

2

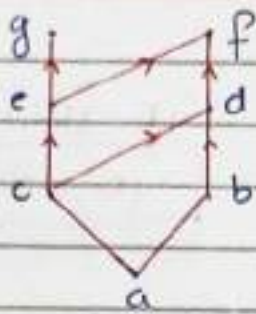
1

L.E: 1

G.E: NA

L.E: NA

G.E: NA

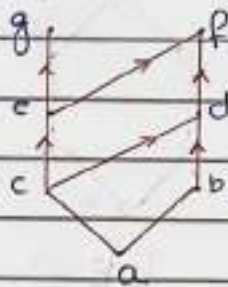


(A, R) poset

Upper bound:

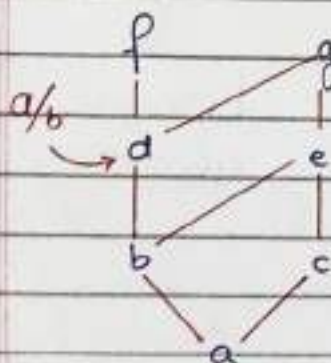
BCA

x is upper bound of B if:
all elements $\in B \leq x \in A$.

 $b \leq e$ $c \leq e$ (T) $b \leq e$ (F)eg: $B = \{b, c\}$ $\rightarrow$ UBs of $\{b, c\}$ in $\{d, f\}$ $b \leq d$ | $b \leq f$ $c \leq d$ | $c \leq f$ $c \leq d$ | $c \leq f$ $B = \{a, b\}$ UBs: $\{b, f, d\}$ $B = \{e, b\}$ UBs: f $B = \{c, d\}$ UBs: $\{a, f\}$ $B = \{a, g\}$ UBs: g 

(A, R) poset

BCA

Lower bound: x is called lower bound if $x \in A \leq$ all elements of B $B = \{e, f\}$ LBs of B in $\{e, c, a\}$ $e \leq e$ | $c \leq e$ | $a \leq e$ $B_1 = \{g, f\}$ LB: $\{e, c, a\}$ $B_2 = \{e, b\}$ LB: $\{a\}$ Greatest/Lower Bound: GLB $\rightarrow x \in A \leq$ all elements of B a/b will be uniqueall lower
bounds
of B

$B = \{f, g\}$ lower bound: $\{a, b, a\}$
 d is a/b of $\{f, g\}$.

$(D_{12}, |)$

$a/b(2,3) = 1$

$a/b(2,6) = 2$

$a/b(3,6) = 3$

$a/b(2,12) = 2$

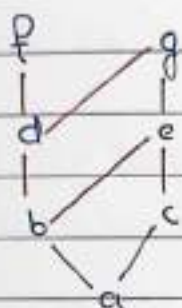
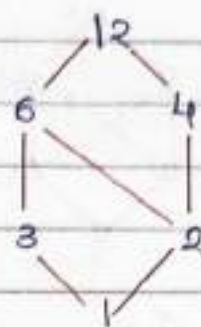
$\hookrightarrow a/b$

$2 \leq 2 \quad | \quad 1 \leq 2$

$2 \leq 12 \quad | \quad 1 \leq 12$

$l/b: a/b \{1, 2\}$

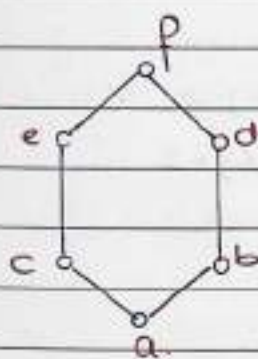
$\hookrightarrow a/b \rightarrow \{2\}$

Lower/Upper Bound : LUB $\hookrightarrow$ all elements $\leq x \in A$ $x \in A =$ all up $\hat{c}$ of B.

$B = \{a, c\}$

$\hookrightarrow$ U.B of $\{a, c\} = \{e, g\}$

$\hookrightarrow$ LUB: $\{c\}$

 (A, R) lattice

$a/b(a, b) = a$

$l/b(a, b) = b$

$b \leftarrow l/b$

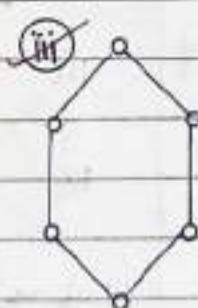
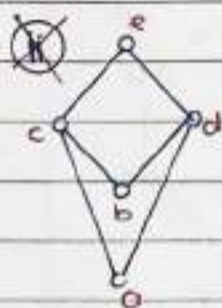
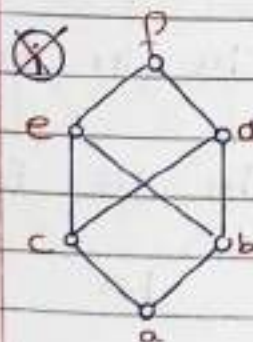
$a \leftarrow a/b$

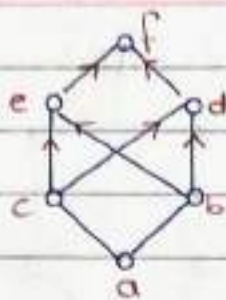
$l/b(b, c) = f$

$a/b(bc) = a$

$l/b(e, b) = f$

$a/b(e, b) = a$

 (A, R) poset $\hookrightarrow$ when a/b and l/b exist for all pairs $\rightarrow (A, R)$ lattice



(i) x

$$\text{lub}(c, d) = \text{NF}$$

$$\hookrightarrow \text{ub}\{c, d\} \rightarrow \{e, d, f\}$$

check e)	check d)	check f)
$\times e \leq d$	$d \leq edf$	$f \leq edf$
$e \leq e(T)$	$d \leq e(F)$	$f \leq e(F)$
$e \leq d(F)$		

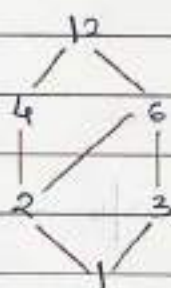


(ii) x

$$a/b(c, d) =$$

$$\text{lub}(c, d) \rightarrow \{a, b\}$$

check(a)	check(b)
$ab \leq a$	$ab \leq b$
$a \leq a(T)$	$ab \leq b(F)$
$b \leq a(F)$	



(D₁₂, 1)

$$\text{lub}(4, 6) = 12$$

$$a/b(4, 6) = 2$$

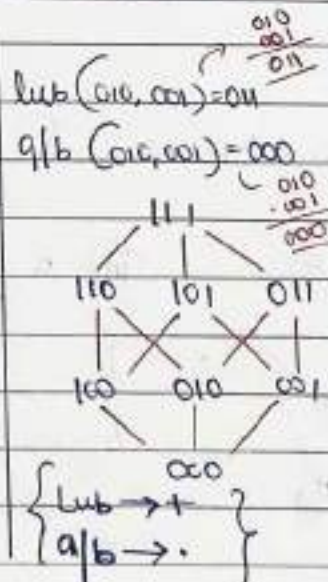
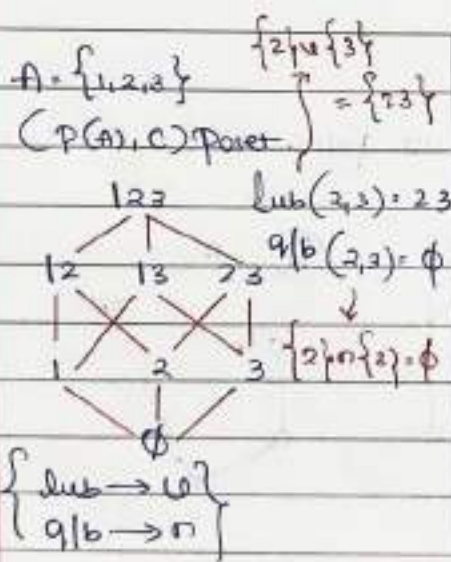
$$\text{lcm}(4, 6) = 12$$

$$\text{gcd}(4, 6) = 2$$

(D_n, 1)

$$a/b \rightarrow \text{gcd}$$

$$\text{lub} \rightarrow \text{lcm}$$



$\vee$	$\cup$	$+$
$\wedge$	$\cap$	$\cdot$
T	v	1
F	ϕ	0

* Every TOSET is lattice
(chain)

$[L, \vee, \wedge]$ lattice

($\vee$) lub

($\wedge$) g/b

1. $a \vee a = a \quad a \wedge a = a$

$\text{lub}(a, a) = a \quad \text{g/b}(a, a) = a$

2. $a \vee b = b \vee a \quad a \wedge b = b \wedge a$

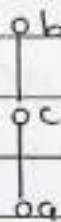
$\text{lub}(a, b) = \text{lub}(b, a) \quad \text{g/b}(a, b) = \text{g/b}(b, a)$

3. $a \vee (b \wedge c) = (a \vee b) \wedge c \quad a \wedge (b \vee c) = (a \wedge b) \vee c$

$\text{lub}(a, \text{lub}(b, c)) = \text{lub}(\text{lub}(a, b), c)$

4. $a \vee (a \wedge b) = a \quad a \wedge (a \vee b) = a$

(3) $a \vee (b \wedge c) = (a \vee b) \wedge c$



$a \vee (b \wedge c) = (a \vee b) \wedge c$

$\hookrightarrow \text{lub}(b, c)$

$\hookrightarrow a \vee b$

$\text{lub}(a, b)$

$\hookrightarrow b$

$\hookrightarrow \text{lub}(a, b)$

$\hookrightarrow b \wedge c$

$\text{lub}(b, c)$

$\hookrightarrow b$

$a \vee (a \wedge b) = a \quad a \wedge (a \vee b) = a$

Case 1:

Case 2:

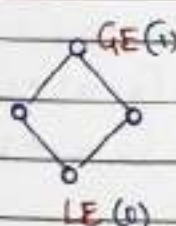


$a \vee (a \wedge b) = a$

$\hookrightarrow \text{g/b}(a, b)$

$\hookrightarrow (a \vee a) = a$

Bounded Lattice:



Lattice $[L, \vee, \wedge]$

$(G.E \& L.E)$ are existing

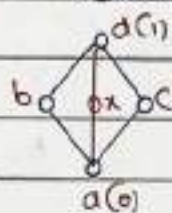
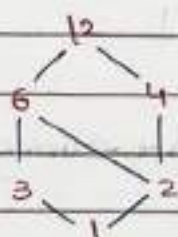
Bounded Lattice
 $[L, \vee, \wedge, 1, 0]$

$(Z^+, \leq) \rightarrow \text{Poset} \rightarrow \text{lattice}$ $\hookrightarrow G.E \rightarrow N.A$ $L.E \rightarrow 1$

* Every finite lattice can be bounded lattice.

 $(Z, \leq) \rightarrow \text{Poset} \rightarrow \text{lattice}$ $\hookrightarrow G.E \rightarrow N.A$ $L.E \rightarrow N.A$

$a+b=1$	$a \vee b = G.E$	$1 \rightarrow G.E$
$a \cdot b = 0$	$a \wedge b = L.E$	$0 \rightarrow L.E$
a, b are complement to each other		

Complement lattice: $a+b=1$ $a \cdot b = 0$ $\text{lub}(a, b) = G.E(1)$ $\text{glb}(a, b) = L.E(0)$ $\hookrightarrow$ Bounded lattice, atleast 1 element complement exist for all elements. $\text{lub}(b, c) = d$ $\text{glb}(b, c) = a$ $\text{lub}(\) = G.E$ $\text{glb}(\) = L.E$ $\text{lub}(b, c) = d(1) \quad b^1 = c$ $\text{glb}(b, c) = a(0)$ $b^1 = \{c\}$ $a^1 = d$ $\text{lub}(a, d) = d(1) \quad a+d=1$ $\text{glb}(a, d) = a(0) \quad a \cdot d = 0$ (D_{12,1}) Complement lattice? $3^1 = ?$ $3+ = 12$ $3 \cdot = 1$ $\therefore 3^1 = 4$ $\text{Lcm}(3, 4) = 12$ $\text{HCF}(3, 4) = 1$ $6^1 = N.A$ $6+ = 12$ $\text{lcm}(6, 12) = 12 \checkmark$ $\text{HCF}(6, 12) = 6 \times$ $\therefore$ It is not complement lattice.

$$\begin{cases} a \cup b = \{1, 2, 3\} \\ a \cap b = \emptyset \end{cases}$$

$$\{1\} \cup \{2,3\} = \{1,2,3\}$$

$$\{1\} \cap \{2,3\} = \emptyset$$

$$f_1^{-1} = \{2, 3\}$$

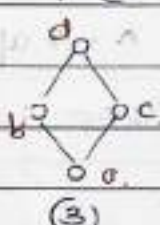
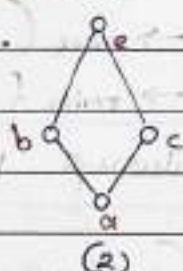
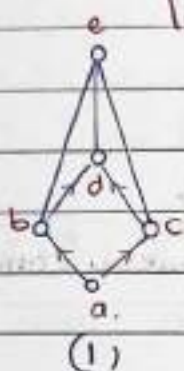
$$\text{lub}(b, c) = d$$

$$a|b(b,c) = a$$

$$\text{lub}(b, c) = e$$

$$a|b \rightarrow (b,c) = a.$$

B is sublattice
of A

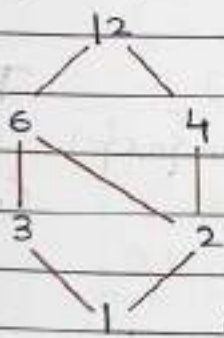


- 1) $B \subseteq A$
- 2) alt of lub of each pair of B must be same as in A

Distributive Notice:

$\begin{cases} a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c) \checkmark & \wedge \rightarrow a|b \\ a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c) \checkmark & \vee \rightarrow \text{sub} \end{cases}$

(D_{12,1}) Distributive Lattice / Not Complement Lattice



$$6 \vee (4 \wedge 2) = (6 \vee 4) \wedge (6 \vee 2)$$

$$\downarrow$$

$$\gcd(4, 2)$$

$G \vee 2$
 $\hookrightarrow \text{sub}(G, 2)$

$$G' = NA$$

$$\text{Luro}(G, 2) = 6$$

$$\ln(6.4) \wedge \ln(6.2)$$

$$12 \wedge 6$$

$$\hookrightarrow \gcd(6, 12) = 6$$

$$(D_{n,1}) \rightarrow \text{Lattice}$$

↳ distributive law

$$\left\{ \begin{array}{l} a/b \rightarrow \text{gcd} \\ \text{lub} \rightarrow \text{lcm} \end{array} \right\}$$

$$(P(A), \subseteq)$$

→ Distribuzione Latice

$$\lim \rightarrow 0 \rightarrow \gamma$$

$$a(b \rightarrow n \rightarrow \wedge)$$

eg: $(D_{60}, |)$
 $(\{1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30, 60\}, |)$

$$10 \vee (12 \wedge 15) = (10 \vee 12) \wedge (10 \vee 15)$$

$$\hookrightarrow a/b(12, 15)$$

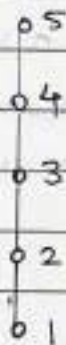
$$\hookrightarrow 10 \vee 3$$

$$\hookrightarrow \text{lcm}(10, 3) = 30$$

$$(\{1, 2, 3, 4, 5\}, \leq) \quad \vee \rightarrow \text{lub} \rightarrow \max(\dots)$$

$$\wedge \rightarrow a/b \rightarrow \min\{\dots\}$$

Every TOSet is always Distributive Lattice.



$$\text{lub}(1, 2) = \max(1, 2) = 2$$

$$a/b(1, 2) = \min(1, 2) = 1$$

$$1 \vee (2 \wedge 3) = (1 \vee 2) \wedge (1 \vee 3)$$

$$\max(1, \min(2, 3)) = (\max\{1, 2\}, \max\{1, 3\}) \min$$

$$\hookrightarrow \max(1, 2) = 2 = \min(2, 3) = 2$$



$$\vee \rightarrow \text{lub} \rightarrow \max$$

$$\wedge \rightarrow a/b \rightarrow \min$$

$$a \vee (b \wedge c)$$

$$(a \vee b) \wedge (a \vee c)$$

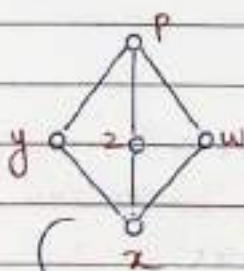
$$\max(a, \min\{b, c\})$$

$$\min(\max\{a, b\}, \max\{a, c\})$$

$G/E/L/E$

Banded

Complement



Not distributive
Lattice.

$$y \wedge (z \vee w) = (y \wedge z) \vee (y \wedge w)$$

$$\text{lub}(z, w)$$

$$\hookrightarrow y \wedge p$$

$$a/b(y, p)$$

$$\downarrow$$

$$y$$

$$a/b(y, z) \vee a/b(y, w)$$

$$\downarrow \quad \downarrow$$

$$x \vee x$$

$$\downarrow$$

$$x$$

$$y \neq x$$

Complement ✓
Distributive ✗

Boolean Algebra = Lattice $\wedge$ Complement $\wedge$ Distributives.

$$a+a=a$$

$$a+b=1$$

$$a+(b \cdot c) = (a+b)(a+c)$$

$$a \cdot a = a$$

$$a \cdot b = 0$$

$$a \cdot (b+c) = (a \cdot b) + (a \cdot c)$$

$$a+b = b+a$$

$$a \cdot b = b \cdot a$$

$$a+(b+c) = (a+b)+c$$

$$a \cdot (b \cdot c) = (a \cdot b) \cdot c$$

$$a+0=a$$

$$a \cdot 1=a$$

→ $(Q, *)$ is Group

1. Closed: $a \in Q, b \in Q, a * b \in Q$

2. Associative: $a * (b * c) = (a * b) * c$

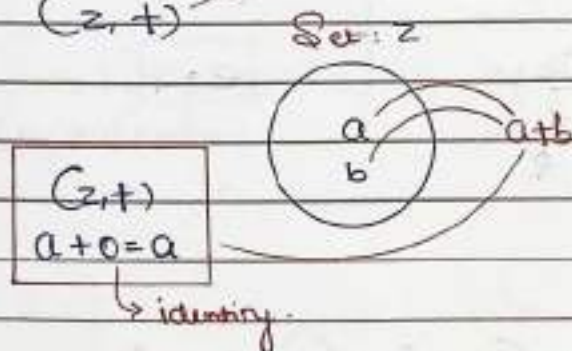
3. Identity: $a * e = a$.

→ identity.

$*$ is any random
operation.

Closed:

$(Q, *)$ → $a \in Q, b \in Q, a * b \in Q$
 $(Q, +)$ → $a \in Q, b \in Q, a + b \in Q$
 $(Z, +)$ → $a \in Z, b \in Z, a + b \in Z$



Associative:

$$a+(b+c) = (a+b)+c$$

$(D_{12}, q|b)$ $(D_{n,1})$

(Set Operation)

$a \in D_{12}, b \in D_{12}$

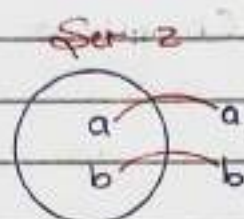
$gcd(a,b) \in D_{12}$

$(\{1, 2, 3, 4, 6, 12\} q|b) \rightarrow$ Associative

$(\{1, 2, 3, 4, 6, 12\} gcd)$

$$\rightarrow (G, *)$$

$$(2, 1)$$



$$a/b \notin 2$$

$$\rightarrow (G, *)$$

$$(A, \times) \quad A = \{2^n \mid n \in \mathbb{Z}\}$$

Associative:

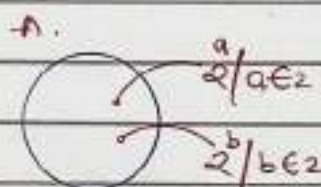
$$2^a \cdot (2^b \cdot 2^c) = (2^a \cdot 2^b) \cdot 2^c$$

$$2^a (2^{b+c})$$

$$2^{a+b+c} = 2^{a+b+c}$$

$$2^a \times 2^b$$

$$\{2^{a+b} \mid a \in \mathbb{Z}, b \in \mathbb{Z}, a+b \in \mathbb{Z}\}$$



$$2^a \times 2^0 = 2^a$$

Identity

$$\rightarrow (R, +) \quad a+b = \frac{ab}{2}$$

$$a + (b+c) = (a+b) + c$$

$$a + \left(\frac{bc}{2}\right) = \left(\frac{ab}{2}\right) + c$$

$$\frac{a \left(\frac{bc}{2}\right)}{2} = \frac{\left(\frac{ab}{2}\right) c}{2}$$

Associative:

$$a * e = a$$

$$\frac{ae}{2} = a$$

$$e = 2$$

unique

$$\rightarrow (A, \times)$$

$$A = \{1, \omega, \omega^2\}$$

$$\omega^3 = 1$$

$$1 \times (\omega \times \omega^2) = (1 \times \omega) \times \omega^2$$

$$1 \times \omega^3 = \omega \times \omega^2$$

$$1 \times 1 = \omega^3$$

$$1 = 1$$

$$\omega \times 1 = \omega$$

$$\omega^2 \times 1 = \omega^2$$

→ $(A, \times)$ $i^2 = -1$
 $A = \{1, i, -i, -1\}$

$$i \times i = -1$$

Closed.

Associative.

$$1 \times (i \times -i) = (1 \times i) \times -i$$

$$1 \times (-1) = i \times -i$$

$$1 \times (-(-1)) = -i^2$$

$$1 = -(-1)$$

$$1 = 1$$

→ (Set, Operation) $A = []_{2 \times 2}$ $B = []_{2 \times 2}$ $A \times B \neq B \times A$

$(M_{2 \times 2}, \times)$ $A \times B = []_{2 \times 2} \in M_{2 \times 2}$

$(\{ []_{2 \times 2}, []_{2 \times 2}, \dots \}, \times)$ $A \cdot (B \cdot C) = (A \cdot B) \cdot C$

$$A \times I = A$$

→ (Set, $*$)
 $(\{0, 1, 2, 3, 4, 5\}, \oplus_6)$

Operation: addition mod 6
 $\oplus_6$

$$a \oplus_6 b = (a+b) \bmod 6$$

$$2 \oplus_6 3 = (2+3) \bmod 6 = 5 \bmod 6 = 5$$

$$3 \oplus_6 (4 \oplus_6 5) = (3 \oplus_6 4) \oplus_6 5$$

$$3 \oplus_6 3 = 1 \oplus_6 5$$

$$0 = 0 \text{ (Associative)}$$

$$3 \oplus_6 0 = 3$$

→ $A = \{1, 2, 3\}$

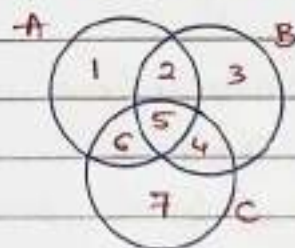
$(P(A), \Delta)$ or $(2^A, \Delta)$

$\oplus$

$(P(A), \oplus)$ $(2^A, \oplus)$

$$A \oplus B = (A \cup B) - (A \cap B)$$

$$A \cap (B \Delta C) = (A \Delta B) \Delta C$$



$$A \oplus B = (A \cup B) - (A \cap B)$$

$$A \oplus \phi = A$$

$$(A \cup \phi) - (A \cap \phi)$$

$$= A$$

$$A = \{1, 2, 3\}$$

$1 \rightarrow 1$	$1 \rightarrow 1$	$1 \rightarrow 2$	$1 \rightarrow 2$	$1 \rightarrow 3$	$1 \rightarrow 3$
$2 \rightarrow 2$	$2 \rightarrow 3$	$2 \rightarrow 1$	$2 \rightarrow 3$	$2 \rightarrow 1$	$2 \rightarrow 2$
$3 \rightarrow 3$	$3 \rightarrow 2$	$3 \rightarrow 3$	$3 \rightarrow 1$	$3 \rightarrow 2$	$3 \rightarrow 1$
f_1	f_2	f_3	f_4	f_5	f_6

(Set, Operation)

 $(\{f_1, f_2, f_3, f_4, f_5, f_6\}, \circ)$

$$f_5 \circ f_1 = f_5$$

(Set, Operation)

 $(\{f_1, f_2, f_3, f_4, f_5, f_6\}, \circ)$ Closed

$$f_2 \in A$$

$$f_3 \in A$$

$$f_2 \circ f_3 = f_5$$

$$1 \rightarrow 2 \rightarrow 3$$

$$2 \rightarrow 1 \rightarrow 1$$

$$3 \rightarrow 3 \rightarrow 2$$

$$f_3 \circ f_2$$

$$f_2 \circ (f_3 \circ f_5) = (f_2 \circ f_3) \circ f_5$$

$$1 \rightarrow 3 \rightarrow 3$$

$$2 \rightarrow 1 \rightarrow 2$$

$$3 \rightarrow 2 \rightarrow 1$$

$$f_2 \circ f_5$$

Group $(G, +)$ 1. > Closed $\rightarrow a \in G, b \in G, a+b \in G$ 2. > Associative $\rightarrow a+(b+c) = (a+b)+c$ 3. > Identity $\rightarrow a+e = a$ 4. > Inverse $\rightarrow a+\bar{a} = e$ $(\mathbb{Z}, +)$ ✓

1. Closed

2. Associative

3. Identity $a+0 = a$ ($e=0$) unique4. Inverse $a+\bar{a} = e$

$$2+(-2) = 0$$

 $(\mathbb{Z}, \times)$ ✗ (not group)

1. Closed

2. Associative

3. Identity $a \times 1 = a$ 4. Inverse $a \times \frac{1}{a} = 1$

$(\mathbb{Q}, +)$ $(\mathbb{Q} \neq 0, \times) \rightarrow \text{Group}$

1. Closed
2. Associative
3. $a \times 1 = a$ (Identity)
4. $a \times \frac{1}{a} = 1$ ($a \neq 0$)

Group

Infinite Groups

 $(\mathbb{Z}, +)$ $(\mathbb{Q} \neq 0, \times)$

Finite Groups

 $(\{1, \omega, \omega^2\}, \times)$ $(\{1, i, -i, -1\}, \times)$ Cayley Table

$*$	e_1	e_2	...
e_1			
e_2			
$\vdots$			

 $\omega^3 = 1$ $(\{1, \omega, \omega^2\} | 2)$ $\omega^2 \neq 1 = \omega^2$

$\times$	1	ω	ω^2
1	1	ω	ω^2
ω	ω	ω^2	1
ω^2	ω^2	1	ω

$$\begin{cases} 1' = 1 \\ \omega' = \omega^2 \\ \omega^2' = \omega \end{cases}$$

1. Closed
2. Associative
3. Identity
4. Inverse

$1 \rightarrow 1$	$1 \rightarrow 1$	$1 \rightarrow 2$	$1 \rightarrow 2$	$1 \rightarrow 3$	$1 \rightarrow 3$
$2 \rightarrow 2$	$2 \rightarrow 3$	$2 \rightarrow 1$	$2 \rightarrow 3$	$2 \rightarrow 1$	$2 \rightarrow 2$
$3 \rightarrow 3$	$3 \rightarrow 2$	$3 \rightarrow 3$	$3 \rightarrow 1$	$3 \rightarrow 2$	$3 \rightarrow 1$
f_1	f_2	f_3	f_4	f_5	f_6

 $f_4 \text{ of } f_5 = f_1$

$\circ$	f_1	f_2	...
f_1			
f_2			
$\vdots$			

	2	3	1
3	1	2	
1	2	3	

$\oplus_6$	0	1	2	3	4	5
0	0	1	2	3	4	5
1	1	2	3	4	5	0
2	2	3	4	5	0	1
3	3	4	5	0	1	2
4	4	5	0	1	2	3
5	5	0	1	2	3	4

 $(\{0, 1, 2, 3, 4, 5\} \oplus_6)$

1. Closed ✓
2. Associative ✓
3. Identity ✓
4. Inverse ✓

$$\begin{aligned} 0' &= 0 & 2' &= 4 & 4' &= 2 \\ 1' &= 5 & 3' &= 4 & 5' &= 1 \end{aligned}$$

Subgroup: H is called Subgroup of G if

(i) $H \subseteq G$

(ii) H should also satisfy:

a. Closed

c. Identity

b. Associative

d. Inverse

eg:

$$\{ \{0, 1, 2, 3, 4, 5\} \oplus 6 \} \quad H = \{1, 3, 5\}$$

$$\rightarrow H \subseteq G \checkmark$$

$$\rightarrow \text{identity } \times$$

$\rightarrow$ not subgroup because identity element is absent.

$$\text{eg: } \{ \{0, 1, 2, 3, 4, 5\} \oplus 6 \} \quad H = \{0, 3, 5\}$$

$$\rightarrow H \subseteq G \checkmark$$

$$\rightarrow \text{Closed } \times$$

- not group

- not closed

- not subgroup.

$$\text{eg: } \{ \{0, 1, 2, 3, 4, 5\} \oplus 6 \} \quad H = \{0, 2, 4\}$$

$$\rightarrow H \subseteq G \checkmark$$

$$\rightarrow H \text{ is also Group.}$$

	0	2	4
0	0	2	4
2	2	0	4
4	4	0	2

Theorem: if H is Subgroup of G , then $\frac{|G|}{|H|}$. And vice versa not true.

$$\rightarrow H = \{0, 2, 4\} \quad |G| = 6$$

$$H \text{ is subgroup of } G \rightarrow \frac{|G|}{|H|} = \frac{6}{3}$$

Every Group Contains 2 trivial Subgroup:

1) $e \in G$

2) $G \subseteq G$

closed

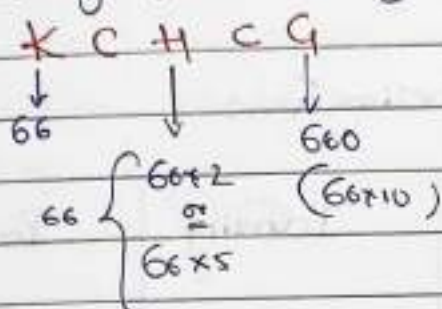
associative

identity $e + e = e$

inverse $e + e = e$

- Q1. If $|G| = 84$, then what will be maximum size of subgroup? $\rightarrow 84$
 - what will be the maximum size of proper subgroup? $\rightarrow 42$

G be the group with Subgroup $-H \& K$. $|G| = 660$ $|K| = 66$



* Subgroup of Cyclic group is also cyclic group



$$2^1 = 2$$

$$2^2 = 2 \oplus 2 = 4$$

$$2^3 = 2 \oplus 2 \oplus 2 = 0$$

$$4 \oplus 2 = 0 \quad \{0, 2, 4\} - \text{Subgroup}$$

$$2^4 = 0 \oplus 2 = 2, \quad 2 \text{ has generated } \{0, 2, 4\}$$

$$\langle 2 \rangle = \{0, 2, 4\}$$

$$3^1 = 3$$

$$3^2 = 3 \oplus 3 = 0$$

$$3^3 = 0 \oplus 3 = 3$$

$$\langle 3 \rangle = \{0, 3\}$$

$$1^1 = 1$$

$$1^2 = 1 \oplus 1 = 2$$

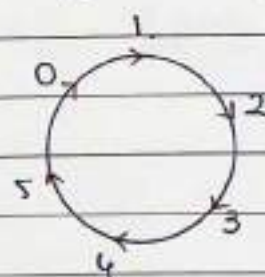
$$1^3 = 2 \oplus 1 = 3$$

$$1^4 = 3 \oplus 1 = 4$$

$$1^5 = 4 \oplus 1 = 5$$

$$1^6 = 5 \oplus 1 = 0$$

$$\langle 1 \rangle = \{0, 1, 2, 3, 4, 5\}$$



$$5^1 = 5$$

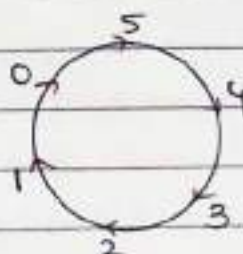
$$5^2 = 5 \oplus 5 = 4 \quad \langle 5 \rangle = \{5, 4, 3, 2, 1, 0\}$$

$$5^3 = 4 \oplus 5 = 3$$

$$5^4 = 3 \oplus 5 = 2$$

$$5^5 = 2 \oplus 5 = 1$$

$$5^6 = 1 \oplus 5 = 0$$



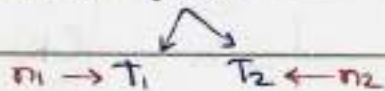
- * In previous example '1' has generated all value/elements in the group.
- * 1 is called "Generator".
- * Group $\rightarrow$ Generator $\rightarrow$ Cyclic Group.

Group + Commutative = Abelian Group.

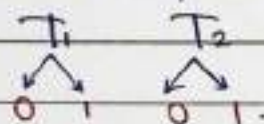
Closed	Associative	Identity	Inverse
Algebraic Structure.			
Semi Group.			
Monoid			
Group.			

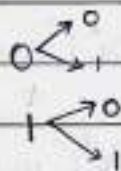
COMBINATORICSProduct Rule:

Given a Task.

 $k=0$ for $i=1$ to n_1 for $j=1$ to n_2 $k=k+1$ $k=n_1 \times n_2$ $i=1, j=1$ to n_2 $i=2, j=1$ to n_2 $\vdots$ $j=1$ to $n_2, i=n_1$ n_1 ways to do task T_1 n_2 ways to do task T_2 Each of n_1 ways we can perform n_2 both the tasks are happening simultaneously.

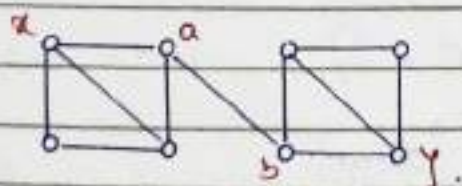
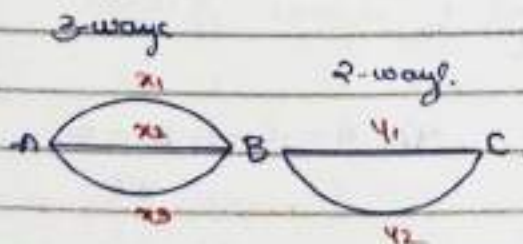
Q1. How many code we can generate if we have 2 bits?



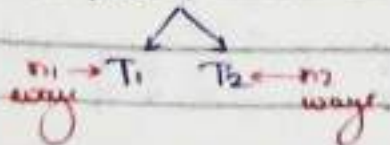
$$\begin{cases} 00 \\ 01 \\ 10 \\ 11 \end{cases}$$


Q2. How many ways we can generate a number plate if number plate consist of four characters formed by 2 digit.

$$26 \times 26 \times 26 \times 26 \times 10 \times 10 = 26^4 \times 10^2$$

 $x \rightarrow a \rightarrow b \rightarrow y$ 3 ways $\times 1 \times 3 = 9$ paths from $x \rightarrow y$. $A \rightarrow C$ $3 \times 2 = 6$ ways.Sum Rule:

Task T



Both tasks are happening simultaneously.

 $k=0$ for $i=1$ to n_1 : $k=k+1$ for $i=1$ to n_2 : $k=k+1$ $k=n_1+n_2$

Q3. How many number plates we can generate if 4 characters followed by 2 digit or 3 digit?

---|--- (OR) ---|---

$$26^4 \cdot 10^2 + 26^4 \cdot 10^3 = 26^4 (10^2 + 10^3)$$

Q4. How many ways we can generate if one or two or 3 or 4 characters followed by 1 or 2 or 3 or 4 digit?

1C	1	
(or)	(or)	$26(10+10^2+10^3+10^4)$
2C	2	+
(or)	(or)	$26^2(10+10^2+10^3+10^4)$
3C	3	+
(or)	(or)	$26^3(10+10^2+10^3+10^4)$
4C	4	+

$$26^4(10+10^2+10^3+10^4)$$

$$\Rightarrow (26 + 26^2 + 26^3 + 26^4) \times (10 + 10^2 + 10^3 + 10^4)$$

Q5. 1) A—B—C : 3 ways × 2 ways = 6 ways



2) A—C



$$6 \text{ ways} + 2 \text{ ways} = 8 \text{ ways}$$

3) A—B—C—B—A

$$3 \times 2 \times 2 \times 3 = 6 \times 6$$

$$= 36 \text{ ways}$$

4) A—C—B—A

$$(3 \times 2 + 2) \times (3 \times 2 + 2)$$

$$= 8 \times 8 = 64 \text{ ways}$$

Q6. How many 4 digit integer contains exactly one 6.

$$\begin{array}{ccccccc} \overset{6}{\downarrow} & \overset{10}{\downarrow} & \overset{10}{\downarrow} & \overset{10}{\downarrow} & \overset{10}{\downarrow} & \overset{6}{\downarrow} & \\ 9 \cdot 9 \cdot 9 & + & 8 \cdot 9 \cdot 9 & + & 8 \cdot 9 & + & 8 \cdot 9 \cdot 9 = 9 \cdot 9 \cdot 9 + 3 \cdot (8 \cdot 9 \cdot 9) \end{array}$$

n objects $\rightarrow r$ places.

$$n \times (n-1) \times \dots \times (n-r+1) \left\{ \begin{array}{l} (n-r)! \\ (n-r)! \end{array} \right.$$

$${}^n P_r \Rightarrow \frac{n!}{(n-r)!} \left\{ \begin{array}{l} \text{Permutation.} \end{array} \right.$$

eg: $\{a, b, c\}$ $\rightarrow$ This is an example of combination

$${}^n C_r \Rightarrow \frac{n!}{r!(n-r)!}$$

$\rightarrow \{a, b\}$
 $\rightarrow \{b, c\}$
 $\rightarrow \{a, c\}$

Combination: Selecting 2 elements is same as shifting 2 lines.

$\{a, b, c\}$

Place lines $\rightarrow {}^3 C_2$

$$110 - \{a, b\} \rightarrow \{1, 1, -\}$$

$$011 - \{b, c\} \rightarrow \{-, 1, 1\}$$

$$101 - \{a, c\} \rightarrow \{1, -, 1\}$$

eg. A man which contains 3 containers, A, P, O how many ways can choose 4 fruits?

4 A	3A 1O	2A 2O	2A 1O 1P
4 O	3A 1P	2A 2P	2O 1A 1P
4 P	3O 1A	2P 2O	2P 1A 1O
	3O 1P		
	3P 1O		
	3P 1A		

$$3 + 6 + 3 + 3 \Rightarrow 15$$

$$3A 1O - \{A \bar{A} \bar{A} | \bar{O} | \}$$

$$2A 2O - \{A \bar{A} | \bar{O} \bar{O} | \} \quad A | O | P$$

$\vdots$

$\vdots$

4 fruits + 2 lines = 6 places, lines = 2

$$\therefore {}^6 C_2 = 15$$

Place
C lines

When done manually.

Q1. How many ways we can distribute 4 chocolate among 2 girls?

1	2
4	0
3	1
2	2
1	3
0	4

$\rightarrow \{ \overline{000} | 0 \}$ Place C line
 $\Rightarrow {}^5C_1 \Rightarrow \underline{\underline{5}}$

5 ways

Q2. How many ways we can distribute 10 chocolate among 3 students?

1	2	3
00	00	00
00	00	00
0000	00	0000

${}^{12}C_2 = \underline{\underline{66}}$ ways

How many non negative solutions are possible:

$$x_1 + x_2 + x_3 = 10 \quad x_i \geq 0$$

$$x_1 = 4 \quad x_2 = 2 \quad x_3 = 4$$

$$x_1 = 3 \quad x_2 = 3 \quad x_3 = 4 \quad \text{Ans} = \underline{\underline{{}^{12}C_2}}$$

Q3. How many ways we can distribute 10 chocolate among 10 students, each child gets at least 1 chocolate?

1	2	3
0	0	0
00	00	00
0	0	0

1 chocolate already given.

$$10 - 3 = 7 \Rightarrow \underline{\underline{{}^{7+2}C_2}}$$

— how many non-negative solutions?

$$x_1 + x_2 + x_3 = 10 \quad x_i \geq 1$$

$$y_i = x_i - 1 \quad x_i - 1 \geq 0$$

$$y_i + 1 = x_i \quad y_i$$

$$y_1 + 1 + y_2 + 1 + y_3 + 1 = 10 \quad y_i \geq 0$$

$$y_1 + y_2 + y_3 = 10 - 3 \quad y_i \geq 0$$

$$= \underline{\underline{7}}$$

— If 20 chocolate are there

$$x_1 + x_2 + x_3 = 20$$

$$x_1 \geq 2 \quad x_2 \geq 3 \quad x_3 \geq 4$$

$$x_1 - 2 \geq 0 \quad x_2 - 3 \geq 0 \quad x_3 - 4 \geq 0$$

$$y_1$$

$$y_2$$

$$y_3$$

$$y_1 + 2 = x_1, \quad y_2 + 3 = x_2, \quad y_3 + 4 = x_3$$

$$y_1 + 2 + y_2 + 3 + y_3 + 4 = 20$$

$$y_1 + y_2 + y_3 = 20 - 9$$

$$= \underline{\underline{11}}$$

$$\therefore \text{Ans} = \underline{\underline{{}^{11+2}C_2}} \Rightarrow \underline{\underline{{}^{13}C_2}}$$

- If 40 chocolate are there and 10 students, 10 chocolate to be distributed among first 3 students.

$$x_1 + x_2 + x_3 + \dots + x_{10} = 40 \quad x_i \geq 0$$

$$x_1 + x_2 + x_3 = 10$$

$$\boxed{x_1 + x_2 + x_3} + \boxed{x_4 + x_5 + \dots + x_{10}} = 40$$

$$\hookrightarrow 10+2C_2$$

$$\hookrightarrow 30+6C_6$$

$$\therefore \text{Ans} = 12C_2 \times 36C_2$$

- Q4. How many ways we can distribute m identical balls among n boxes such that each box contains atleast k balls ($m \geq kn$)?

Ans

$$\boxed{k} \mid \boxed{k} \mid \boxed{k}$$

$$k+k+k \dots = n \cdot k$$

after distribution

$$\text{Remaining balls} = m - nk$$

$$\text{line} = n-1$$

$$\text{Ans} = m - nk + n - 1 C_{n-1}$$

- Q5. How many ways we can distribute 10 Roses, 15 daffodils among 2 girls?

Ans

10 Roses

$$\begin{array}{c|c} \text{♀} & \text{♀} \\ \hline \circ \circ \circ \circ \circ & \circ \circ \circ \circ \circ \end{array}$$

$$11C_1$$

15 Daffodils

$$\begin{array}{c|c} \text{♀} & \text{♀} \\ \hline \circ \circ \circ \circ \circ & \circ \circ \circ \circ \circ \\ \circ \circ \circ \circ \circ & \circ \circ \circ \circ \circ \end{array}$$

$$16C_1$$

$$\text{Ans} = 11C_1 \times 16C_1$$

$$\rightarrow x_1 + x_2 + x_3 \leq 10, \quad x_i \geq 0$$

$$x_1 + x_2 + x_3 = 0$$

$$x_1 + x_2 + x_3 = 1$$

$$\vdots$$

$$10$$

$$x_1 + x_2 + x_3 + 0 = 10$$

$$x_1 + x_2 + x_3 + 0 = 9$$

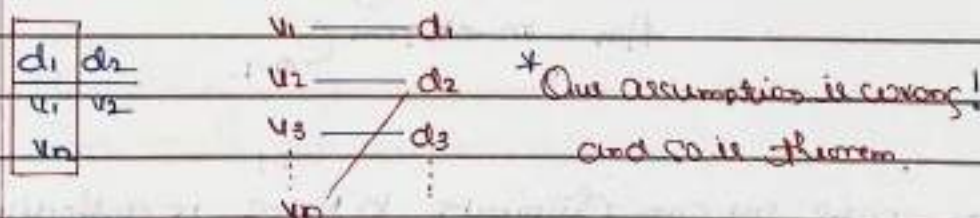
$$\vdots$$

$$\Rightarrow 10+3C_3 \Rightarrow 13C_3$$

GP	atleast 2	H_1	H_2	H_3	H_4		atleast 4	atleast 3
	✓	$\begin{smallmatrix} \circ & \circ \\ \circ & \circ \end{smallmatrix}$	$\circ$			$(1,1)$	✓	✓
	✓		$\begin{smallmatrix} \circ & \circ \\ \circ & \circ \end{smallmatrix}$	$\circ$		$(1,4)$	✓	✓
	✓	$\begin{smallmatrix} \circ & \circ \\ \circ & \circ \end{smallmatrix}$	$\circ$			$(3,2)$	✗	✓
Some balls →	✓			$\begin{smallmatrix} \circ & \circ \\ \circ & \circ \end{smallmatrix}$	$\circ$	$(3,2)$	✗	✓
	✓	$\begin{smallmatrix} \circ & \circ \\ \circ & \circ \end{smallmatrix}$	$\circ$	$\circ$		$(3,1,1)$	✗	✓
	✓		$\begin{smallmatrix} \circ & \circ \\ \circ & \circ \end{smallmatrix}$	$\circ$	$\circ$	$(3,1,1)$	✗	✓
	✓		$\circ$	$\circ$	$\begin{smallmatrix} \circ & \circ \\ \circ & \circ \end{smallmatrix}$	$(1,1,3)$	✗	✓
	✓	$\begin{smallmatrix} \circ \\ \circ \end{smallmatrix}$	$\circ$	$\circ$	$\circ$	$(2,1,1,1)$	✗	✓
							✗	✗

Theorem: In simple graph atleast 2 vertices will have same degree. ($n \geq 2$)

Assumptions: Let's take all vertices will have different degree.
 Degree Possibilities: $\{1, 2, 3, 4, \dots, n-1\}$ Total different degrees = $n-1$.
 Total vertices = n



Note: If we have $n+1$ pigeons, n holes, then some of the hole contains atleast 2 pigeons.

Q1. 1500 flights are taking off in a single day. Prove that atleast 2 flights are taking off in a single min.

Ans. 1 Day = 24 hrs
 $24 \times 60 = 1440 \text{ minutes}$

m_1	m_2	m_3	$\dots$	$m_n(1440)$
$\circ$	$\circ$	$\circ$	$\circ$	
$\circ$				

Q2. What will be min no of students required such that atleast 2 students will have same month's birthday.

m_1	m_2	$\dots$	m_{12}
$\circ$			
$\circ$			

Q3. What will minimum no. of students requires such that atleast 6 will have same grade in school, where it has five different grade system.

Ans

atleast 6	A	B	C	D	E
	o	o	o	o	o
	o	o	o	o	o
	o	o	o	o	o
	o	o	o	o	o
	o	o	o	o	o
	o	o	o	o	o

$5 \times 5 + 1$
 $\Rightarrow \underline{26}$

Q4. What will be the minimum no. of cards will be drawn such that atleast 3 belong to same suit?

atleast 3	H	D	S	C
	o	o	o	o
	o	o	o	o
	o	o	o	o

$2 \times 4 + 1$
 $\Rightarrow \underline{9}$

Q5. In a graph of 9 vertices, Prove that some of the vertices will contain atleast 4 degree if sum of degree of all vertices is atleast 27.

Ans

atleast 4	v_1	v_2	...	v_9
	o	o		o
	o	o	...	o
	o	o	...	o
	o			

$4 \times 3 = 27$

$\sum d(v_i) = \text{even}$
 $\sum d(v_i) = \text{atleast } 27$

Q6. What will be min no. of students require atleast 9 will be born on same month?

Ans

$12 \times 8 + 1 \Rightarrow \underline{97}$

Q7. A bag contains 6 Red, 8 blue, 10 Green, 15 white, 20 yellow min no of balls such that atleast 6 balls of same color.

Ans

atleast 6	6R	8B	10G	15W	20Y
	o				
	o				
	o				
	o				
	o				

$5 \times 5 + 1 \Rightarrow \underline{26}$

Ans

$$6 \times 5 + 8 + 1 \Rightarrow \underline{\underline{39}}$$

Collected by boys 2 & 5
- this is y - (205).

$$\rightarrow |A_1 \cup A_2 \cup \dots \cup A_n| = \sum a_i - \sum a_i a_j + \sum a_i a_j a_k - \dots + (-1)^{n+1} |A_1 \cap A_2 \cap \dots \cap A_n|$$

Q12 No. of elements which are not divisible by 2, 3 or 5 in a set $\{1, \dots, 120\}$?

Ans 33

Q13 How many elements are $\div$ by 4, 6, or 8 in a set $\{1, \dots, 100\}$?

$$\begin{aligned} \text{Ans } n(4 \text{ or } 6 \text{ or } 8) &= n(4) + n(6) + n(8) - n(4 \cap 6) - n(4 \cap 8) - n(6 \cap 8) + n(4 \cap 6 \cap 8) \\ &\Rightarrow \left\lfloor \frac{100}{4} \right\rfloor + \left\lfloor \frac{100}{6} \right\rfloor + \left\lfloor \frac{100}{8} \right\rfloor - \left\lfloor \frac{100}{\text{LCM}(4,6)} \right\rfloor - \left\lfloor \frac{100}{\text{LCM}(4,8)} \right\rfloor - \left\lfloor \frac{100}{\text{LCM}(6,8)} \right\rfloor + \left\lfloor \frac{100}{\text{LCM}(4,6,8)} \right\rfloor \Rightarrow \underline{\underline{33}} \end{aligned}$$

Q14 A, B, C, P Intersection of any two of these 4 sets contains 12 elements

$$n(A) = 42$$

Intersection of any three of these contains 8 elements.

$$n(B) = 36$$

$$A \cup B \cup C \cup D = A + B + C + D - 4C_2 + 4C_3 - 4$$

$$n(C) = 28$$

$$n(D) = 24$$

$$\Rightarrow 130 - 6 \times 12 + 4 \times 8 - 4 \Rightarrow \underline{\underline{86}}$$

eg: $A_1: x_1 + x_2 + x_3 = 11$ $x_1 \geq 4$ $7 + 2C_2 = 9C_2$	$A_2: x_1 + x_2 + x_3 = 11$ $x_2 \geq 5$ $6 + 2C_2 = 8C_2$	$A_3: x_1 + x_2 + x_3 = 11$ $x_3 \geq 7$ $4 + 2C_2 = 6C_2$
--	--	--

$$\begin{aligned} A_1 A_2: x_1 + x_2 + x_3 = 11 \\ \left. \begin{array}{l} x_1 \geq 4 \\ x_2 \geq 5 \end{array} \right\} \begin{array}{l} 2 + 2C_2 \\ = 4C_2 \end{array} \end{aligned}$$

$$\begin{aligned} A_1 A_3: x_1 + x_2 + x_3 = 11 \\ \left. \begin{array}{l} x_1 \geq 4 \\ x_3 \geq 7 \end{array} \right\} \begin{array}{l} 0 + 2C_2 = 2C_2 \end{array} \end{aligned}$$

$$\begin{aligned} A_2 A_3: x_1 + x_2 + x_3 = 11 \\ \left. \begin{array}{l} x_2 \geq 5 \\ x_3 \geq 7 \end{array} \right\} \begin{array}{l} \text{Ans} = 0 \end{array} \end{aligned}$$

$$A_1 A_2 A_3: \begin{array}{l} * x_1 + x_2 + x_3 = 11 \\ \{x_1 \geq 4, x_2 \geq 5, x_3 \geq 7\} \end{array} \quad {}^{13}C_2 - [\text{Ans}]$$

$$\text{Ans} = 0$$

$$\begin{aligned} A_1 \cup A_2 \cup A_3 &= A_1 + A_2 + A_3 - A_1 A_2 - A_2 A_3 - A_1 A_3 + A_1 A_2 A_3 \\ &= 9C_2 + 8C_2 + 6C_2 - 4C_2 - 2C_2 - 0 + 0 \end{aligned}$$

$$\Rightarrow 72$$

$$\text{Total} = {}^{13}C_2 \therefore {}^{13}C_2 - 72 \Rightarrow 78 - 72 \Rightarrow \underline{\underline{6}}$$

Derangement: Nothing is at its place
(Dn) (n > 1)

$$D_1 = 0$$

$$D_2 = 1 \left\{ \frac{1}{2} \times 2 \right\} \text{ Only possibility of derangement}$$

$$D_3 = 3! - (\text{at least 1 element @ right})$$

$$D_4 = 4! - (\text{at least 1 element @ right})$$

$$= \text{Total Arrangement} - (A_1 \cup A_2 \cup A_3 \cup A_4)$$

$$A_1 \cup A_2 \cup A_3 \cup A_4 = A_1 + A_2 + A_3 + A_4 - \sum A_i A_j + \sum A_i A_j A_k - |A_1 A_2 A_3 A_4|$$

$$= 4! - \left(4! - \frac{4!}{2!} + \frac{4!}{3!} - \frac{4!}{4!} \right) \Rightarrow \underline{9}$$

$$D_n = n! \left[1 - \frac{1}{2!} + \frac{1}{3!} - \frac{1}{4!} + \dots + \frac{(-1)^n}{n!} \right]$$

$$D_1 = 0 \quad D_5 = 44$$

$$D_2 = 1 \quad D_6 = 265$$

$$D_3 = 2$$

$$D_4 = 9$$

$$D_n \approx 0.367 n! \quad (n \geq 7)$$

expansion

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

$$e = 2.7$$

$$1/e = 0.36$$

$$e^{-1} = 0.36$$

$$e^{-1} = 1 + \frac{(-1)}{1!} + \frac{(-1)^2}{2!} + \frac{(-1)^3}{3!} + \dots$$

$$0.36 = 1 - 1 + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \frac{1}{5!} + \dots$$

$$0.36 = \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \frac{1}{5!} + \dots$$

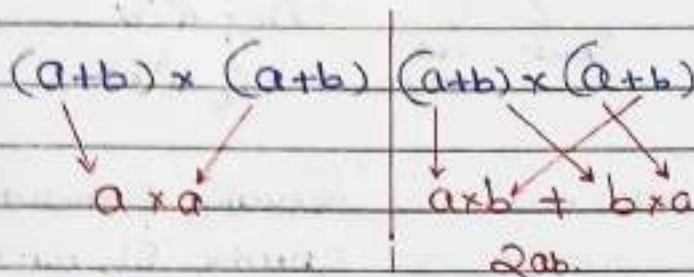
$$D_n = n! \left[\frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \dots + \frac{(-1)^n}{n!} \right]$$

$$D_n \approx n! (0.36)$$

COMBINATORICS (Continued).

$$(a+b)^2 = a^2 + 2ab + b^2$$

$$= 1.a^2 + 2.ab + 1.b^2$$



* $2C_1$
 $\rightarrow 2$ ways we can take ab out.

$$(a+b)$$

a or b

$1.a^2 = 1$ way
 we can take a^2 out.

* 2 boxes $(a+b)$
 how many ways we can take
 '1b' out from 2 boxes.
 $(_+b) \times (_+b)$
 $\Rightarrow 2C_1 ab$

$$(a+b)^2 = a^2 + 2ab + b^2$$

$$= 1.a^2 + 2ab + 1.b^2$$

$$\Rightarrow {}^2C_0 \cdot a^2 b^0 + {}^2C_1 ab + {}^2C_2 a^0 b^2$$

$\downarrow$
 2 boxes
 1b out.

$\downarrow$
 2 boxes
 2b's out.

$\rightarrow$ Out of 2 boxes how many
 ways we take '2 b' at same
 time:

$$(a+b) \times (a+b)$$

$\rightarrow b \times b \Rightarrow {}^2C_2$

* Power of a is decreasing & power
 of b is increasing.

* Power of a and b will always be 'n'.

$$\Rightarrow (a+b)^n = {}^nC_0 a^n b^0 + {}^nC_1 a^{n-1} b^1 + {}^nC_2 a^{n-2} b^2 \dots {}^nC_n a^0 b^n$$

$$(a+b)^n = \sum_{i=0}^n {}^nC_i \times a^{n-i} \times b^i$$

Note:

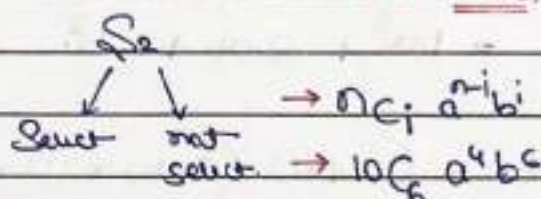
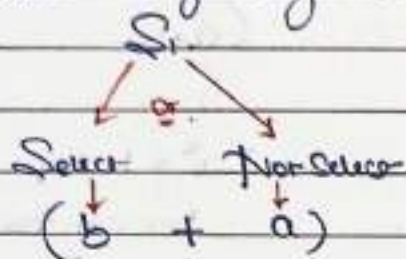
$${}^nC_x = {}^nC_{n-x}$$

$\rightarrow$ binomial coefficient: no. of ways we can
 take b^i outside in i
 boxes.

$\{a, b, c\} \rightarrow$ no. of ways to take
 1 element out = 3C_1 } no. of ways
 we left 2 elements
 inside.

Q1. How many ways we can select 6 students in a class of 10?

Ans: $^{10}C_6$



$$(a+b) \times (a+b) \times (a+b) \dots$$

$\downarrow$
 $b \times b \times b$

* Selecting 6 students means
Selecting 6b out of 10b.

$$(a+b)^n$$

$$(a+b)^n = \sum_{i=0}^n {}^nC_i a^{n-i} b^i$$

$$\Rightarrow \underline{\underline{{}^{10}C_6 a^4 b^6}}$$

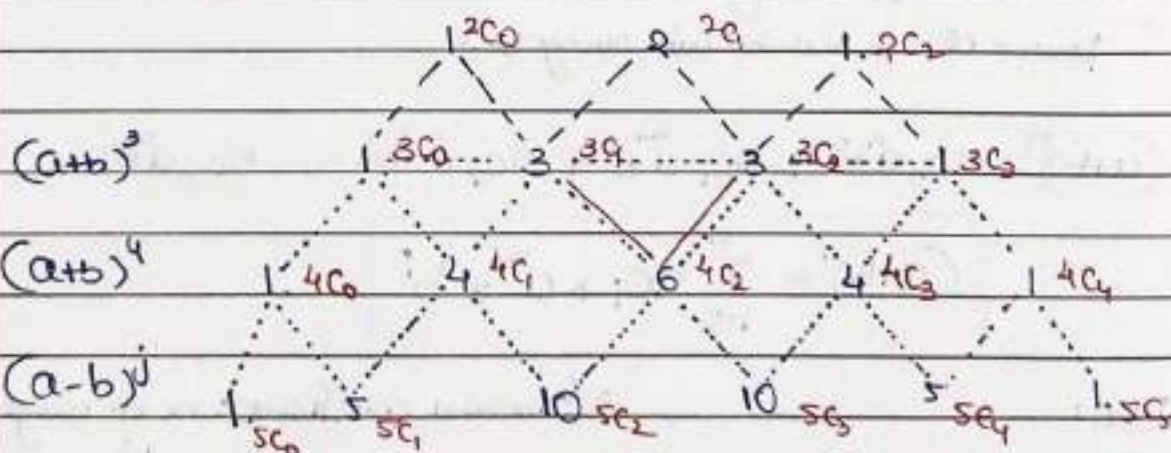
* Selecting 6 student
* Coefficient of b^6 .

eg: $(a+b)^n = \sum_{i=0}^n {}^nC_i a^{n-i} b^i$

$a=1, b=1$

$$(1+1)^n = \sum_{i=0}^n {}^nC_i (1)^{n-i} (1)^i$$

$$2^n = \sum_{i=0}^n {}^nC_i = \boxed{{}^nC_0 + {}^nC_1 + {}^nC_2 + \dots + {}^nC_n = 2^n}$$



$$3C_2 + 3C_1 = 4C_2$$

$$3 + 3 = 6$$

$$\Rightarrow {}^nC_x + {}^nC_{x+1} = {}^{n+1}C_{x+1}$$

$$\rightarrow (x+y)^{30} \text{ co-efficient of } y^{10} = {}^{30}C_{10} = {}^{30}C_{20}$$

$$\Rightarrow {}^{30}C_{10} 2^{20} y^{10}$$

$$\rightarrow (2x+3y)^{30} \text{ co-efficient of } y^{10} = {}^{30}C_{10} 2^{20} 3^{10}$$

$${}^{30}C_{10} (2x)^{20} (3y)^{10} \Rightarrow {}^{30}C_{10} 2^{20} 3^{10} 2^{20} y^{10}$$

Extended Binomial Coefficients:

$$\rightarrow {}^{11}C_2 = \frac{(11)!}{2! \times 9!} = \frac{11 \times 10}{2 \times 1} \Rightarrow {}^{11}C_2 = \frac{11 \times 10}{2 \times 1}$$

$$\rightarrow {}^{11}C_3 = \frac{11 \times 10 \times 9}{3 \times 2 \times 1}$$

$${}^nC_k = \frac{n(n-1)(n-2)\dots(n-k+1)}{k!}$$

$$-{}^nC_k = \frac{(-n)(-n-1)(-n-2)\dots(-n-k+1)}{k!}$$

$$= \frac{(-n-k+1)(-n-2)(-n-3)\dots(-n)}{k!} = (-1)^k \frac{(n+k-1)(n+k-2)\dots(n)}{k!}$$

$$= (-1)^k \frac{(n+k-1)!}{k! \times (n-1)!}$$

$$\left. \begin{array}{l} \frac{(n-1)!}{(n-1)!} \left\{ \begin{array}{l} \div \text{ by } n! \\ \text{by } (n-1)! \end{array} \right\} \end{array} \right\}$$

$$\boxed{-{}^nC_k = (-1)^k \cdot {}^{n+k-1}C_k} \rightarrow \frac{(-1)^k (n+k-1)!}{k! \times (n+k-1-k)!} = \frac{(-1)^k (n+k-1)!}{k! (n-1)!}$$

$$\text{eg: } -{}^{11}C_2 = {}^{12}C_2 \quad | \quad -{}^{11}C_2 = -{}^{13}C_2 \quad | \quad -{}^1C_0 = 1 \quad | \quad -{}^1C_1 = -1$$

$$-{}^{11}C_2 = -{}^nC_k \begin{cases} \rightarrow \text{even (total +ve)} \\ \rightarrow \text{odd (total -ve)} \end{cases}$$

$$n=11 \quad k=2 = (-1)^k {}^{n+k-1}C_k \\ = (-1)^2 \Rightarrow {}^{12}C_2$$

Extended Binomial Co-efficients: (Continued)

$$-n C_k = (-1)^k n+k-1 C_k$$

even odd

Co-efficient $\rightarrow$ +ve. Co-efficient $\rightarrow$ -ve.

eg: $-11 C_2 = +$

$-11 C_3 = -$

$$(a+b)^n = \sum_{i=0}^n n C_i a^{n-i} b^i$$

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + x^4 \dots$$

$$\frac{1}{1-x} = (1-x)^{-1} = (1+(-x))^{-1} = n C_0 a^{n-0} b^0 + n C_1 a^{n-1} b^1 + n C_2 a^{n-2} b^2 \dots$$

$$\left. \begin{array}{l} a=1 \\ b=-x \\ n=-1 \end{array} \right\}$$

$$(a+b)^n = -1 C_0 (1^{-1-0}) (-x)^0 + -1 C_1 (1^{-1-1}) (-x)^1$$

$$\downarrow \qquad \qquad \downarrow$$

$$1 \times 1 \times 1 + (-1) \times 1 \times (-x)$$

$$\rightarrow 1 + x + x^2 + (-1) \times (-x)$$

$$\frac{1}{1-ax} = (1+(-ax))^{-1}$$

$$= -1 C_0 (1^{-1-0}) (-ax)^0 + -1 C_1 (1^{-1-1}) (ax)^1 + -1 C_2 (1)^{-1-2} (-ax)^2$$

$$1 + (-1) \times 1 \times (-1) (ax) + (ax)^2 + (ax)^3$$

$$\Rightarrow \frac{1}{1-ax} = 1 + ax + (ax)^2 + (ax)^3 + (ax)^4 \dots$$

$$\rightarrow a=1, \frac{1}{1-x} = 1 + x + x^2 \dots$$

$$\rightarrow a=2, \frac{1}{1-2x} = 1 + 2x + (2x)^2 + (2x)^3 \dots$$

$$\Rightarrow \frac{1}{1+ax} = (1+ax)^{-1} = -1 C_0 (1^{-1-0}) (ax)^0 + -1 C_1 (1)^{-1-1} (ax)^1$$

$$\frac{1}{1+ax} = 1 - ax + (ax)^2 - (ax)^3 + (ax)^4 \dots$$

Q1. How many ways to distribute 10 similar coins among 3 children such that each child gets atleast 2 coins and atmost 4 coins?

Ans. atleast: 2 coins. } $2 \leq x_i \leq 4$
 atmost: 4 coins. }

0	0	0
2	2	2
3	3	3
4	4	4

 Giving 2 coins is represented as x^2 .

0	0	0
x^2	x^2	x^2
x^3	x^3	x^3
x^4	x^4	x^4

 One possible soln.
 $= 2 + 4 + 4 = 10$ ($2 \leq x_i \leq 4$)
 Co-efficient of $x^{10} = ?$

$$(x^2 + x^3 + x^4) \times (x^2 + x^3 + x^4) \times (x^2 + x^3 + x^4)$$

$$x^2 + x^3 + x^4 = x^2(1 + x + x^2)$$

$$x^2 + x^3 + x^4 = x^2(1 + x + x^2)$$

$$e_1 + e_2 + e_3 = 10$$

→ $(x^2 + x^3 + x^4)^3$
 * take x^2 common
 $(x^2)^3 [1 + x + x^2]^3 \rightarrow x^6 \left[\frac{1-x^3}{1-x} \right]^3 = x^6 (1-x^3)^3 (1-x)^{-3}$
 $= x^6 [1 - 3x^3 + 3x^6 - x^9] [1-x]^{-3}$
 (expanding the equation).
 $\Rightarrow x^6 [1 - 3x^3 + 3x^6 - x^9] \left[{}^{-3}C_0 1^3 (-x)^0 + {}^{-3}C_1 1^2 (-x)^1 + {}^{-3}C_2 1^1 (-x)^2 + \dots \right]$
 Co-efficient of x^{10}
 $\Rightarrow [x^6 - 3x^9 + 3x^{12} - x^{15}] [\dots]$
 $\Rightarrow x^6 [-3C_4 (1) (-x)^4] - 3x^9 [-3C_1 1^3 (-x)^1]$
 $\Rightarrow x^6 (15x^4) - 3x^9 (3x) \Rightarrow (15-9)x^{10}$
 $\Rightarrow 6$

Ans=6

Q2. How many ways to distribute 12 coins among 4 students where each student gets atleast 2 coins and atmost 5 coins?

Ans. $(x^2 + x^3 + x^4 + x^5)^4$ - children.
 atleast atmost
 Co-efficient of $x^{12} = ?$
 $(x^2 + x^3 + x^4 + x^5)^4$
 * take x^2 common
 $(x^2)^4 [1 + x + x^2 + x^3]^4 \Rightarrow x^8 \left[\frac{1-x^4}{1-x} \right]^4$

$$\Rightarrow x^8 \frac{(1-x^4)^4}{(1-x)} = x^8 (1-x)^4 (1-x)^{-1}$$

$$\Rightarrow x^8 (1-4x^4+6x^8 \dots) (1-x)^{-4}$$

$$(x^8 - 4x^{12} + \dots) (1-x)^{-4}$$

Ans = 31

$$\Rightarrow x^8 (-4C_4 \cdot 1 (-x)^4) - 4x^{12} \Rightarrow \underline{31x^{12}}$$

Q3. How many ways we can distribute 15 coins among 4 children such that each student gets at least 2 coins & at most 6 coins?

Ans $(x^2 + x^3 + x^4 + x^5 + x^6)^4$ children

at least at most Co-efficient of x^{15}

$$(x^2 + x^3 + x^4 + x^5 + x^6)^4$$

$$(x^2)^4 (1 + x + x^2 + x^3 + x^4)^4$$

$$x^8 \left(\frac{1-x^5}{1-x} \right)^4 = x^8 (1-x)^4 (1-x)^{-4}$$

$$x^8 (-4C_4 (-x)^4) - 4x^{13} (-4C_2 \cdot 1 (-x)^2) = x^8 (1-4x^5+6x^8 \dots) (1-x)^{-4}$$

$$(10C_3 x^{15} - 4 \times 8C_2 x^{15}) = (x^8 - 4x^{13} + \dots) (1-x)^{-4}$$

Ans. $\Rightarrow 80$

eg: x^{12}

$$(x^3 + x^4 + x^5 + x^6 \dots)^3$$

$$(x^3)^3 \cdot (1 + x + x^2 + \dots)^3$$

$$x^9 \left(\frac{1}{1-x} \right)^3 \cdot x^9 \cdot (1-x)^{-3} \rightarrow x^9 (-3C_3 (-x)^3)$$

$$= (-1)^3 \cdot 3+3-1 C_3 (-1)^3 x^3$$

$$= 5C_3 x^{12}$$

$\Rightarrow 10$

Q4. There are 300 identical envelopes

(25 envelope/package $\rightarrow$ 4 students)

at least $\frac{150}{25} \rightarrow 6$

$$\text{Almost } 1000 = \frac{1000}{25} = 40$$

$$\text{Coefficient of } x^{100} \\ (x^6 + \dots + x^{40})^4$$

$$(x^6)^4 [1 + x + \dots + x^{34}]^4$$

$$x^{24} \left[\frac{1-x^{35}}{1-x} \right]^4$$

$$x^{24} [1-x^{35}]^4 [1-x]^{-4} \Rightarrow x^{24} [1 - 4x^{35} + 6x^{70} - 4x^{105}]$$

$$(x^{24-4} \binom{4}{6} x^{96}) - 4x^{59}$$

$$\Rightarrow -4 \binom{4}{6} (x^5) \Rightarrow +6x^{94} \left(-4 \binom{4}{26} (-x)^{26} \right)$$

Generating Functions

$$\langle a_0, a_1, a_2, a_3, \dots \rangle$$

$$G(x) = a_0 x^0 + a_1 x^1 + a_2 x^2 + \dots \quad \text{closed form}$$

$$G(x) = \sum_{i=0}^{\infty} a_i x^i$$

$$\langle 1, 1, 1, 1, \dots \rangle \\ a_0 \ a_1 \ a_2 \ a_3 \ \dots$$

$$G(x) = a_0 x^0 + a_1 x^1 + a_2 x^2 + a_3 x^3 + a_4 x^4 \\ = 1x^0 + 1x^1 + 1x^2 + 1x^3 + 1x^4 \\ = 1 + x + x^2 + x^3 + x^4 \dots$$

$$G(x) = \frac{1}{1-x}$$

$$\rightarrow \frac{1}{1-x} = 1 + x + x^2 + x^3 + x^4 \dots$$

$$\frac{d}{dx} \left(\frac{1}{1-x} \right)$$

$$= \frac{-1}{(1-x)^2} \frac{d}{dx} (-x)$$

$$= \frac{-1}{(1-x)^2} (-1) = \frac{1}{(1-x)^2}$$

$$\frac{d}{dx} \left(\frac{1}{x} \right) = -\frac{1}{x^2}$$

$$\frac{1}{(1-x)^2} = 0 + 1 + 2x + 3x^2 + 4x^3 + 5x^4 \dots$$

$$\rightarrow 1, -1, 1, -1, 1, \dots$$

$$G(x) = \frac{1}{1+x}$$

eg: $0, 0, 0, 6, -6, 6, -6$

$$C(x) = a_0x^0 + a_1x^1 + a_2x^2 + a_3x^3 \dots$$

$$G(x) = \frac{6x^3}{1+x}$$

$$6x^3 - 6x^4 + 6x^5 - 6x^6 \dots$$

take $6x^3$ common,

$$6x^3 [1 - x + x^2 - x^3 + x^4 - x^5 \dots]$$

$$\Rightarrow \underline{6x^3}$$

$$1+x$$

eg: $0, 0, 1, a, a^2, a^3, \dots (a \neq 0) \rightarrow \frac{x^2}{1-ax}$

$$1, 0, 1, 0, 1, \dots \rightarrow \frac{1}{(1-x^2)}$$

7, 8, 9, 10, ...

$$G(x) = a_0x^0 + a_1x^1 + a_2x^2 + a_3x^3 + a_4x^4 \dots$$

$$= 7x^0 + 8x + 9x^2 + 10x^3 \dots$$

$$= 7 + 8x + 9x^2 + 10x^3$$

$$= 1 + 6 + 6a + 2a + 6a^2 + 3a^2 + 6a^3 + 4a^3.$$

$$\rightarrow \underline{6 + 6x + 6x^2 + \dots} + \boxed{1 + 2x + 3x^2 + 4x^3 + \dots}$$

$$\hookrightarrow \frac{6}{1-x}$$

$$\frac{C_1}{(1-x)^2}$$

$$\Rightarrow \frac{6}{1.x} + \frac{1}{(1-x)^2}$$

$$\rightarrow G_1(x) = \frac{x^3}{1-x^2}$$

$$= x^2 [1 + x^2 + x^4 + x^6 \dots]$$

$$= x^3 + x^5 + x^7 + x^9 \dots$$

$$C(x) = 0 \cdot x^0 + 0 \cdot x^1 + 0 \cdot x^2 \dots \Rightarrow \langle 0, 0, 0, 1, 0, 1, 0 \dots \rangle$$

$$\rightarrow G(x) = \frac{1}{3x} = \frac{1}{3[1-x/3]} = \frac{1}{3} \times \frac{1}{(1-x/3)}$$

$$= \frac{1}{3} \left[1 + \left(\frac{x}{3}\right) + \left(\frac{x}{3}\right)^2 + \left(\frac{x}{3}\right)^3 \dots \right]$$

$$= \frac{1}{3} + \frac{1}{3} \times \frac{1}{3} x + \frac{1}{3} \times \frac{1}{3^2} x^2 + \frac{1}{3} \times \frac{1}{3^3} x^3 + \dots$$

$$\Rightarrow \left\langle \frac{1}{3}, \frac{1}{9}, \frac{1}{27}, \dots \right\rangle$$

$$\rightarrow G(x) = \frac{1}{1-x} + 3x^2 - 11$$

$$= [1 + x + x^2 + x^3 + x^4 + \dots] + [3x^2 - 11]$$

$$= (1 - 11) + x + x^2 + \dots + x^6 + (x^7 + 2x^7) + \dots$$

$$= -10 + \dots + 4x^7 + \dots$$

$$a_0 = -10 \quad \langle -10, 1, 1, \dots, 4, 1, \dots \rangle$$

Recurrence Relation

In a colony, there are 5 bacteria which is at present at time 0. Bacteria are increasing 2 times as the previous what will be the total bacteria at time 100?

a_0 : no. of bacteria at time 0 $\rightarrow 5$

a_{100} : no. of bacteria at time 100 $\rightarrow$

Initial Condition: $a_0 = 5$

$$a_1 = 2 \times a_0$$

$$a_2 = 2 \times a_1 = 2(a_0 \times 2)$$

$$= 2^2 \cdot a_0$$

$$a^3 = 2 \times a_2$$

$$= 2(2^2 \cdot a_0)$$

$$a^3 = 2^3 \cdot a_0 \quad \therefore \underline{a_n = 2^n \cdot a_0 \rightarrow O(2^n)}$$

$$\text{Type-1: } a_n = d \cdot a_{n-1}$$

$a_0 = \text{initial condition}$

$$\boxed{a_n = d^n \cdot a_0}$$

$$\rightarrow (a_{n+2}) = (3a_{n+1} - 2a_n) \quad (n \geq 0)$$

$$\sum a_n \cdot 2^{n+2} = 3 \cdot \sum a_{n+1} \cdot 2^{n+2} - 2 \sum a_n \cdot 2^{n+2}$$

$$\sum_{n \geq 0} a_{n+2} \cdot 2^{n+2} = 2 \cdot 2 \sum_{n \geq 0} a_{n+1} \cdot 2^{n+1} - 2^2 \sum_{n \geq 0} a_n \cdot 2^n$$

$$G(x) - a_0 x^2 - a_1 x = 3x(G(x) - a_0 x) - 2x^2 G(x)$$

$$G(x) - 0x^0 - x = 3x(G(x)) - 2x^2 G(x)$$

$$G(x) - x = (3x - 2x^2) G(x)$$

$$G(x) + (3x - 2x^2) G(x) = x$$

$$G(x) [1 - 3x + 2x^2] = x$$

$$G(x) = \frac{x}{1 - 3x + 2x^2}$$

$$\rightarrow G(x) = \sum_{i=0}^{\infty} a_i x^i = G(x) = \sum_{n=0}^{\infty} a_n x^n$$

$$G(x) = a_0 x^0 + a_1 x^1 + a_2 x^2 + \dots$$

$$G(x) = a_0 x^0 + a_1 x^1 + \sum_{n=2}^{\infty} a_n x^n$$

$$\rightarrow G(x) = \frac{x}{1 - 3x + 2x^2} = \frac{x}{(1-x)(1-2x)} = \left\{ \frac{A}{1-x} + \frac{B}{1-2x} \right\} = \dots$$

$\frac{1 - 3x + 2x^2}{1 - x - 2x + 2x^2}$	$= A(1-x) + B(1-2x)$
$\frac{(1-x) - 2x(1-x)}{(1-x)(1-2x)}$	$= A(-2x) + B(-Bx)$
	$= \frac{(-2Ax - Bx)}{(1-x)(1-2x)}$

$$G(x) = \frac{A}{1-x} + \frac{B}{1-2x} \quad A = -1, B = 1$$

$$= \frac{-1}{1-x} + \frac{1}{1-2x}$$

$$G(x) = \left(\frac{1}{1-2x} \right) - \left(\frac{1}{1-x} \right)$$

$$\sum_{n=0}^{\infty} a_n x^n = \sum_{n=0}^{\infty} 2^n x^n - \sum_{n=0}^{\infty} 1^n x^n$$

$$a_n = 2^n - 1$$

$$\frac{1}{1-2x} = 1 + (2x) + (2x)^2 + (2x)^3 + \dots$$

$$= \sum_{n=0}^{\infty} 2^n x^n$$

$$\frac{1}{1-x} = 1 + x + x^2 + \dots$$

$$= \sum_{n=0}^{\infty} 1^n x^n$$

$$a_n = 2n + 3$$

$$\sum a_n x^n = 2 \cdot \sum n x^n + 3 \sum x^n \quad (x, 2x^2)$$

$$G(x) = 2 \cdot \frac{x}{(1-x)^2} + \frac{3}{(1-x)}$$

$$\sum_{n \geq 0} n x^n$$

$$\frac{x}{(1-x)^2} = 0x^0 + 1x^1 + 2x^2 + 3x^3 + \dots$$

$$= x + 2x^2 + 3x^3 + 4x^4 + \dots$$

$$x(1 + 2x + 3x^2 + 4x^3 + \dots)$$

$$a_{n+2} = 3a_{n+1} - 2a_n$$

$$x^2 - 3x + 2$$

$$x^2 - 3x + 2 = 0$$

$$\text{Roots} = 2, 1$$

$$n=0$$

$$a_0 = 2^0 C_1 + 1^0 C_2$$

Characteristic Equation: $a_n = 2^n C_1 + 1^n C_2$

$$a_n = 2^n \cdot 1 - 1^n$$

$$a_n = 2^n - 1$$

$$a_n = 5a_{n-1} - 6a_{n-2}$$

$$x^2 - 5x + 6$$

$$x^2 - 5x + 6$$

$$\text{Roots: } 3, 2$$

Characteristic Equation: $a_n = 3^n C_1 + 2^n C_2$

Type-2:

$$a_n = x a_{n-1} + y a_{n-2}$$

$$\text{Roots: } R_1, R_2$$

Characteristic Eq: $a_n = (R_1)^n C_1 + (R_2)^n C_2$

Type-3

$$a_n = x a_{n-1} + y a_{n-2}$$

$$\text{Roots: } R, R$$

C.E: $a_n = R^n C_1 + n R^n C_2$

$$a_n = a_{n-1} + a_{n-2}$$

$$a_0 = 0, a_1 = 1$$

$$x^2 = x + 1$$

$$x^2 - x - 1 = 0$$

$$= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\rightarrow \frac{1 \pm \sqrt{5}}{2}$$

$$\text{C.E} = a_n = \left(\frac{1+\sqrt{5}}{2}\right)^n C_1 + \left(\frac{1-\sqrt{5}}{2}\right)^n C_2$$

$$C_1 = \frac{1}{\sqrt{5}} \quad C_2 = -\frac{1}{\sqrt{5}}$$

$$a_n = \frac{1}{\sqrt{5}} \left(\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n \right)$$

$$a_n = 6a_{n-1} - 9a_{n-2}$$

$$CE: a_n = 3^n C_1 + n \cdot 3^n C_2$$

$$n=0$$

$$a_0 = 3^0 C_1 + 0 \cdot 3^0 \cdot C_2$$

$$1 = 1 \cdot C_1$$

$$C_1 = 1$$

$$n=1$$

$$a_1 = 3^1 C_1 + 1 \cdot 3^1 C_2$$

$$2 = 3 \cdot 1 + 3 C_2$$

$$-1 = 3 C_2 \quad C_2 = -1/3$$

$$a_n = 3^n + n \cdot 3^n \left(-\frac{1}{3}\right)$$

Type 4:

$$a_n = x a_{n-1} + y a_{n-2} + z a_{n-3}$$

Roots: R_1, R_2, R_3

$$CE: a_n = R_1^n C_1 + R_2^n C_2 + R_3^n C_3$$

Roots: R, R, R_1

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Roots: R, R, R_1

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Euler-totient-Function: $\phi(n)$

no. of relative prime less than n

$$\phi(4) = 2$$

$$1. \gcd(1, 4) = 1 \checkmark$$

$$2. \gcd(2, 4) = 2 \times$$

$$3. \gcd(3, 4) = 1 \checkmark$$

$$4. \gcd(4, 4) = 4 \times$$

$$\phi(3) = 2$$

$$\begin{array}{c} \checkmark \quad \checkmark \quad \times \\ 1 \quad 2 \quad 3 \\ \underbrace{\hspace{1cm}} \\ 2 \end{array}$$

$$\gcd(2, 3) = 1$$

Relative prime

Coprime

$$\gcd(a, b) = 1$$

$$\gcd(3, 5) = 1$$

3, 5 are relative prime

$$\phi(12) = 4$$

$$1 \checkmark \quad 5 \checkmark \quad 9 \times$$

$$2 \times \quad 6 \times \quad 10 \times$$

$$3 \times \quad 7 \checkmark \quad 11 \checkmark$$

$$4 \times \quad 8 \times \quad 12 \times$$

$$\phi(12) = \phi(3 \times 4)$$

$$= \phi(3) \times \phi(4)$$

$$\gcd(3, 4) = 1$$

condition

- to be

satisfied

$$= 2 \times 2 \Rightarrow 4$$

$$\phi(12)$$

$$\phi(3) \cdot \phi(4)$$

$$\gcd(3, 4) = 1$$

$$\phi(A \cdot B) = \phi(A) \cdot \phi(B)$$

$$\gcd(A, B) = 1$$

$$\phi(7) = 6$$

1
2
3
4
5
6
7

$$\gcd(1, 7) = 1$$

$$\phi(p) = p-1 \quad \{ p \text{ is prime no.} \}$$

$$\begin{aligned} \phi(49) &= \text{Total elements} - \text{non relative prime} \\ &= 49 - 7 = 49 - \left(\frac{49}{7}\right) \end{aligned}$$

$$\gcd(7, 49) = 7 \quad (\text{non relative prime})$$

$$\phi(49) = 49 - \frac{49}{7}$$

$$\phi(7^2) = 7^2 - \frac{7^2}{7}$$

$$\phi(p^a) = p^a - \frac{p^a}{p} \quad \begin{cases} a \geq 0 \\ p \text{ prime} \end{cases}$$

$$n = p_1^a \cdot p_2^b \cdot p_3^c \dots \quad \begin{cases} a, b, c \dots \geq 0 \\ p_1, p_2, p_3 \dots \\ \text{prime no} \end{cases}$$

$$4 = 2^2 \cdot 3^0 \cdot 5^0 \dots$$

$$5 = 2^0 \cdot 3^0 \cdot 5^1 \dots$$

$$n = 2^k y \quad \begin{cases} k \geq 0 \\ y \rightarrow \text{odd no.} \end{cases}$$

$$n = p_1^a \cdot p_2^b \cdot p_3^c \dots$$

$$\phi(n) = \phi(p_1^a \cdot p_2^b \cdot p_3^c \dots)$$

$$= \phi(p_1^a) \times \phi(p_2^b) \times \phi(p_3^c)$$

$$= \left(p_1^a - \frac{p_1^a}{p_1}\right) \times \left(p_2^b - \frac{p_2^b}{p_2}\right) \times \left(p_3^c - \frac{p_3^c}{p_3}\right)$$

$$= p_1^a \left(\frac{p_1-1}{p_1}\right) p_2^b \left(\frac{p_2-1}{p_2}\right) p_3^c \left(\frac{p_3-1}{p_3}\right)$$

$$\phi(n) = n \frac{(p_1-1)}{p_1} \frac{(p_2-1)}{p_2} \frac{(p_3-1)}{p_3} \dots$$

$$\text{eg: } \phi(110) \quad \phi(n) = n \frac{(p_1-1)}{p_1} \frac{(p_2-1)}{p_2} \frac{(p_3-1)}{p_3}$$

$$110 = 2 \cdot 5 \cdot 11$$

$$= 110 \frac{(2-1)}{2} \frac{(5-1)}{5} \frac{(11-1)}{11}$$

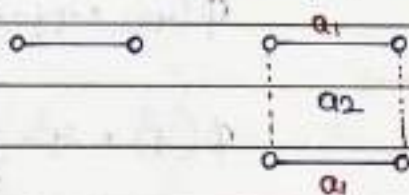
$$= 1 \cdot 4 \cdot 10 = 40$$

Recurrence Relations.homogeneous

$$a_{n+1} = 6a_n - 7a_{n-2} + f(n)$$

$$f(n) = 0$$

$$a_n = 2a_{n-1} + f(n)$$



$$a_2 = 2a_1 + 2^1$$

non-homogeneous

$$a_n = \lambda a_{n-1} + \mu a_{n-2} + f(n)$$

$$f(n) = 2^n$$

$$f(n) = n \cdot 2^n$$



$$a_3 = 2a_2 + 2^2$$

$$a_2 = 2a_1 + 2^1$$

$$a_3 = 2a_2 + 2^2$$

$$a_{n+1} = 2a_n + 2^n$$

$$f(n) = 2^n$$

$$\rightarrow a_n - 3a_{n-1} = 5(7^n) \quad a_0 = 2$$

$$\text{Total Soln} = a_n = a_n^h + a_n^p$$

$$a_n - 3a_{n-1} = 0$$

$$\lambda - 3 = 0$$

$$\lambda = 3$$

Roots:

$$a_n^h = 3^n C$$

$$a_n^p = A(7^n)$$

$$\frac{A(7^n)}{7^n} - \frac{3A(7^{n-1})}{7^{n-1}} = \frac{5(7^n)}{7^n}$$

$$\frac{A(7^n)}{7^n} - 3A = \frac{5(7^n)}{7^n}$$

$$a_n = -\frac{27}{4} \cdot 3^n + \frac{35}{4} \cdot 7^n$$

$$7A - 3A = 5 \cdot 7$$

$$4A = 35$$

$$A = 35/4$$

$$a_n^p = \frac{35}{4}(7^n)$$

$$a_n = a_n^h + a_n^p$$

$$a_n = 3^n C + \frac{35}{4} \cdot 7^n$$

$$n=0$$

$$a_0 = 3^0 C + \frac{35}{4} \cdot 7^0$$

$$2 = \frac{C + 35}{4}$$

$$\rightarrow a_{n+1} = 2a_n + 2^n \quad a_0 = 0$$

$$a_n = a_n^{(h)} + a_n^{(p)}$$

$$a_{n+1} = 2a_n$$

$$a = 2$$

$$\text{Root: } 2$$

$$a_n^{(h)} = 2^n C$$

$$a_n^{(p)} = A n (2^n)$$

$$\frac{A(n+1) \cdot 2^{n+1}}{2^n} = \frac{2(A n 2^n)}{2^n} + \frac{2^n}{2^n}$$

$$a_n^{(p)} = \frac{1}{2} n (2^n)$$

$$a_n = a_n^{(h)} + \frac{1}{2} n (2^n)$$

$$2A(n+1) = 2An + 1$$

$$2An + 2A = 2An + 1$$

$$\underline{2A = 1}$$

$$A = 1/2, \quad a_n = 2^n C + \frac{1}{2} n (2^n)$$

$$0 = 2^0 C + \frac{1}{2} \cdot 0 (2^0)$$

$$C = 0$$

$$a_n = \frac{1}{2} n (2^n)$$

$$\boxed{a_n = n \cdot 2^{n-1}}$$

$$a_n - 3a_{n-1} = 5 (3^n) \quad a_0 = 0$$

Homogeneous:

$$a_n - 3a_{n-1} = 0$$

$$\lambda - 3 = 0$$

$$\text{Root: } 3$$

$$a_n^{(h)} = 3^n C$$

$$a_n^{(p)} = A n (3^n)$$

$$A n (3^n) - 3A(n-1) 3^{n-1} = 5 \cdot 3^n$$

$\div$ by 3.

$$\frac{A n (3^n)}{3^n} - \frac{3A(n-1) 3^{n-1}}{3^n} = \frac{5 \cdot 3^n}{3^n}$$

$$A n - A(n-1) = 5$$

$$A n - A n + A = 5$$

$$\text{Root: } 3 \quad f(n) = 5 (3^n)$$

$$\rightarrow a_n^{(p)} = A n (3^n)$$

$$a_n^{(p)} = 5 n (3^n)$$

$$\boxed{a_n = 3^n \cdot 2 + 5 n \cdot 3^n}$$

$$a_n = a_n^{(h)} + a_n^{(p)}$$

$$a_n = 3^n C + 5 n (3^n)$$

$$a_0 = 0$$

$$0 = 3^0 C + 5 \cdot 0 (3^0)$$

$$C = 2$$

$f(n)$	$a_n^{(p)}$
5	A

Root: 3	$f(n)$	$a_n^{(p)}$
Root: 3	5 (3^n)	A (3^n)
Root: 3	5 (3^n)	A n (3^n)
	5 (3^n)	A n^2 (3^n)

$f(n)$	$a_n^{(p)}$
n	A n + b
n^2	A n^2 + B n + C
n (3^n)	A n + b (3^n)

$$\text{Root: } 3, 3$$

$$a_n^{(h)} = 3^n C_1 + 3^n n C_2$$

$$a_{n+2} - 6a_{n+1} + 9a_n = 3 (2^n) + 7 (3^n)$$

$$\text{Root: } 3, 3$$

$$a_n^{(h)} = 3^n C_1 + n 3^n C_2$$

$$a_n^{(p)} = A (2^n) + B n^2 (3^n)$$